

UNIVERSIDADE FEDERAL FLUMINENSE

MATHEUS DE CASTRO RIBEIRO

Nuclear Magnetic Resonance reservoir
characterization assisted by computational
simulations.

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Tese de Doutorado apresentada ao Programa de Pós-Graduação em Computação da Universidade Federal Fluminense como requisito parcial para a obtenção do Grau de Doutor em Computação. Área de concentração: Ciência da Computação.

Orientador:

Prof. Dr. Ricardo Leiderman

Coorientador:

Prof. Dr. Jefferson Gonçalves Filgueiras

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"E agora eu fico pensando: as pessoas que vão viver daqui a cem, duzentos anos, as pessoas para quem nós estamos, agora, abrindo caminho, será que elas vão falar bem de nós?"

— Anton Tchêkhov, *Tio Vânia*

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Resumo

A Ressonância Magnética Nuclear (RMN) é uma ferramenta poderosa para caracterizar meios porosos, oferecendo entendimento sobre o comportamento de fluidos em espaços porosos. Este trabalho implementa simuladores baseados em imagens para experimentos de RMN utilizando o método Random Walk, com foco na caracterização de rochas reservatório via difusometria e relaxometria. Desafios computacionais são tratados por meio de paralelização massiva baseada em GPU e modelagem de dados para acesso otimizado à memória.

Na difusometria, o coeficiente de difusão aparente é analisado ao longo do tempo utilizando a técnica de Gradiente de Campo Pulsado. Os comportamentos assintóticos de curto e longo prazo são usados para estimar a razão superfície-volume e a tortuosidade, correlacionando esses parâmetros com a permeabilidade absoluta em meios granulares sintéticos e imagens segmentadas de arenitos e carbonatos. Os resultados das simulações concordam bem com experimentos em arenitos, porém mostram desvios nos carbonatos, possivelmente devido à presença de heterogeneidades como a porosidade vugular.

Na relaxometria, investigamos o impacto de gradientes internos de campo magnético nos tempos de relaxação T_2 , influenciados por altos contrastes de susceptibilidade magnética. Para incorporar o mecanismo de relaxação difusiva, o fluxo de trabalho envolve o cálculo de mapas espaciais do campo magnético utilizando o método de elementos finitos baseado em imagens, seguido por simulações de Random Walk para modelar a dinâmica dos spins. Simulações realizadas em imagens segmentadas de modelos sinterizados de esferas de vidro com inclusões de óxidos de ferro e arenitos revelam desafios na interpretação de T_2 como indicador do tamanho de poros. Além disso, distribuições quantitativas de gradientes internos obtidas de simulações multi- τ de RMN mostram uma concordância variada com os dados experimentais, falhando em capturar gradientes internos de maior magnitude associados a poros menores. Por fim, propomos uma abordagem para incorporar os efeitos de difusão restrita na análise.

Palavras-chave: Ressonância magnética nuclear, Random walk, Meios porosos, Auto-difusão molecular, Gradientes pulsados de campo magnético, Relaxação por difusão.

Abstract

Nuclear Magnetic Resonance (NMR) is a powerful tool for characterizing porous media, offering insights into fluid behavior in pore spaces. This work implements image-based simulators of NMR experiments using the Random Walk method, focusing on reservoir rock characterization via diffusometry and relaxometry. Computational challenges are addressed through GPU-based massive parallelization and data modeling for optimized memory access.

In diffusometry, the apparent diffusion coefficient is analyzed over time using the Pulsed-Field Gradient NMR technique. Short and long-time asymptotics estimate surface-to-volume ratio and tortuosity, correlating these parameters with permeability in synthetic granular media and segmented images of sandstones and carbonates. Simulation results agree well with experiments for sandstones but show deviations in carbonates, possibly due to heterogeneities such as vugular porosity.

In relaxometry, we investigate the impact of internal magnetic gradients on T_2 relaxation times, influenced by high magnetic susceptibility contrasts. To incorporate the diffusive relaxation mechanism, the workflow involves computing spatial magnetic field maps using an image-based finite element method, followed by random walk simulations to model spin dynamics. Simulations performed on segmented images of sintered glass bead models with iron oxide inclusions and sandstone reveal the challenges in interpreting T_2 as a pore size indicator. Also, quantitative internal gradients distributions from multi- τ NMR simulations show mixed agreement with experimental data, failing to capture high-magnitude internal gradients linked to smaller pores. Last, we propose an approach to incorporate the effects of restricted diffusion into the analysis.

Keywords: Nuclear magnetic resonance, Random walk, Porous media, Molecular self-diffusion, Pulsed magnetic field gradient, Diffusion relaxation.

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Chapter 1

Introduction

A good understanding of the pore structure of reservoir rocks is crucial when studying the economic viability of a particular oil field in the petroleum industry. In this context, ^1H Nuclear Magnetic Resonance (NMR) offers valuable information about the ^1H on the fluid molecules filling the pore space. It is, therefore, a robust tool to evaluate reservoir rock properties. Several deliverables including porosity, micro-connectivity, pore size distribution, wettability and permeability can be extracted from NMR laboratory measurements from rock plug samples or from in situ NMR logging tools [24, 73, 88].

The Pulsed Field Gradient Spin Echo (PFGSE) is a well-established NMR technique to assess molecular diffusion in porous media. First proposed by Stejskal and Tanner [76], this technique is based on the application of radiofrequency (r.f.) excitation pulses followed by narrow gradient pulses with constant duration, δ , and amplitude, g , separated by a given observation time, Δ , (see Figure 1.1). The resulting time-dependent magnetic field gradient encodes the spatial position of nuclear spins, allowing the measurement of the diffusion coefficient [14, 64].

In the presence of motion restrictions, the diffusion coefficient presents a time dependence with very predictable asymptotic behaviors for both short and long observation times [70]. These trends allow geometrical characterization of the pore space: from short observation times, upon which surface relaxation effects are negligible, it is possible to estimate pore sizes [56]; from the long-time behavior, as the diffusion coefficient reaches a plateau, it is possible to extract information on the connectivity and the tortuosity of the pore network [72]. Further, these inferred parameters can also be used to evaluate the absolute permeability of reservoir rocks from prediction models based on the well-known Kozeny-Carman equation [30].

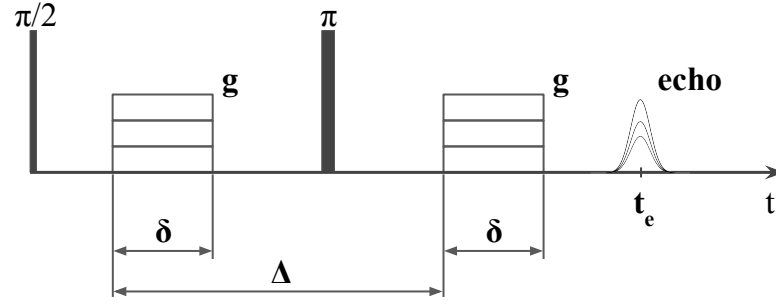


Figure 1.1: **PFGSE NMR:** The Pulsed Magnetic Field Gradient Spin Echo (PFGSE) NMR technique to assess molecular diffusion in porous media. It encodes the spatial position of nuclear spins based on its effective magnetic field. This is accomplished by applying finite narrow gradient pulses, $\mathbf{g}$, with a constant duration, δ , and amplitude, resulting in a time-dependent magnetic field gradient. An initial gradient pulse is applied after a radio frequency, r.f., excitation pulse ($\pi/2$), linearly modifying the local magnetic field. Its effect is sensed by the spins aligned with the $x'y'$ plane. After a given observation time interval, Δ , a refocusing pulse (π) is applied followed by a *reading* gradient pulse. The application of the second gradient pulse aims to restore the phases that were spread after the initial gradient pulse.

However, the PFGSE technique has some practical drawbacks despite its recognized value. Notably, when performing each diffusion measurement, it demands repeating the same sequence applying varying wave vectors $\mathbf{k}$, which can require several hours in order to complete the entire procedure. In addition, the effects of transversal or longitudinal relaxation and signal noise account for another major disadvantage, as it can often forbid accurate diffusion measurements before the asymptotic tortuosity limit is reached. In that sense, a simulation of the PFGSE NMR experiment to measure diffusion in digital porous media that is at the same time reliable and efficient is highly desirable to help the interpretation of experimental results.

Molecular self-diffusion also plays a crucial role in other NMR experiments. In relaxometry studies, such as estimating the characteristic transverse relaxation times, T_2 , of a specific formation, there is an enhanced signal relaxation due to the interaction of magnetized spins with spatial inhomogeneities of the static magnetic field within the porous space [15, 32, 12]. These internal magnetic field gradients become more pronounced when a significant contrast in magnetic susceptibility between the fluid and the materials composing the rock matrix is present [52, 71, 17].

In this context, the Carr-Purcell-Meiboom-Gill pulse sequence (CPMG) is arguably the most widely used NMR technique for acquiring the T_2 decay. This pulse sequence enables reducing the acquisition interval to minimize the effects of internal gradients.



Figure 1.2: **Micro-computed tomographic image:** Austin Chalk 3D model. In this example, the image is acquired at a resolution of $1\ \mu\text{m}/\text{voxel}$, representing a volume of around $1\ \text{mm}^3$ of the original sample segmented in porous (black) and solid (gray) phases.

However, this acquisition interval can only be shortened to a limit defined by the NMR tool [40, 41, 17]. Moreover, significant gradients in the sample can greatly affect the interpretation of relaxation time distributions derived from T_2 decay [54]. In low-field scenarios, high intensity gradients may indicate that the sample is abundant in clays or other materials with high iron concentrations [1]. The multi- τ method, which involves executing and processing multiple CPMG sequences with varying echo intervals, is a well-established method for quantitatively assessing the presence and impact of these internal gradients [40].

Recent technological advances allowed 3D imaging techniques to reach high-quality resolutions in the micrometre scale leading to great developments of image-based computational simulations in porous media. In this context, x-ray micro computed tomography (μ -CT) is the most established technique that allows spatial description of the pore space. Figure 1.2 shows a 3D model constructed from segmented micro-computed tomographic images of an Austin Chalk carbonate sample. The highly favorable operational cost comprehends a major advantage for current prominence of digital petrophysics.[25, 68].

Regarding time-domain NMR, trustworthy simulations open the door for better comprehension of the inherent mechanisms of magnetic resonance phenomena. Also, it enables a deeper understanding of the effects caused by parameters of the NMR physical experiments. The Random Walk (RW) is arguably the most frequently adopted method when

simulating NMR relaxometry and diffusometry due to its ability to emulate fluid molecules Brownian motion. Compared to other methods such as Finite Elements Method (FEM), Finite Differences Method (FDM), or Lattice-Boltzmann Method (LBM), the Random Walk method offers some significant computational advantages: i) its formulation and implementation are considerably simpler; ii) the numerical stability of the solution can be controlled through proper statistical representation; and iii) it requires less computational time to obtain the NMR response over time. Research works using these other techniques can be found in [57, 59, 10, 11].

For that matter, the simulation of transverse relaxation times (T_2) magnetization decay is the most usual application [47, 78, 80]. In the PFGSE context, RW is used in simulations of the NMR response in digitally generated porous media [82] and periodic microstructures [8]. Also, analytical solutions for the PFGSE NMR signal in simple geometries such as isolated pores [13, 55] and periodic arrays of spherical grains [9] can be found. Other NMR related problems such as finding a representative elementary volume (REV) for NMR T_2 simulations [33] are solved by using the RW method. It also allows defining and solving inverse problems such as estimating the surface relaxivity of rock samples from experimental NMR data [5, 48].

Furthermore, efforts have been put in order to recover some of the aforementioned geometrical parameters. RW simulations are used to estimate the tortuosity of synthetic media [44, 60, 86] and microtomographic images of rock samples [38], or used in combination with experimental data to estimate pore sizes in [6, 49]. In fact, for diffusional tortuosity estimation from digital porous media, other image-based software that employs varying computational techniques can be found in [18, 28, 85].

The investigation of NMR signal relaxation in the presence of internal magnetic field gradients has also been the main focus of much research found in the literature. Many of these works use computational tools for a better understanding of the physical phenomenon. In this context, numeric simulations confirm theoretical predictions regarding the influence of susceptibility-induced gradients in transverse relaxation rates [52]. While an analytic-numeric method is used to calculate local magnetic fields for small susceptibility contrasts in periodic structures [71]. Also, an analytic-numeric solution for computing the internal field gradients in periodic porous media using diffusion eigenstates in Fourier representation is presented [23]. The interpretation of restricted diffusion and magnetization decay in rocks is explored, showing the correlation between pore sizes and internal gradients. Inversion methods are proposed to obtain the internal field gradients distribu-

tion [1]. In addition, RW simulations of NMR relaxation-diffusion in a collection of x-ray digital images of reservoir rocks help quantifying the influence of restricted diffusion and internal gradients [2]. The impact of internal magnetic field gradients in the T_2 interpretability is also discussed [54]. The authors highlight that at higher magnetic fields, T_1 may be a more reliable pore size indicator.

RW simulations of spins under a uniform magnetic field gradient are performed to study the dephasing effect [87]. Some theoretical predictions are confirmed such as the occurrence of the motional averaging and the localization regimes. Connolly et al [17] propose a combination of finite element simulations to measure the background magnetic field and RW to simulate the NMR response in segmented micro-computed tomographic images of sandstone rocks. The simulation results are compared with the experimental measurements, showing a relatively good agreement. From simulations of the relaxation mechanisms on real pore structures obtained from micro-computed tomographic images, intrinsic magnetic inhomogeneities in complex porous media are explored [34]. The authors uncovered a transition to a pre-asymptotic localization regime at high inter-echo spacings with a logarithmic relaxation rate dependence. A strong agreement with experimental NMR results is obtained by exploring a dual-scale approach to incorporate clay microstructures in sandstone samples [20]. Computational simulations are used to explore the effects of heterogeneously distributed surface relaxivities and internal gradient distributions in digital rock cores with mixed compositions [45]. Advanced Brownian dynamics and spin theories for modeling relaxation enhanced by magnetic field inhomogeneities in irregular porous geometries are combined in a simulation framework [16].

In this work, we present a methodology to perform image-based simulations of the NMR response in digital porous media. In particular, we show the details of implementing a simulation of both the PFGSE and the CPMG pulse sequences. The RW method is employed to simulate the molecular self-diffusion, while important considerations regarding the algorithm design and implementation are provided. In that sense, we indicate how to treat the boundaries of the image and interface interactions at the pore walls. For the image boundaries, we apply symmetric conditions in the form of straightforward mapping functions, while, for the interface interactions, adequate penalties are applied whenever a particle hits a pore wall. By following these instructions, we can simulate the PFGSE procedure and reconstruct the NMR signal, estimating the diffusion coefficient for a collection of observation times or compute the transversal relaxation times decays. For the CPMG sequence, we also present a two-step workflow to incorporate the effects of enhanced relaxation due to internal gradients. This workflow is composed of computing

the background magnetic field and then incorporating the spin dephasing that comes from interactions with the internal gradients, computed in the first step. We also show how to simulate a routine to estimate the internal gradient distribution, based on the multi- τ method. We indicate the stages of the RW algorithm that are the most appropriate for massive parallel computation in Graphics Processing Units (GPU). Then, a series of numerical experiments are performed.

Regarding diffusion simulations, our code is validated in media with analytical solutions such as isolated simple pores and periodic arrays of spherical grains. During this stage, we also repeat the experiments adopting varying simulation parameters such as the number of particles, the length of the walk, and the image resolution, in order to investigate their impact on the quality of the results. We then generate a set of synthetic 3D models and infer geometrical information regarding the pore network's average pore size and tortuosity from the diffusivity measurements obtained from simulations. We compare the results to other numerical tools and then analyze them to independent absolute permeability measurements. After visual inspection, we use linear regression on the inferred data to recover exponential absolute permeability prediction functions. We perform simulations from μ -CT images of sandstones and carbonate samples, inferring the geometric properties of the samples from the apparent diffusion coefficient data, and exploring the correlation with absolute permeability. Last, we compare the simulation results with NMR experimental data. We investigate the root cause of some deviations found in carbonate samples. Our results show that they may be caused by the presence of vugular porosity.

Regarding T_2 decay simulations, our main focus is to explore the effect of relaxation due to diffusion in the presence of internal magnetic field gradients. In that sense, we start by validating our simulation results with analytic predictions in pores with simple geometries. The magnetic field computation is compared to analytic-numeric predictions on periodic porous media, whereas the dephasing routine is compared to the relaxation prediction of unrestricted media under a uniform gradient. Then, a series of numerical experiments are performed. We first simulate the multi- τ procedure in segmented micro-computed tomographic images. We present results from simulations conducted in sintered media and a collection of sandstone samples. Quantitative distributions of the internal magnetic field gradients are then estimated from simulations. These results are compared with the NMR experimental data. To our knowledge, this is the first time in the literature that this simulation approach has been applied to three-dimensional segmented images of reservoir rocks. Finally, we explore the effects of restricted diffusion in the multi- τ

methodology, applying a different approach to take this diffusion regime into account.

This work is organized as follows. In Chapter 2, we provide a description of the theory behind acquiring geometrical information from the restricted diffusion of fluid molecules in porous media, and of the NMR procedure that enables these measurements. We also show in this chapter the theory behind the diffusional relaxation mechanism as well as the NMR technique to quantitatively estimate the internal gradients in a sample. Then, the algorithm to simulate the PFGSE and CPMG NMR experiments are described in Chapter 3. In Chapter 4, we present and discuss the results of our computational diffusion experiments. We first validate our code, comparing our results with some well-established analytical solutions found in the literature. Then, we estimate the surface-to-volume ratio and tortuosity of a set of digitally generated synthetic porous media. We also fit absolute permeability data as a function of porosity, surface-to-volume ratio or tortuosity. Last, we present the results from simulations in 3D images of reservoir rock samples. These results are used to explore correlations with absolute permeability, while further analysis of the pore space is accomplished by comparing them to NMR experimental data. In Chapter 5, we present and discuss the results of our numerical experiments of NMR transversal relaxation. We start by validating our code and comparing simulation results with some well-established analytical solutions found in the literature. Then, we present the results from simulations in 3D x-ray images of sintered media and reservoir rock samples. These results are used to explore the effects of the enhanced relaxation introduced by magnetized spins interaction with magnetic field internal gradients. The quantitative distribution of internal gradients is estimated from simulations of the multi- τ procedure. Last, we propose and investigate an alternative approach that considers the effect of motion restriction in these measurements. Finally, we conclude and present future works in Chapter 6.

Chapter 2

Theory

In this chapter, we present the theory behind inferring geometrical parameters of a restrictive medium from molecular diffusivity measurements. We present the basic concepts of the physical NMR experiment based on the pulsed magnetic field gradient technique that allows such measurements. We also present the theory behind the potential impact of molecular self-diffusion in NMR relaxometry. In this context, we are particularly interested in exploring the effects of the diffusion relaxation mechanism when strong magnetic field internal gradients are present in the porous space of a reservoir rock.

2.1 Diffusion propagator and time-dependent diffusion coefficient

Molecular self-diffusion refers to the random, thermally-driven motion of individual molecules within a homogeneous system, such as a liquid or gas, without any net mass transport. This phenomenon, intimately related to Brownian motion, was first theoretically described by Einstein in his work on the molecular-kinetic theory of heat. In this work, the relationship between the diffusion coefficient, D , and thermal energy is established [26].

The precise determination of the self-diffusion coefficient, D_0 , of a given material is critical for understanding its transport properties. In this context, fluid molecule self-diffusion is governed by Fick's law and the following condition [55]

$$\frac{\partial G}{\partial t} = D_0 \nabla^2 G, \quad (2.1)$$

$$G|_{t=0} = \delta_D(\mathbf{r} - \mathbf{r}'). \quad (2.2)$$

In Eq. (2.1), G is the diffusion propagator, solution to this differential problem. In a

nutshell, $G(\mathbf{r}, \mathbf{r}', t)$ is the conditional probability that at time t , a molecule that started at position $\mathbf{r}$ is at position $\mathbf{r}'$. Also, δ_D in the initial condition (2.2) is the Dirac delta function and it is basically stating that the initial position of every particle is known.

The presence of motional restrictions adds a boundary condition that must be satisfied at the interface, Σ ,

$$\frac{\partial G}{\partial n} + \frac{\rho}{D_0} G|_{\mathbf{r} \in \Sigma} = 0, \quad (2.3)$$

where $\mathbf{n}$ is an unitary normal vector at $\mathbf{r}$, pointing outwards, and ρ is the absorption strength in the interface. If $\rho = 0$, there is no interface absorption, and Eq. (2.3) collapses to blind reflection at Σ . In the context of NMR, ρ can be seen as the surface relaxivity [13].

We use in this work Einstein's definition for the time-dependent diffusion coefficient, $D(t)$. In this context, $D(t)$ is well-defined for observation times that are orders of magnitude above the typical timescale of underlying molecule collisions as the mean squared displacement of fluid molecules grows linearly with time [70]

$$D(t) = \frac{\langle [\mathbf{r}'(t) - \mathbf{r}(0)]^2 \rangle}{2dt}. \quad (2.4)$$

The brackets operator $\langle \dots \rangle$ computes the ensemble average among the particles in the system, which means that the term $\langle [\mathbf{r}'(t) - \mathbf{r}(0)]^2 \rangle$ represents the mean squared displacement. Also, d in Eq. 2.4 corresponds to the dimensionality of the problem.

Fluid molecules self-diffusion is basically distinguished in two types, or regimes, which will somehow explain the typical behavior of the diffusion coefficient in short and long time ranges. In the first regime, molecules experience unrestricted diffusion and $D(t)$ assumes the constant value D_0 if we consider a free media without obstacles. In the second, restricted diffusion is perceived if there is any kind of obstruction to random Brownian motion, implying a time dependence on the apparent diffusivity.

A clear distinction between these two regimes is essential for short time inferences. In this particular range, fluid molecules can be distinguished in two fractions. The first comprehends molecules that are close to obstacles and thus experience motional restrictions. The second fraction is composed of molecules located far enough from the walls so that they do not perceive any motion restriction, behaving as in a free medium. For a better visualization of this distinction, see Figure 2.1. From this scenario, as time evolves and thus a larger distance is covered by molecules, the fraction experiencing the walls increases. As a consequence, $D(t)$ deviates from D_0 and also decreases its value. This

deviation from unrestricted (free) self-diffusion can be expressed as [56]

$$\frac{D(t)}{D_0} \approx 1 - \frac{4}{9\sqrt{\pi}} \sqrt{D_0 t} \frac{S}{V}, \quad (2.5)$$

allowing direct estimation of the surface-to-volume ratio of the pore space, S/V . Notice that in the current interval of observations for short times, the effect of interface absorption can be neglected. However, this assumption no longer holds as time increases [74].

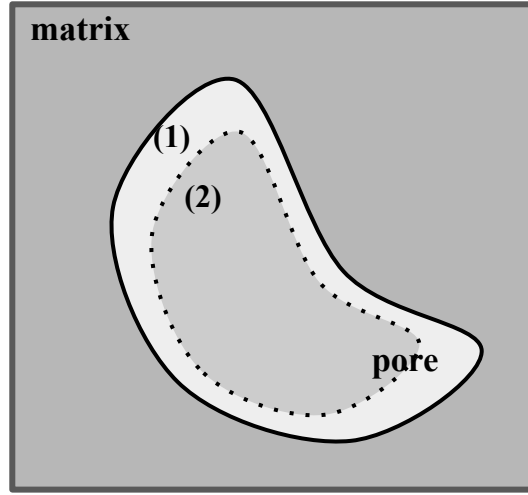


Figure 2.1: **Molecular self-diffusion:** distinct diffusion regimes are seen by molecules in short-time observations. Molecules close to the walls (region 1, in lighter gray) experience restrictive diffusion, while those that are far enough from the obstacles (region 2, in darker gray) behave as in a free medium. As the time evolves, the volume fraction corresponding to restricted diffusion increases.

For observations in the long time range, we expect the molecules to have diffused throughout the entire heterogeneity of the confining medium. In this context, the time-dependent diffusion coefficient asymptotically reaches a plateau, and this finite value is known as the effective diffusion coefficient, D_e . For media with low heterogeneity, it does not take long for D_e to be reached, as the molecules only need to diffuse distances a few times greater than the characteristic structural length of the medium [55]. Figure 2.2 illustrates the difference between molecules occupying isolated pore spaces and well-connected pore networks.

The tortuosity factor, τ , is a parameter of the porous medium that captures the connectivity inside the pore space and is somehow related with the effective diffusion coefficient, D_e . In a nutshell, a large tortuosity indicates longer paths are required for fluid molecules to move across two connected points in the pore network. We are mainly interested in diffusional tortuosity in this work, but there are a handful other tortuosity

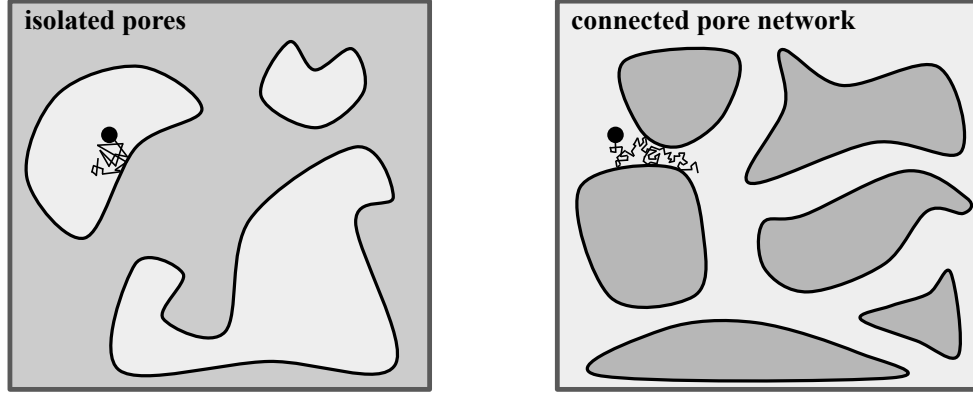


Figure 2.2: **Molecular self-diffusion:** The different scenarios experienced by molecules trapped in isolated pores and molecules occupying a well-connected porous medium.

definitions such as hydraulic, thermal, geometrical etc. In fact, correlations between these tortuosities must be taken with a pinch of salt, as each definition can describe a specific transport process [30]. In a particular porous medium, both D_e and τ are related by the expression [70]

$$\frac{D_e}{D_0} = \frac{1}{\tau} = \frac{1}{\phi F}. \quad (2.6)$$

where F is the geometrical factor and ϕ is the porosity of the medium.

Furthermore, the behavior of the diffusion coefficient in the short and long time ranges can be interpolated by the Padé approximation given by [70]

$$\frac{D(t)}{D_0} = 1 - (1 - \tau^{-1}) \frac{\frac{4}{9\sqrt{\pi}} \sqrt{D_0 t} \frac{S}{V} + (1 - \tau^{-1}) \frac{D_0 t}{D_0 \theta}}{(1 - \tau^{-1}) + \frac{4}{9\sqrt{\pi}} \sqrt{D_0 t} \frac{S}{V} + (1 - \tau^{-1}) \frac{D_0 t}{D_0 \theta}}. \quad (2.7)$$

Here, the fitting parameter θ has dimension of time, and it is interpreted as the typical time required for the diffusion coefficient to reach the tortuosity limit.

Figure 2.3 shows a typical time-dependent diffusion coefficient measurement. It also illustrates the short and long times asymptotic behaviors and the Padé approximant that interpolate these two regions.

2.2 Pulsed magnetic field gradient amplitudes

First proposed by Stejskal and Tanner [76], the Pulsed Magnetic Field Gradient Spin Echo (PFGSE) is a well-established NMR technique to assess molecular diffusion in porous media. This technique aims to encode the spatial position of each nuclear spin based on

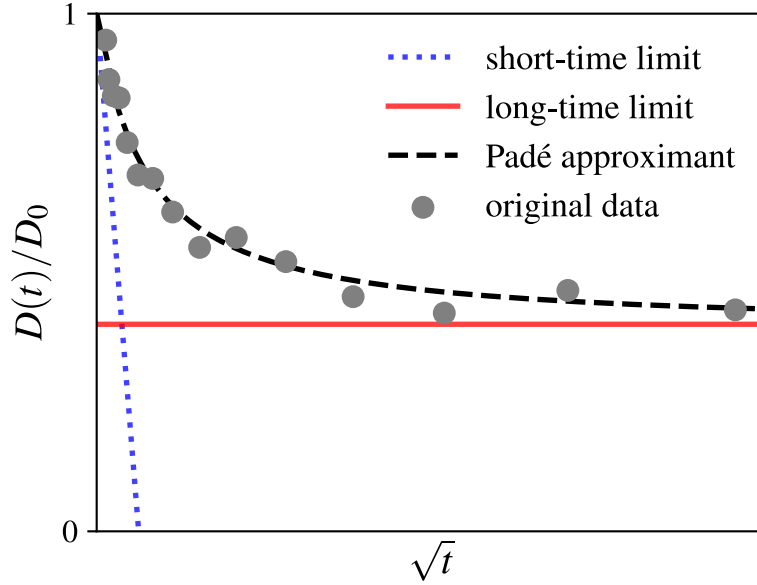


Figure 2.3: **Molecular self-diffusion:** illustration of a typical collection of measurements of the apparent diffusion coefficient, $D(t)/D_0$, in a medium with a connected pore space. Also shown the asymptotic behaviors for short and long observation time ranges and the Padé approximant that interpolate these two intervals.

its effective magnetic field. This can be accomplished by applying finite narrow gradient pulses, $\mathbf{g}$, with a constant duration, δ , and amplitude, g , resulting in a time-dependent magnetic field gradient [64]. An initial gradient pulse is applied after a radio frequency, r.f., excitation pulse ($\pi/2$), linearly modifying the local magnetic field. Its effect is sensed by the spins aligned with the $x'y'$ plane. After a given observation time interval, Δ , a refocusing pulse (π) is applied followed by a *reading* gradient pulse. The application of the second gradient pulse aims to restore the phases that were spread after the initial gradient pulse. However, molecular self-diffusion results in a loss of coherence because spins moving throughout regions with distinct magnetic fields inevitably lose their relative coherence [14]. This is illustrated in Figure 2.4 by the red and blue vectors. The dotted lines represent the magnetized spins motion occurred between the gradient pulses. As these spins occupy different positions, their phase is not completely restored after the second gradient pulse. This pulse sequence and the resulting phase shifts due to spins translational motion between the two gradient pulses are shown in Figure 2.4. Alternative schemes of excitation and refocusing r.f. and gradient pulses can be employed in other applications, depending on the properties and sensitivity of the sample to the investigated phenomena [65].

In this context, a spin initially at point $\mathbf{r}$ and found at point $\mathbf{r}'$ after the second

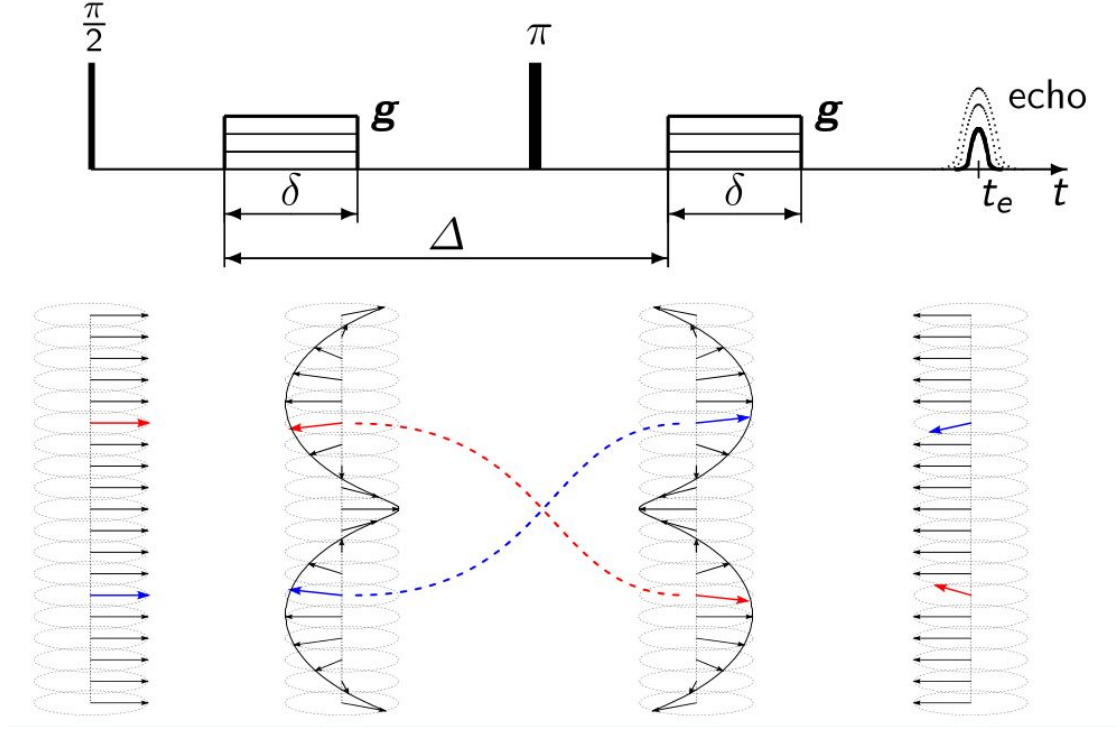


Figure 2.4: **PFGSE NMR:** schemes for pulse sequence and phase shifts due to molecular self-diffusion between gradient pulses. Source: [69]

gradient pulse experience a phase shift, $\Delta\Phi$, given by [14]

$$\Delta\Phi = \mathbf{k} \cdot (\mathbf{r} - \mathbf{r}'). \quad (2.8)$$

Here, the wave vector $\mathbf{k}$ is defined as $\mathbf{k} = \gamma\delta\mathbf{g}$, where γ is the gyromagnetic ratio of the nucleus ($\gamma = 2.67 \times 10^8$ rad/Ts for ^1H). Ideally, for $\mathbf{k}$ to be a constant finite value, it is required that $\delta \rightarrow 0$ and $\mathbf{g} \rightarrow \infty$. Yet, both the magnitude and the width of the gradient pulse are limited by the NMR instrument. In practice, δ is set much smaller than Δ in order to minimize diffusion during the application of the gradient pulse [74].

For a given measurement, the PFGSE amplitude, $M(\mathbf{k}, \Delta)$, is recovered by applying a double Fourier transform of the diffusion propagator, $G(\mathbf{r}, \mathbf{r}', \Delta)$, inside the pore space [55]

$$M(\mathbf{k}, \Delta) = \frac{1}{V_p} \int_{V_p} \int_{V_p} G(\mathbf{r}, \mathbf{r}', \Delta) e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} dV dV'. \quad (2.9)$$

In Eq. (2.9), V_p is the porous volume. The recorded coherence loss depends on the applied gradient pulse and, as a consequence, on the wave vector $\mathbf{k}$.

The echo attenuation under distinct gradient pulse configurations can be used to infer

the diffusion coefficient as function of the observation time. In that sense, in the small wave vector $\mathbf{k}$ range, the normalized PFGSE amplitude, $M(\mathbf{k}, \Delta)/M(\mathbf{k} = 0, \Delta)$, decays exponentially in $\mathbf{k}^2$ at a rate that is proportional to the time-dependent diffusion coefficient as [55]

$$\frac{M(\mathbf{k}, \Delta)}{M(0, \Delta)} \approx e^{-D(\Delta)k^2\Delta}. \quad (2.10)$$

The application of logarithm to both sides of Eq. (2.10) lead to the Stejskal-Tanner equation [76]

$$\log \left(\frac{M(\mathbf{k}, \Delta)}{M(0, \Delta)} \right) \approx -D(\Delta)(\gamma\delta g)^2(\Delta - \delta/3). \quad (2.11)$$

Notice that in this equation, a term was added to account for spin dephasing that occurs from diffusion in the direction of the applied gradient during the pulse application ($\delta/3$). Finally, from Eq. (2.11), a linear regression can be applied to recover $D(\Delta)$.

2.3 Diffusion relaxation in NMR T_2 decay

Molecular self-diffusion also plays a crucial role in other NMR-related procedures such as NMR relaxometry. In this context, measuring the transverse relaxation time, T_2 , in particular, is arguably the most popular for studying petrophysics of reservoir rocks. From the T_2 exponential decay many pore space properties such as the porosity, the wettability, the pore size distribution and the permeability of fluid-saturated rock samples can be inferred [92, 62].

The Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence is the standard procedure used to measure the evolution of magnetization $M(x, y)$ in a system and, consequently, the transverse relaxation time, T_2 [15]. Figure 2.5 illustrates this sequence, which essentially consists of an initial $\pi/2$ pulse followed by a series of π pulses separated by an interval t_E (the echo time). Echoes are formed between the π pulses. This sequence gained popularity in part due to its ability to maximize the signal-to-noise ratio [48]. Figure 2.6 illustrates the typical T_2 decay obtained from a CPMG sequence and its corresponding relaxation time distribution. This distribution is obtained from an inversion procedure of the magnetization signal decay. In this regard, Tikhonov regularization is a well established mathematical technique used to stabilize the solution of ill-posed problems [83]. It can be applied to solve for T_2 while mitigating the potential effects caused by noise in the acquisition signal.

The characteristic relaxation time, T_2 , of a fluid confined in a porous system is com-

monly described by contribution of three independent relaxation mechanisms [92]:

$$\frac{1}{T_2} = \frac{1}{T_{2,B}} + \frac{1}{T_{2,S}} + \frac{1}{T_{2,D}}. \quad (2.12)$$

Each term of the RHS of Eq. 2.12 is described as follows:

- $T_{2,B}$ is the bulk relaxation, resulting from spin-spin interactions within the saturation fluid. It is an intrinsic characteristic of the fluid and can be determined by the experimental conditions. As an example, $T_{2,B}$ takes no longer than 2.2 seconds at 2 MHz for pure water at room temperature.
- $T_{2,S}$ is the surface relaxation, occurring due to interactions between fluid magnetized spins and ions from minerals on the pore walls, i.e., at the fluid/solid interface. Its intensity depends on factors including pore geometry and the mineral composition of the solid phase. In many scenarios of interest, the surface relaxation is the dominant relaxation mechanism and it can be approximated by $\rho_2 \frac{S}{V}$, when in fast diffusion regime [14, 12].

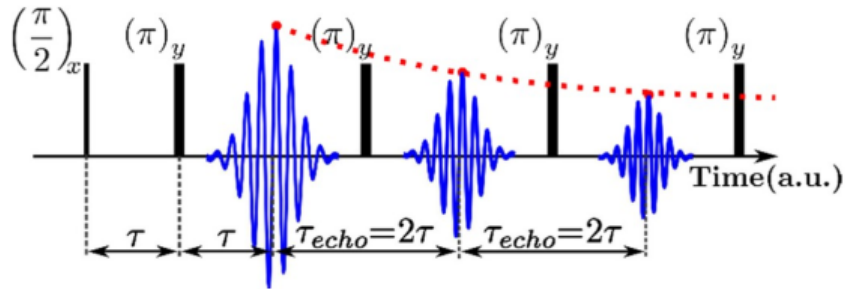


Figure 2.5: Schema for CPMG pulse sequence to measure the NMR transversal relaxation times, T_2 . Source: [48].

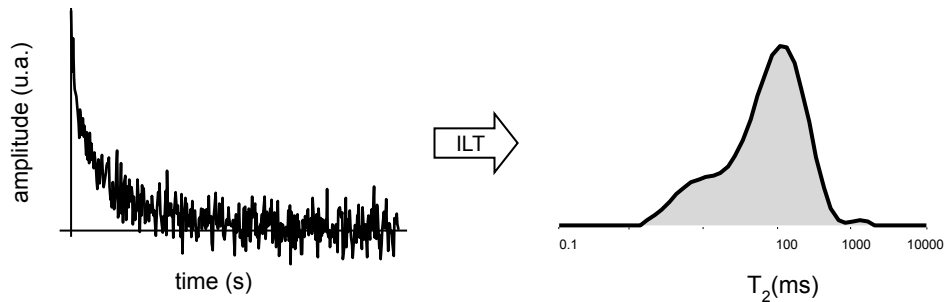


Figure 2.6: A typical T_2 exponential decay and the corresponding relaxation times distribution.

- $T_{2,D}$ is the enhanced relaxation arising from the interaction of magnetized spins in the fluid with spatial inhomogeneities in the static background magnetic field within the pore space, i.e., internal gradients. It introduces an additional relaxation that depends not only on the properties of the saturated fluid and the magnitude of magnetic field gradients within the pore space, but also on tool related parameters such as the observation time between acquisitions, t_E . Assuming an unrestricted diffusion regime, this term can be modeled as $\frac{1}{12}D_0\gamma^2g^2t_E^2$, where g is magnitude of the static magnetic field internal gradient [40, 14].

Thus, with the terms described above, Eq. 2.12 can be expanded as [84]:

$$\frac{1}{T_2} = \frac{1}{T_{2,B}} + \rho_2 \frac{S}{V} + \frac{1}{12}D_0\gamma^2g^2t_E^2. \quad (2.13)$$

Note that the equation above is a simplified view of a somewhat complex system. In a realistic scenario, it is unreasonable to expect only a single surface-to-volume ratio or that a single internal gradient acts within the porous space. In fact, we expect a spatial distribution of pores with varying sizes (and possibly different surface relaxation factors). Additionally, the magnetic field is likely to vary spatially according to factors such as the geometry and mineral composition found in the surroundings of a particular point in space. These factors do not invalidate the equation above but add a complexity layer to the problem.

In the NMR experiment, a constant external magnetic field, B_0 , is applied. However, different materials interact with this field in distinct ways. The physical property that describes a material's ability to interact with this background field is known as magnetic susceptibility, χ_m . The contrast between the magnetic susceptibility of the fluid and the rock matrix can induce significant internal gradients [40]. Furthermore, the spatial distribution of materials with high magnetic susceptibility contrast can amplify inhomogeneities in the resulting static magnetic field, leading to higher internal gradients [17].

For sedimentary rocks at low magnetic field conditions, this is observed more frequently in sandstones than in carbonate samples. Generally speaking, sandstones contain higher concentrations of high magnetic susceptibility minerals in their composition, such as clays (illite, chlorite etc). Figure 2.7 shows a quantitative study performed by Hürlimann [40] that demonstrates this empirical observation. In a set of 321 rock samples from sedimentary basins, they found out that sandstones typically exhibit higher magnetic susceptibility (roughly an order of magnitude higher), which tends to generate stronger

internal gradients.

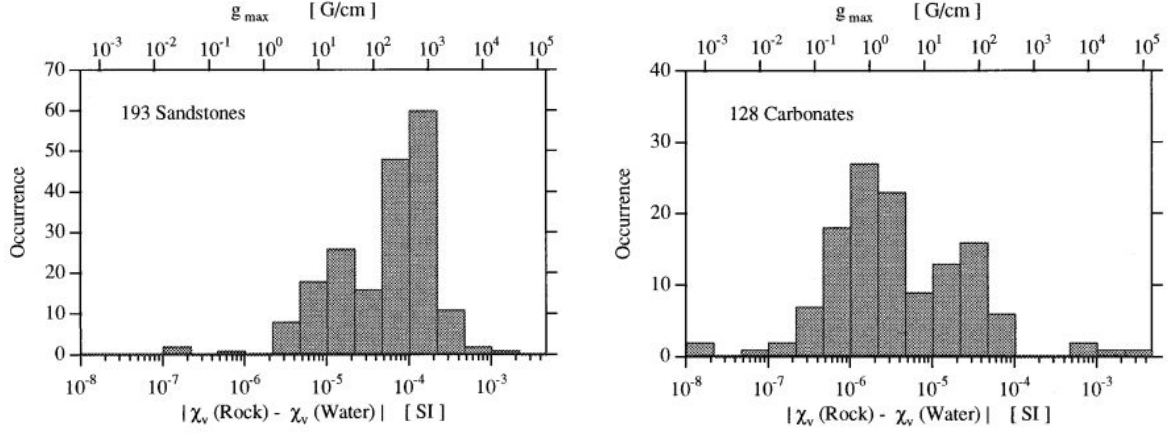


Figure 2.7: Quantitative study comparing sandstones and carbonates magnetic susceptibilities for 321 samples. Sandstone samples present magnetic susceptibilities an order of magnitude higher than carbonates, which leads to higher internal gradients. Source: [40].

In this context, an enhanced relaxation is observed due to the interaction of magnetized spins with these internal gradients. Similar to what was observed in the PFG experiment for estimating diffusion, the interaction of magnetized spins with a background magnetic field causes dephasing in their magnetic moment. During the spin's trajectory, over the time interval between measurements, this dephasing can be described by the following integral equation [87]:

$$\phi(t) = \gamma \int_0^{\tau/2} B[r(t)]dt - \gamma \int_{\tau/2}^{\tau} B[r(t)]dt, \quad (2.14)$$

where ϕ is the phase of a particle at a particular time, t , and B is the magnitude of the magnetic field, measured at the position occupied by the particle, r .

It is crucial to note that the Equation 2.14 is formed of two integral parts because the CPMG sequence includes a phase inversion (π pulse) during the acquisition. This inversion is crucial for the diffusion relaxation mechanism to be considered negligible when the magnetic field is homogeneous. Under this condition, all the phase accumulated in the first half of the trajectory would be canceled by the phase accumulated in the second half. Thus, at the time of acquisition, we would expect a zero net phase, and its contribution to signal relaxation would also be zero. Another interesting hypothetical situation to understand the phenomenon would be the case where the spin follows a *pendular* trajectory; that is, the same path taken in the first half is retraced in the second

half. As a result, the final position would match the initial one, and all positions and possible fluctuations in the magnetic field experienced during the initial path would also be experienced on the return. In this scenario, the dephasing would also be zero.

However, we know that spins exhibit random Brownian motion. Thus, even if at the end of the trajectory these spins return to their initial position, it is more likely that they traverse distinct regions, each exhibiting different magnetic field magnitudes. Therefore, in an environment where the background magnetic field is not homogeneous, the accumulated dephasing will not be zero, and its contribution to signal relaxation depends on factors such as the intensity of the fluctuations experienced.

Another important factor to consider from the above equation and previous discussion is that the longer the interval between acquisitions, the greater the probability that the spins will experience distinct trajectories and fluctuations in the background magnetic field. In this sense, it is important to note that the CPMG sequence incorporates mechanisms to minimize the relaxation introduced by the presence of these internal gradients. By reducing the echo interval, t_E , we restrain the effect of the internal gradients on signal relaxation [15]. Decreasing the magnitude of the static magnetic field B_0 is also a technique employed, as the gradients increase linearly with B_0 . However, even with these two feasible actions to mitigate the effects of diffusion relaxation, there may be samples where the internal gradients are high enough to influence the observed decay. Additionally, t_E and B_0 can only be reduced up to a certain limit dictated by the NMR tool and the system's configurations.

An implication of the undesirable influence caused by diffusion relaxation can be the distortion in the interpretation of the relaxometry experiment. As mentioned earlier, since the dominant relaxation mechanism is modeled by $\rho_2 \frac{S}{V}$, it is a common practice to correlate the distribution of T_2 relaxation times with a distribution of pore sizes in the sample. By introducing a new relevant mechanism into the transverse relaxation processes, this interpretation might be compromised.

Another interesting point regarding the presence of internal gradients in sedimentary rock samples relates to the effectiveness of the diffusion experiments. Whether it is the previously mentioned PFG or other well-logging experiments to estimate the diffusion coefficient of the confined fluid, as long as they utilize techniques related to the application of gradients by the acquisition tool. In this scenario, if the internal gradients exceed the gradient applied by the logging tool, or are even of similar magnitudes, any measurement obtained will therefore lose its value. However, for experiments conducted in the

field, there is a mitigating factor. Under acquisition conditions (higher temperatures and pressures), although the diffusion coefficient of the fluid generally increases, the magnetic susceptibility of the sample as a whole decreases. Consequently, the tendency is for the intensity of the internal gradients to decrease [40].

2.3.1 Diffusional relaxation regimes

In simple systems, characterized by a single dominant geometry, the decay resulting from the interaction of magnetized spins with an internal gradient, g , can be analyzed using three characteristic lengths [40]:

- L_S is the structural length, which is the characteristic size of the pores.
- L_D is the diffusion length. The typical distance a molecule travels over a time interval, t . For CPMG, the characteristic time is the echo interval, t_E , and

$$L_D = \sqrt{D_0 \times t_E}. \quad (2.15)$$

- L_G is the dephasing length. Basically, it is the distance required for the spin to experience a dephasing of 2π , and

$$L_G = \sqrt[3]{\frac{D_0}{\gamma g}}. \quad (2.16)$$

In general terms, the shortest of these three lengths indicates the dominant regime experienced by the system. When L_D is the shortest, the time allowed for particles to diffuse is so short that the spins travel a distance insufficient to experience significant restriction or dephasing. In this regime, typically called either the asymptotic short-time or free diffusion regime, we can approximate the magnetization for the N^{th} echo as:

$$\frac{M(Nt_e)}{M_0} = e^{-\frac{1}{12}D_0\gamma^2g^2t_E^2Nt_e} \quad (2.17)$$

When L_S is the shortest of the characteristic lengths, the spins "perceive" the entire geometry of the porous medium, and the relaxation effect resulting from the internal gradient is an average of the fluctuations occurring within the confined space. For this reason, the regime under these conditions is the so-called *motional averaging regime*. Finally, when L_G is the shortest of the characteristic lengths, we have the *localization regime* [13, 40, 87].

2.3.2 Multi- τ experiment for internal gradients estimation

First proposed by Hürlimann [40], the multi- τ procedure is based on applying multiple series of CPMG experiments using different echo times, t_E , to estimate a quantitative distribution of *effective* internal gradients in a specific porous medium. These *effective* gradients can be seen as the average internal gradients experienced by magnetized spins traveling in the pore space during each inter-echo interval. To this end, we assume that the system operates within the asymptotic short-time regime. We also assume that the porous medium is composed of a distribution of pore sizes and effective gradients acting within it [54, 29]. In this scenario, we can model the transverse magnetization as:

$$\frac{M(t, t_E)}{M_0} \approx \sum_i h(T_2) \exp\left(-\frac{t}{T_2}\right) \times \int_{g_{\min}}^{g_{\max}} f(g_{\text{eff}}) \exp\left(-\frac{1}{12} D_0 \gamma^2 g_{\text{eff}}^2 t_E^2 t\right) dg_{\text{eff}}, \quad (2.18)$$

where $h(T_2)$ is the distribution of relaxation times, T_2 , and $f(g_{\text{eff}})$ is the distribution of effective internal gradients.

As previously discussed, at low-fields, the term related to the gradient effect is negligible when compared to the term associated with surface relaxation when the smallest possible echo time is applied. Assuming that these two terms are also decoupled, i.e., there is minimal correlation between them, by dividing the magnetization signals for a given t_E by the decay observed adopting the smallest echo time, $t_{E,\min}$, we obtain the ratio:

$$\frac{M(t, t_E)}{M(t, t_{E,\min})} \approx \int_{g_{\min}}^{g_{\max}} f(g_{\text{eff}}) \exp\left(-\frac{1}{12} D_0 \gamma^2 g_{\text{eff}}^2 t_E^2 t\right) dg_{\text{eff}} \quad (2.19)$$

It is noteworthy that the ratio $\frac{M(t, t_E)}{M(t, t_{E,\min})}$ is independent of the effects of surface relaxation, being a function that solely depends on the term $t_E^2 t$. This new decay can be solved similarly to estimating the distribution of transverse relaxation times, T_2 . The difference is that here we want the distribution of effective internal gradients, $f(g_{\text{eff}})$. It is worth mentioning that, as in the previous case, this is an ill-posed problem that requires regularization to find a solution. It is often referred to in the industry as an Inverse Laplace Transformation (ILT), although, in fact, the procedure to find this solution is better defined as a multiexponential regression. Tikhonov regularization can be applied to solve for $f(g_{\text{eff}})$ while mitigating the effects of noise present in the signal [83]. Thus, selecting an appropriate regularizer is crucial for ensuring the quality of the result.

A final point worth mentioning is the influence of the selected value of t_E on these results. The larger the echo interval, the more pronounced the effect of diffusion relaxation, allowing the method to capture internal gradients of smaller magnitudes. On the other hand, as the echo interval increases, the decay of the magnetization signal also becomes more pronounced. Consequently, fewer acquisitions can be obtained until the measurement is completely dominated by the noise, resulting in the ratio calculated in Eq. (2.19) losing its meaning. For this reason, the multi- τ procedure is performed for different echo times in this work. Thus, the resulting curve is constructed from the initial acquisitions for each value adopted for t_E , i.e., those who have not yet been affected by noise. By doing so, we can cover a wider range of effective internal gradients.

Chapter 3

Methods

This chapter describes the algorithm for simulating the NMR experiments in 3D images representing digital porous media for reservoir characterization. We show how to simulate the diffusion experiment, based on the PFGSE technique, and the transverse relaxation decay, based on the CPMG NMR signal. We indicate how to treat image boundaries and interface interactions. Also, we highlight computational aspects and optimizations adopted to enhance the performance of our implementation.

3.1 Random walk simulations

The foundation for simulating NMR experiments in porous media is modeling the molecular self-diffusion propagator [8, 9]. In that sense, in this work, the particle motion is simulated through random walks and the porous medium is described by three-dimensional segmented binary images. In these images, each voxel belongs unequivocally to the porous or to the matrix phase.

We begin the RW simulations by uniformly positioning a large group of particles, or walkers, throughout the pore space. These walkers represent a sample of the magnetized fluid spins. The simulation then starts and, for each discrete time step (δt), each walker changes its position by taking a step in a random direction. This step has a fixed spatial length, ϵ , that is correlated to δt so that [9]

$$\epsilon = \sqrt{6D_0\delta t}. \quad (3.1)$$

In this work, we restrict particle movement to the Cartesian directions ($\pm\Delta x$, $\pm\Delta y$ or $\pm\Delta z$), i.e., the next position a walker can occupy is randomly chosen between one of

its lattice neighbors with an equal probability of $1/6$. This approach is widely used in the literature and is frequently referred to as the lattice random walk method [60, 86, 91]. As it will be later shown, this restriction allows us to compute the particle motion with simple bitwise operations, drastically reducing the computational cost of this repetitive task.

3.1.1 Boundary treatment

Applying an adequate boundary treatment in RW simulations is essential for obtaining reliable results. We highlight two scenarios that require special attention in our implementation. First, when a walker crosses the image limits and, second, when it hits a pore wall. Each scenario requires a specific approach, which is described as follows.

3.1.1.1 Image limits

For the first scenario, periodic boundary conditions are applied whenever a walker tries to escape the image limits (see Figure 3.1). Here, we assume that the digital image is a Representative Elementary Volume (REV) of the underlying porous medium. The same approach can be found in [60, 86]. In the context of porous media, a REV is the smallest volume required to represent the property of interest, such as porosity, permeability, or saturation [4]. Every reachable position in space is mapped to local coordinates within the finite domain of the image. Moreover, a symmetric extended space ensures continuity at the borders. In that sense, a simple conversion from global (infinite porous medium) to local (finite image representation) reference systems is required. With that in mind, Algorithm 1 presents a straightforward function that maps the position of walkers trying to *escape* the image to the position in the opposite border. The spatial coordinates of a walker are thus allowed to grow indefinitely in absolute values by adopting these image boundary rules.

Notably, the aforementioned approach allows removing the confinement effects that appear if other restrictive boundary treatments are employed. Alternatively, a *no-flux* boundary rule could be applied meaning that walkers would be prohibited from crossing the image limits. Yet, in this approach, each border of the image acts as an artificial barrier for particle motion, leading to an underestimation of the apparent diffusivity, something we want to avoid. With that in mind, we assume that the digital image is a REV and discussions regarding the confinement effects are out of the scope of the present work. Nonetheless, these effects are considered in other works such as [51, 91].

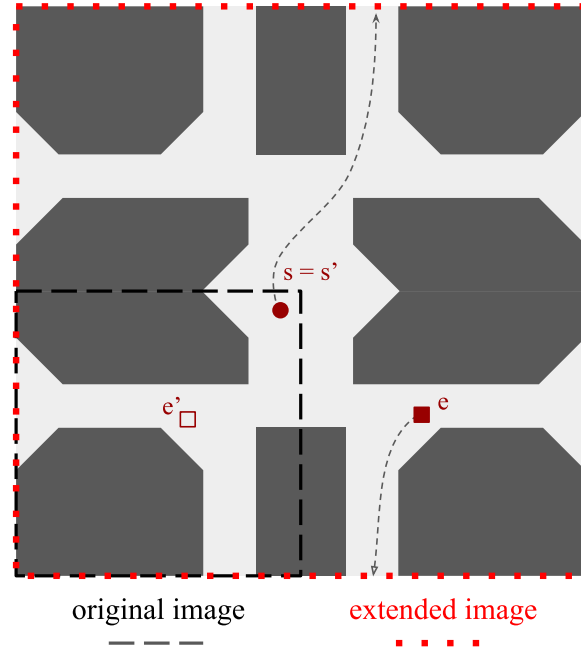


Figure 3.1: **Boundary treatment:** 2D illustration of how image limits are treated by application of periodic boundaries. The extended space repeats infinitely and guarantees continuity, avoiding confinement effects. This way, every point in the path of a particle is mapped to a pixel in the original image. E.g., a particle starting at $\mathbf{s}$ finishes its movement at a point outside the image domain. The end point is mapped to $\mathbf{e}$ (in the extended image domain), and $\mathbf{e}$ is mapped to $\mathbf{e}'$ (in the original image domain).

Algorithm 1: Image position mapping.

input : walker spatial coordinate $\mathbf{r}$, image dimension dim .
output: walker image coordinate $\mathbf{r}^*$.

```

1 if  $r \geq 0$  then
2    $c \leftarrow \mathbf{r} \pmod{dim}$ 
3    $p \leftarrow (\mathbf{r}/dim) \pmod{2}$ 
4 else
5    $c \leftarrow \mathbf{r} \pmod{dim} + dim$ 
6    $p \leftarrow ((1 - (\mathbf{r} + 1)/dim) \pmod{2})$ 
7  $n \leftarrow (p + 1) \pmod{dim}$ 
8  $\mathbf{r}^* \leftarrow c \times (p - n) + p \times (dim - 1)$ 
9 return  $\mathbf{r}^*$ 

```

3.1.1.2 Interface interactions

For the second scenario, pore-matrix interface interactions are treated as follows. Every time a walker hits a pore wall, one of the two possibilities happens:

i) the individual contribution to the NMR signal is lost and the walker is thus assigned as *dead*;

ii) it returns to its previous position with no sensible effect.

In such cases, the probability of death (i.e., the first scenario), γ_d , is given by [9]

$$\gamma_d = \frac{2}{3} \frac{\rho\epsilon}{D_0}. \quad (3.2)$$

Where ρ is the surface absorption factor, ϵ is the length of each simulation step, and D_0 is the bulk self-diffusion coefficient. Note that, in the context of NMR, ρ can be seen as the surface relaxivity. The $2/3$ factor in Eq.(3.2) is in fact a geometric parameter used to conciliate the curvature of the pore wall where surface interaction takes place. Thus, by adopting the value $2/3$ we assume a spherical pore. As shown in [11], this approximation is reasonable for pores in rock samples.

Eq. (3.2) can be implemented in different ways and, in the present work, we decided to treat it stochastically as in [5]. In that sense, walkers carry their current probability of being *alive*, P , and this value is constantly updated as

$$P(t + \delta t) = \begin{cases} (1 - \gamma_d)P(t), & \text{if there is a pore wall hit.} \\ P(t), & \text{otherwise.} \end{cases} \quad (3.3)$$

Here, the term $(1 - \gamma_d)$ can be interpreted as the probability of a walker surviving after an interface interaction. By doing so, when reconstructing the NMR magnetization signal, the individual contributions are weighted by their respective P value. Therefore, the contribution from walkers that have experienced wall collisions decreases, even though the number of walkers is kept constant throughout the simulation. Keeping the number of alive walkers constant is particularly interesting for computation in massive parallel devices such as GPUs. In that sense, avoiding idle threads and conditional branches, typically very expensive in these devices, can greatly improve performance.

Another detail worth mentioning is that, for γ_d to be interpreted as a probability, we must ensure that $\rho\epsilon \ll D_0$, i.e., the fast-diffusion regime is assumed [12]. Otherwise, P could assume negative values, yielding results with no physical meaning.

Algorithm 2 shows the core idea of the RW simulations. In this algorithm, the *PickRandomPositionInHeighborhood* routine will randomly pick a position from one of the walker's immediate neighbors in the lattice grid, while the *TreatWallCollision* routine will apply the pore wall death penalty described in Eq. (3.3).

Algorithm 2: Core random walk algorithm.

```

input : a set of walkers  $W$ ;
         the number of required steps  $N_s$ ;
         the binary image  $I_m$  (pore:  $I_m = 0$ ; solid:  $I_m = 1$ ).
1 for each  $w|w \in W$  do
2   for  $s = 0$  to  $N_s$  do
3      $r_{next} \leftarrow \text{PickRandomPositionInNeighborhood}(w)$ 
4     if  $I_m(r_{next}) = 0$  then
5        $\mathbf{r}_{new} \leftarrow \mathbf{r}_{next}$ 
6     else
7        $w \leftarrow \text{TreatWallCollision}(\mathbf{r}_{next})$ 
8        $\mathbf{r}_{new} \leftarrow \mathbf{r}_{old}$ 

```

3.2 Time-dependent diffusion coefficient estimation

In the present work, we estimate the time-dependent diffusion coefficient by tracking fluid molecules' Brownian motion and simulating the PFGSE NMR experiment in a digital porous medium. The time-dependent diffusion coefficient can be estimated by a straightforward application of Eq. (2.4), where the mean squared displacement is computed as [9]

$$\langle [\mathbf{r}'(t) - \mathbf{r}(0)]^2 \rangle = \frac{\sum_{i=1}^n P_i(t) [\mathbf{r}_i(t) - \mathbf{r}_i(0)]^2}{\sum_{i=1}^n P_i(t)}. \quad (3.4)$$

Here, $\mathbf{r}_i$ is the position of the i th walker ($i = 1, 2, \dots, n$), n is the total number of walkers in the simulation. Also, P_i is the probability of the i th walker being alive. As suggested by Eq. (3.4), it is crucial to keep track of the initial and final positions of every walker in the simulations.

Another approach to estimate the apparent diffusion coefficient, D , for a given observation time, t , is by using a collection of normalized PFGSE amplitudes as described in Eq. (2.10). We compute the PFGSE amplitudes with a discrete version of Eq. (2.9). The total sum of individual contributions to the NMR signal from each walker is [9]

$$M(\mathbf{k}, t) = \frac{\sum_{i=1}^n P(t) \exp \{ -i\mathbf{k} \cdot [\mathbf{r}_i(t) - \mathbf{r}_i(0)] \}}{\sum_{i=1}^n P(t)}. \quad (3.5)$$

The individual magnetization dephases due to displacement in the direction of the applied magnetic field gradient is expressed in the exponential term in Eq. (3.5). By

adopting a large number of walkers, the imaginary part in this summation becomes negligible, simplifying Eq. (3.5) so that

$$M(\mathbf{k}, t) = \frac{\sum_{i=1}^n P(t) \cos \theta}{\sum_{i=1}^n P(t)}. \quad (3.6)$$

where

$$\theta = \mathbf{k} \cdot [\mathbf{r}_i(t) - \mathbf{r}_i(0)]. \quad (3.7)$$

Last, as a collection of PFGSE amplitudes is computed, we can apply linear least squares regression in order to recover the time-dependent diffusion coefficient from Stejskal-Tanner's equation, Eq. (2.11).

In Appendix A we present the workflow for the PFGSE RW simulation algorithm for estimating the time-dependent diffusion coefficient for a given observation time. Unlike the physical experiment that requires a new procedure to measure the PFGSE amplitude for a single wave vector $\mathbf{k}$, the simulation can reduce computational work by applying several $\mathbf{k}$ at once. Also, when estimating the apparent diffusion coefficient for multiple observation times, we can take advantage of the particle motion recorded in previous estimations.

3.3 NMR transverse relaxation (T_2) simulations

In this section, we describe the modeling approach used to simulate the CPMG signal, aimed at measuring the transverse relaxation time, T_2 . Specifically, we seek to incorporate the effect of diffusion relaxation into these simulations. To achieve this, we use a two-step workflow [17]:

i) we calculate the spatially varying static magnetic field that results from the interaction between the external magnetic field, B_0 , introduced by the NMR experiment, and the magnetic susceptibility of the materials composing the sample;

ii) based on the map of the static magnetic field calculated in the previous step, we introduce a routine in the simulations to account for the enhanced relaxation from particles that experience fluctuations in the magnetic field.

Finally, we also describe the computational aspects of the multi- τ procedure implementation for estimating the distribution of effective internal gradients.

3.3.1 Estimating the internal magnetic field gradients

As mentioned above, the first step in our workflow is to calculate the background magnetic field at each point in the domain defined by the image mesh grid. It is important to note that for this problem, unlike the RW simulation, the image voxels are not limited to a binary classification (pore or matrix). Instead, each voxel is assigned a color based on the material identified in phase segmentation. We assume that the input image is one in which each voxel has been unambiguously segmented for a specific material. Note that the segmentation process of the raw grayscale image obtained via micro-computed tomography, which deserves special attention, is beyond the scope of this work. With the images segmented by phase/material, we can assign a specific magnetic susceptibility to each material. The porous phase also has an assigned susceptibility corresponding to the saturating fluid.

To compute the background magnetic field, we solve Maxwell's equations under zero-current conditions [35]:

$$\nabla \cdot \mathbf{B} = 0 \quad (3.8)$$

$$\nabla \times \mathbf{H} = 0, \quad (3.9)$$

where $\mathbf{B}$ is the magnetic flux density and $\mathbf{H}$ is the magnetic field strength. For most materials, these two quantities are correlated by the linear dependence:

$$\mathbf{B} = \mu \mathbf{H}, \quad (3.10)$$

where μ is the magnetic permeability of the material, which is defined as

$$\mu = \mu_0(1 + \chi_m). \quad (3.11)$$

Here, μ_0 is the magnetic permeability of vacuum, a constant value, and χ_m is the magnetic susceptibility of the constitutive material.

In this work, we obtain a numerical solution for the problem described above in complex porous media through an image-based Finite Elements Method (FEM) implementation for the analogous homogenization problem. Details on this implementation can be found in [31, 46]. To the best of our knowledge, this is the first time in the litera-

ture that this implementation has been applied to solve the problem in three-dimensional segmented images. There are three inputs in this approach: the segmented image; the assigned physical property value for each phase; and the applied magnetic static field, B_0 . In this case, the physical property of interest is, of course, the magnetic permeability, calculated from the magnetic susceptibility according to the constitutive equations, Eqs. 3.10 and 3.11.

3.3.2 CPMG simulation

To simulate the CPMG signal, a simple adaptation of the Random Walk Algorithm described in Section 3.1 is required. In this context, it is necessary to define the echo interval, t_E , which determines a specific number of steps per echo, $N_{s,e}$. Once this time interval is set, the algorithm is adapted so that, at every $N_{s,e}$ steps, the contribution of all walkers to the NMR signal is collected to compute the signal for that specific observation time.

The signal loss due to surface relaxation was discussed in Section 3.1.1.2, where its applicability was established and remains valid in this context. Moreover, due to its predictability, the bulk relaxation can be modeled in such a way that its contribution is only applied at the end of the procedure, without losses. This approach enhances the computational efficiency of the implementation while maintaining the simulation's accuracy.

Algorithm 3 shows the adaptation to Algorithm 2 to simulate the CPMG signal. In this algorithm, the *GetWalkerContribution* routine will return the walker's current probability of being alive.

3.3.3 Incorporating diffusion relaxation in simulations

With the spatial map that describes the magnetic field within the porous space, we can incorporate a routine to account for relaxation due to the effect of internal gradients. Based on Eq. (2.14) and considering that each step in our simulation has a fixed time interval, δt , the spin individual phase, ϕ , starting at zero (i.e., $\phi_0 = 0$), at a given instant depends on the cumulative phase from the previous instant, added to the dephasing experienced by the spin during that step, such that [87]:

$$\phi_{i+1} = \phi_i + \gamma B^*(\mathbf{r}_{i+1}) \delta t \quad (3.12)$$

Algorithm 3: Simple CPMG random walk algorithm.

```

input : a set of walkers  $W$ ;
         the number of required echoes  $N_e$ ;
         the number of steps per echo  $N_{s,e}$ ;
         the binary image  $I_m$  (pore:  $I_m = 0$ ; solid:  $I_m = 1$ ).
1 for  $e = 0$  to  $N_e$  do
2    $\mathbf{M}(e) \leftarrow 0$ 
3   for each  $w | w \in W$  do
4     for  $s = 0$  to  $N_{s,e}$  do
5        $r_{next} \leftarrow \text{PickRandomPositionInNeighborhood}(w)$ 
6       if  $I_m(r_{next}) = 0$  then
7          $\mathbf{r}_{new} \leftarrow \mathbf{r}_{next}$ 
8       else
9          $w \leftarrow \text{TreatWallCollision}(\mathbf{r}_{next})$ 
10         $\mathbf{r}_{new} \leftarrow \mathbf{r}_{old}$ 
11    $\mathbf{M}(e) += \text{GetWalkerContribution}(w)$ 

```

Here, $\mathbf{r}_i$ is the position occupied by the walker at the current state and B^* is the background residual magnetic field, given by

$$B^* = B(\mathbf{r}_i) - B_0. \quad (3.13)$$

Thus, due to phase inversion by application of π pulse, the accumulated phase for each signal acquisition, j , is [87]

$$\phi_{j+1} = \phi_j + \sum_{i=0}^{N_{s,e}/2} \gamma B^*(\mathbf{r}_i) \delta t - \sum_{i=N_{s,e}/2}^{N_{s,e}} \gamma B^*(\mathbf{r}_i) \delta t, \quad (3.14)$$

where $N_{s,e}$ is the number of steps required to simulate a echo interval.

At the end of the acquisition echo interval, the magnetization signal is obtained by summation of the contributions from each walker. The relaxation due to magnetic field fluctuations is represented by the cosine of the accumulated phase from each spin particle, such that [45]:

$$\frac{M(t)}{M_0} = \exp\left(-\frac{t}{T_{2,B}}\right) \times \frac{1}{n} \sum_{w=0}^n P_w(t) \cos \phi_w, \quad (3.15)$$

where n is the number of walkers present in the simulation.

Algorithm 4 presents the algorithm to simulate the CPMG signal considering all the

relaxation mechanisms described in Section 2.3. In this algorithm, the *ComputeSpinPhase* computes the spin current phase based on its current position (see Eq. (3.12)), and the *GetWalkerContribution* routine will return the walker's contribution to the NMR signal, which is the product of the current probability of being alive and the cosine of its current phase as described in Eq. (3.15). In addition, Appendix B presents a visual workflow for the CPMG RW simulation algorithm.

Algorithm 4: Complete CPMG random walk algorithm.

```

input : a set of walkers  $W$ ;
        the number of required echoes  $N_e$ ;
        the number of steps per echo  $N_{s,e}$ ;
        the binary image  $I_m$  (pore:  $I_m = 0$ ; solid:  $I_m = 1$ );
        the residual magnetic field spatial map  $B$ 
1 for  $e = 0$  to  $N_e$  do
2    $M(e) \leftarrow 0$ 
3   for each  $w | w \in W$  do
4     for  $s = 0$  to  $N_{s,e}$  do
5        $r_{next} \leftarrow \text{PickRandomPositionInNeighborhood}(w)$ 
6       if  $I_m(r_{next}) = 0$  then
7          $r_{new} \leftarrow r_{next}$ 
8       else
9          $w \leftarrow \text{TreatWallCollision}(r_{next})$ 
10         $r_{new} \leftarrow r_{old}$ 
11        if  $s = N_{s,e}/2$  then
12           $\phi_w = -\phi_w$ 
13           $\phi_w += \text{ComputeSpinPhase}(B(r_{new}))$ 
14         $M(e) += \text{GetWalkerContribution}(w, \phi_w)$ 

```

3.3.4 Simulating multi- τ procedure

In order to simulate the multi- τ , we essentially acquire n decay measurements with varying echo time intervals [17, 45]. These echo time intervals are chosen in such a way that we are able to build the curve describing the multi- τ decay, as defined by Eq. (2.19). As discussed in Section 2.3.2, this decay curve is influenced exclusively by the effect of diffusion relaxation.

An interesting point to consider is that to calculate the ratio described by Eq. (2.19) it is necessary that both the term in the numerator, which is derived from the decay considering t_E , and the term in the denominator, which is based on the decay considering $t_{E,\min}$, must have been collected for the same observation time, t . In experimental measurements,

however, it is quite common that some of the selected values for t_E are not divisible by $t_{E,\min}$ due to technical limitations of the measurement instrument. A few approaches can be applied to address this difficulty: one can interpolate the curve described by the decay, or use the T_2 distribution obtained from the inversion of the measured decay to sample the curve at points not covered by the acquisition echoes. In the computational scenario, however, we are not limited by the technical constraints of the measurement instrument, so we can choose t_E values that are exact multiples of the value set for $t_{E,\min}$ thus avoiding additional data processing that may introduce some errors.

The multi- τ decay curve is then constructed by calculating the ratio using several samples from each echo interval employed. As discussed in Section 2.3.2, this decay is independent of other relaxation mechanisms and can therefore be solved to estimate the distribution of internal gradients in the sample.

The problem of finding the internal magnetic field gradient distribution that better fits the collected data points can be treated as a least squares problem with regularization and the additional constraint that the amplitudes must be non-negative [22]. It can also be seen as a minimization problem as [36]:

$$x_\lambda = \arg \min \left\{ \|Ax - b\|_2^2 + \lambda^2 \|L(x - x_0)\|_2^2 \right\}, \quad (3.16)$$

where λ is a regularization parameter, x is the solution, i.e., the corresponding amplitudes that best fit the acquired data, b . It is also worth mentioning that, in the standard form, L is assumed to be the identity matrix while x_0 is typically set to zero when no further information is known, simplifying Equation 3.16. It is also worth mentioning that, by adopting $x_0 = \vec{0}$, we are applying a specific type of Tikhonov regularization often called *Ridge* Regularization. For each term of matrix A , the multi- τ method modeling the problem leads to

$$a_{i,j} = \exp \left\{ -\frac{1}{12} \gamma^2 D_0 g_{\text{eff},j}^2 t_{e,i}^2 t_i \right\}, \quad (3.17)$$

which can be solved for g_{eff} .

A final consideration is that the regularizer, λ , in Eq. (3.16), is typically chosen by adopting some criteria. In this work, we use the L-curve criterion to select the regularizer that provides the best trade-off between the norm of the solution and its residual [36, 22].

3.4 Performance optimization

In this section, we discuss several design decisions aimed at optimizing and enhancing the performance of the computational implementation for RW simulations. First, we explore the concept of massive parallelization of the algorithm, particularly through the use of graphics processing units (GPUs). We demonstrate the feasibility and advantages of this approach while also addressing its limitations and the strategies employed to overcome them, based on the architecture of these components. Subsequently, we present the modeling technique used to store image data. In this model, binary images are represented by arrays of 64-bit variables, where each variable corresponds to a cubic block of 4^3 voxels from the original image. Each bit within these variables represents one voxel of the respective block. We will demonstrate that a walker located within one of these cubes tends to occupy voxels within the same block, thus remaining there for a significant portion of the simulations. This modeling approach aims to optimize queries to memory addresses. Finally, we will discuss some other general computational aspects.

3.4.1 Algorithm massive parallelization

Our main goal is to simulate a large number of walkers, hundreds of thousands or even millions of particles, to achieve a satisfactory statistical representation of the physical phenomenon. The conventional approach of sequential computation, where each walker's trajectory is simulated one after another, is not particularly advantageous. Spin-spin interactions are modeled by bulk relaxation and can be factored out of the simulations. Therefore, this scenario offers an opportunity for massive parallelization of the algorithm. Each walker follows a completely independent trajectory until acquisition.

However, due to the large number of particles being simulated, achieving full parallelization in a single run is not feasible, mainly because of hardware constraints. Basic parallelization at the CPU level is insufficient since, for instance, the number of available threads on a personal computer is typically limited to a few dozen at best. Furthermore, access to computing clusters, where such capacity could be scaled, is often restricted. Therefore, the optimal platform to exploit latent parallelism is through GPU computation, which, in modern architectures, can provide several thousand cores.

As previously mentioned, the simulation code consists mainly of memory access operations and vector-based mathematical and logical computations. Additionally, the complete independence among the walkers' trajectories creates an ideal scenario for applying

the SIMD (Single Instruction, Multiple Data) computing paradigm illustrated in Figure 3.2. Nevertheless, adapting the computational code for this new context requires specific modifications to align with the operational characteristics of these devices. The approach focuses on parallelizing the trajectories on the GPU, with each walker processed in an individual thread. At the end of the acquisition, a reduction operation is performed to aggregate all individual contributions.

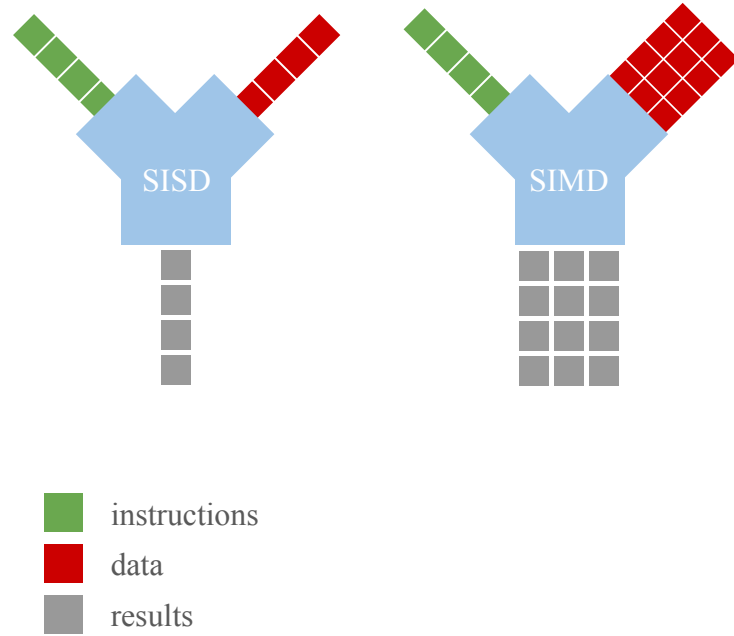


Figure 3.2: Illustration schema for Single Instruction, Single Data (SISC) and Single Instruction, Multiple Data (SIMD) computational paradigms. SISC is a sequential computing paradigm in which a single processor executes one instruction at a time on a single data element, typical of traditional CPUs. In contrast, SIMD applies a single instruction simultaneously to multiple data elements, leveraging data-level parallelism as seen in GPUs and vector processors, making it well-suited for tasks like image processing and matrix operations. The main difference lies in their approach to parallelism: SISC processes data sequentially, while SIMD processes multiple data elements in parallel under the same instruction.

The architecture of GPUs imposes a limitation on this approach due to the division of computational cores into blocks, each with its own shared memory space that is inaccessible to others. To address this, the adopted strategy involves each block aggregating the acquisition data for the particles processed in its designated threads. For instance, suppose the simulation handles a set of 10^6 walkers, and block #1 is responsible for the first 1,000 walkers. After simulating the N steps before the signal acquisition and storing the resulting individual contributions in a vector on the shared memory space accessible only to this specific block, an initial reduce operation is performed on these 1,000 elements.

This summarization, described in [37], is illustrated in Figure 3.3. Once summarized, this value is transferred to a vector in the GPU’s global memory, with indices aligned with the block’s index. Subsequently, a second summarization is performed at the block level on this vector stored in global memory, effectively summarizing data from the entire set of walkers. This second reduction operation can also be parallelized on the GPU. The whole process is shown in Figure 3.4. If the simulation involves more walkers than the GPU’s memory capacity can handle in a single run, a third level of summarization may be introduced at the level of kernel calls.

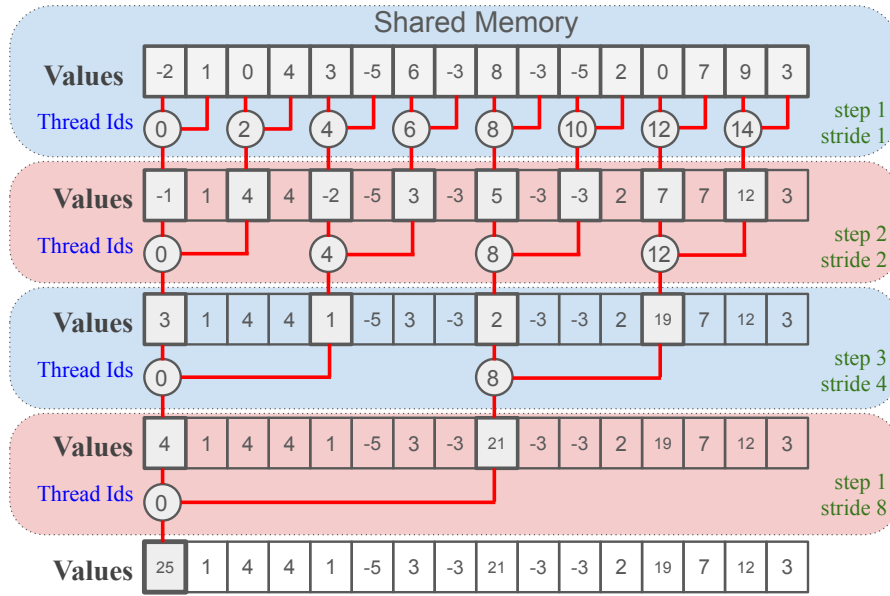


Figure 3.3: Illustration schema of parallel reduction operation for vector summarization in GPU blocks shared memory space. A parallel reduce operation aggregates elements of an array by applying an associative operator in a tree-like pattern, where pairs of elements are combined simultaneously in multiple threads. Instead of processing elements sequentially, the algorithm performs reductions in parallel at each level combining adjacent pairs, then combining the results of those pairs, and so on. The number of elements is halved at each step until a single result remains. This approach reduces the time complexity from linear ($O(n)$) to logarithmic ($O(\log n)$) using sufficient parallel resources.

A final point worth emphasizing concerns the simulations of the CPMG signal. In this case, rather than performing the described cycle for each acquisition, we simulate multiple acquisitions per GPU call. As a result, additional treatment is required during the summarization process to accurately reconstruct many acquisition echoes all at once.

3.4.2 Data structure compression

Suppose we have a three-dimensional image and we divide this image into cubes formed by L_b^3 voxels (see Figure 3.5). Now consider that a walker occupies one of the voxels in one of the cubes that form the original image. As discussed in Section 3.1, in the next step of our simulation, this walker seeks to move randomly to one of the 6 neighboring voxels in its immediate Cartesian vicinity. Depending on the voxel that this walker currently occupies, it belongs to one of the following groups:

- **Group 1:** The walker occupies one of the 8 voxels at the vertices of the cubic block. In the next step of the simulation, the walker has a $3/6$ chance of remaining within the current block.
- **Group 2:** The walker occupies one of the $12 \times (L_b - 2)$ voxels on the edges of the

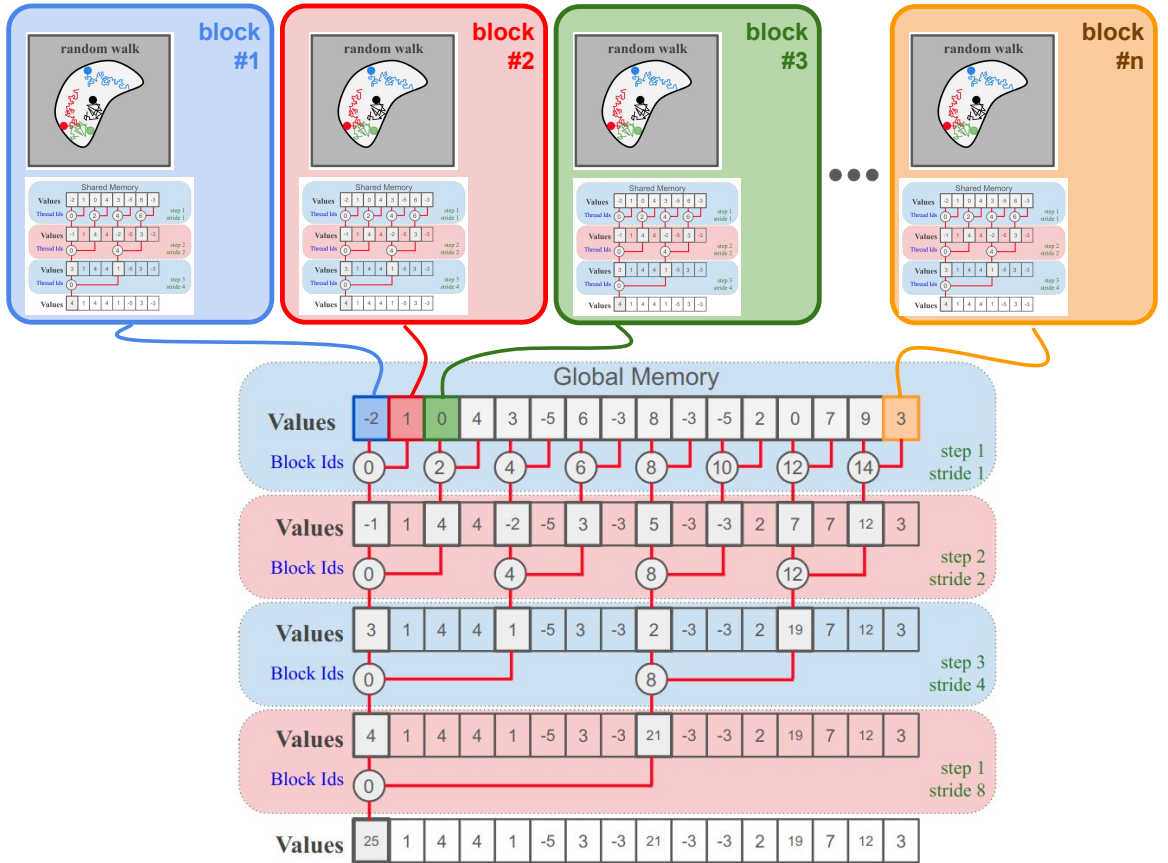


Figure 3.4: Illustration schema of two-level parallel reduction operation for computing a signal acquisition. The first level, performed at each block, summarizes the contribution of a subset of the walkers in the simulation that is related to a GPU block. The contribution from each block is then transferred to the device's global memory space where the second level summarization is performed.

cubic block, excluding the vertex voxels. In the next step of the simulation, the walker has a $4/6$ chance of remaining within the current block.

- **Group 3:** The walker occupies one of the remaining $6 \times (L_b - 2)^2$ voxels on the external faces of the cubic block. In the next step of the simulation, the walker has a $5/6$ chance of remaining within the current block.
- **Group 4:** The walker occupies one of the $(L_b - 2)^3$ voxels in the interior of the cubic block. In the next step of the simulation, the walker has a $6/6$ chance of remaining within the current block.

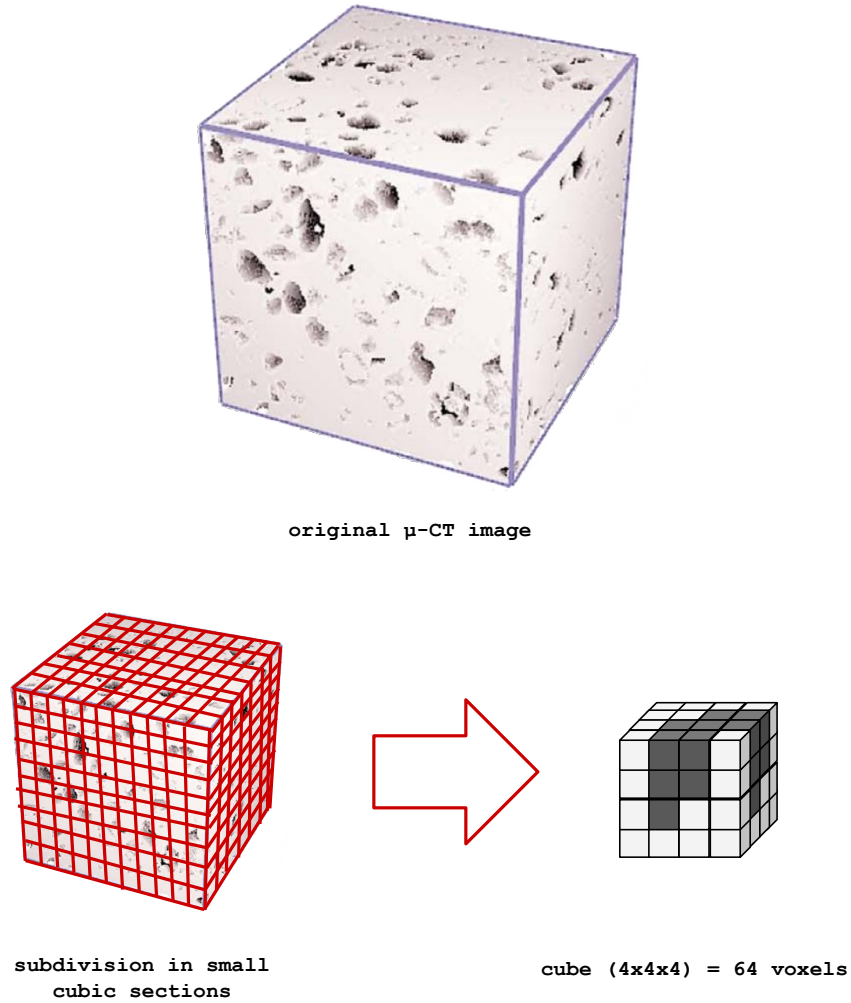


Figure 3.5: Original μ -CT image subdivided in cubic blocks.

Given these circumstances and that $L_b \geq 3$, after some mathematical manipulation, we can prove that the probability of a walker belonging to a given block choosing a new position within the same block is:

$$P(C_h) = \frac{(L_b - 2)^3 + 5(L_b - 2)^2 + 8(L_b - 2) + 4}{L_b^3}, \quad (3.18)$$

Eq. (3.18) shows that for an arrangement where $L_b = 4$, i.e., a cube with 64 voxels ($4^3 = 64$), the probability $P(C_h)$ is 75%. Similarly, for an arrangement with 512 voxels, where $L_b = 8$, the probability becomes 87.5%, and so on.

It is important to note that a RW simulation is a stochastic process in which subsequent positions of a particle depend directly on the previous state. In other words, for an isolated walker, the chance of "escaping" the current block varies according to its position. However, when considering the entire set of walkers, we can state that during the simulation, $P(C_h)$ remains preserved as shown in Table 3.1. On average, in a 4^3 arrangement, walkers will remain in the current block 75% of the time. This is corroborated by the results of subsequent simulations, where we varied the cubic arrangement size, the number of walkers, and the number of steps. In all cases, the average $P(C_h)$ remains consistent with the prediction found in Eq. (3.18).

Table 3.1: Cache hits statistics during RW simulations in varying configurations of number of walkers, number of steps and binary block sizes. Here, cache hits correspond to the situation where a walker chooses its next position within its current block.

n° of walkers	n° of steps	block size	Cache Hits	
			average	std. dev.
10^3	10^3	4^3	0.75	0.02
10^3	10^4	4^3	0.75	0.01
10^3	10^5	4^3	0.75	0.02
10^5	10^1	4^3	0.75	0.17
10^3	10^3	8^3	0.88	0.02
10^3	10^3	16^3	0.94	0.02

We highlight that dividing the image into cubes formed by L_b^3 voxels can represent a significant memory space saving, especially when compared to more naive options such as treating the data as simple RGB or even Grayscale images. For example, consider an image with a volume of about 1000^3 , i.e., 10^9 voxels. In this context, the memory required to store the information of this image using the aforementioned formats is shown in Table 3.2.

We can observe a considerable reduction in memory usage. Grayscale and RGB images use one or three `char` variables, respectively, to store the information for each pixel, which results in significant waste. On the other hand, when using the binary blocks structure, we may encounter situations where the original image dimensions are not necessarily

Table 3.2: Aproximate memory required to store a 3D image with around 1000^3 voxels.

image format	unit value (bits/voxel)	memory size (GB)
RGB	24	3.00
Grayscale	8	1.00
Binary	1	0.12
Binary with blocks	1	0.12*

compatible with the L_b^3 block design. As shown in Figure 3.6, for these cases, we would have idle bits in each Cartesian direction.

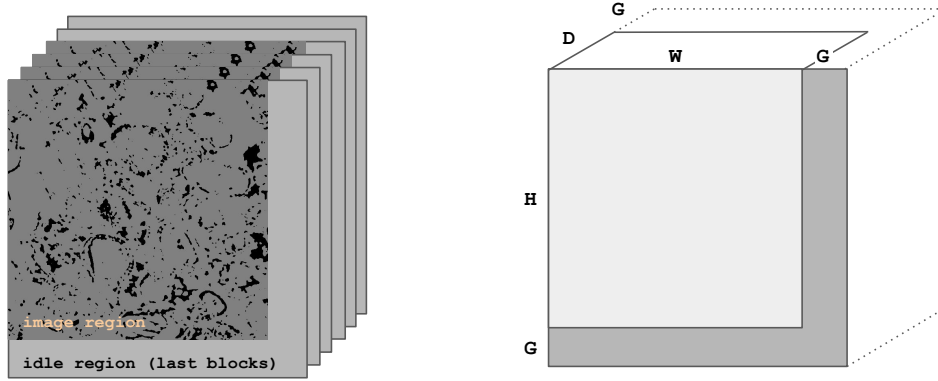


Figure 3.6: Example schema for a HxWxD image represented using the binary blocks data structure. In this representation, G are idle bits necessary to store the image information and preserve the cubic blocks.

In such cases, we must adopt one of the following adjustments to achieve data compatibility: i) Remove voxels from the edges of the image so that the cropped image has dimensions divisible by L_b ; ii) incorporate the idle bits (without relevant information) and apply special treatment to the borders of the image.

After some mathematical manipulation, the idle volume, V_{idle} , can be given as:

$$V_{\text{idle}} = G (HW + (D + G)(H + W + G)), \quad (3.19)$$

where H , W and D are the image original height, width and depth. G corresponds to the relative idle bits in a particular direction. For example, in the x-direction, it can be calculated as the remainder of the integer division of W by L_b . Also, it can assume distinct values for each cartesian direction, but here, for simplification purposes, we assume the

same value for each direction.

Note that, in the worst-case scenario, there is only one useful bit in the last block of each cartesian direction and therefore $G = (L_b - 1)$. In this context, for an image with a total volume of around 1000^3 voxels subdivided into an arrangement of cubic blocks of 4^3 voxels, the idle volume would represent around 9 million bits, which is less than 1% of the useful volume. Therefore, the additional memory cost to implement the binary block structure instead of a simple binary image would be less than 1%, which is clearly negligible.

Thus, we can model the data related to the image volume by subdividing it into 4^3 voxel-blocks. As shown in Figure 3.7, each of these blocks can then be represented as a linearly contiguous 64-bit array. Since the only necessary information is whether a given voxel corresponds to the porous phase or the solid phase, one bit is sufficient to store this information. An `UInt64` variable can also be read as a "64-bit array". Therefore, the image can be represented as an array of `UInt64` variables, where each entry in this array represents one of the 4^3 blocks of the image. In this modeling approach, the memory address corresponding to the voxel of interest for a given walker tends to remain constant during the simulation, optimizing memory access and increasing the occurrence of cache hits.

Note that in the present work, the only information required to perform the Random Walk simulations is whether or not a voxel belongs to the porous phase. However, the proposed data structure cannot be used in more sophisticated simulations involving multiple phases. For instance, in simulations where it would be important to identify the specific mineral phase of a voxel in the solid phase or, in multiphase systems, to determine which fluid is associated with a given voxel in the porous phase. Nevertheless, even in such cases, adaptations could be made to still use the 64-bit integer structure to represent each of these specific phases in binary. Thus, depending on the number of phases involved in the simulation, the proposed structure could still remain advantageous.

The challenge then becomes how to access a specific bit of this `UInt64` variable. The described structure enables the validation of particle movements using bitwise operations, which are typically faster and more efficient than common arithmetic and logical operations. For instance, using bit shifts instead of multiplying an integer by powers of two can result in a 75% reduction in operation time. Modern compilers often perform this kind of optimization automatically though for some operations where bitwise implementation is possible, this may not always hold true. Most of the time, there is a trade-off between

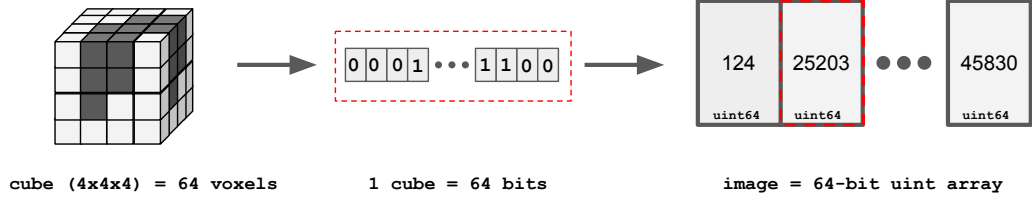


Figure 3.7: **Binary blocks data structure:** each voxel in the 4^3 cube can be represented as a single bit and, therefore, the whole cube can be represented as a 64-bit array. Alternatively, we can also represent this structure as 64-bit integer variable.

optimizing the code and making it easier to read and maintain. In Algorithm 5 we show how to implement bitwise operations to identify the phase of a specific bit in the UInt64 variable representing the current block.

Algorithm 5: Compute particle motion and pore wall collisions using bitwise operation from binary blocks data structure.

input : Each coordinate of the next position x, y, z ;

The binary blocks array B (pore: $B[i] = 0$; solid: $B[i] = 1$);

Binary blocks dimensions B_w, B_h and B_d (width, height and depth).

/* Finding block index, i , in block array. */

1 $X \leftarrow x \gg 2$

2 $Y \leftarrow y \gg 2$

3 $Z \leftarrow z \gg 2$

4 $i \leftarrow (B_h * B_w) * Z + B_w * Y + X$

/* Finding bit index, b_i , in block array entry. */

5 $b_x \leftarrow x \wedge 3$

6 $b_y \leftarrow y \wedge 3$

7 $b_z \leftarrow z \wedge 3$

8 $b_i \leftarrow b_x + (b_y \gg 2) + ((b_z \gg 2) \gg 2)$

/* Finally, checking if the identified bit is pore (0) or wall (1). */

9 return $((B[i] \gg b_i) \wedge 1)$

Thus, by adopting the binary 4^3 voxel-blocks data structure, we have a trade-off at hand: on one hand, we add this small memory overhead and the complexity required to implement bitwise optimizations; on the other hand, we gain performance in memory access.

3.4.3 Final remarks

In conclusion, as the trajectory of each walker is independent of the others in our model, the simulation can be expressively optimized by executing the most costly tasks of the algorithm at the multicore GPU. With that in mind, we structure these routines to exploit thread-level granularity. In this approach, each GPU core is responsible for simulating a unique walker (modern GPU architectures usually possess thousands of cores, in theory allowing as many walkers to be simulated simultaneously). Highly parallelable computations comprehend: walker movement; boundary treatments such as wall collision and image boundary treatment; and spins dephase due to interactions with fluctuations in the background magnetic field. Moreover, in order to compute the mean squared displacement and to reconstruct the PFGSE NMR signal (Eqs. (3.4) and (3.5), respectively), or to compute the CPMG signal acquisitions, it is required to perform a summarization covering the entire population of walkers. For these, we can also enhance performance by employing standard GPU reduction techniques found in [37].

Also, the image data is modeled in a data structure based on 4^3 -voxel blocks. The additional memory space required for adopting this approach is negligible and justified by its potential performance gains. This model structure is selected to optimize memory access, which can be a bottleneck in GPU computations.

Taking all this into consideration, our simulator is written in C++ and GPU kernel routines written in CUDA C. The performance improvement due to GPU parallelization allows us to simulate hundreds of thousands of steps in a large number of walkers in a matter of seconds on a personal computer.

Chapter 4

Diffusion simulations

In this chapter, we present and discuss numerical results obtained from diffusion simulations in three-dimensional digital porous media. We first validate our code, comparing our results with some well-established analytical predictions found in the literature. Then, we create a set of synthetic porous media and estimate their surface-to-volume ratio and tortuosity from the diffusivity measurements. Also, we fit absolute permeability as a function of porosity, surface-to-volume ratio or tortuosity of these synthetic media. We perform simulations from micro-computed tomographic images of sandstones and carbonates. From these results, we explore the correlation between the geometric properties inferred from the apparent diffusion coefficient data and the absolute permeability measured from computational simulations using FEM and from physical experiments performed in the laboratory. Last, we compare the NMR diffusion obtained from simulations with the results obtained from the physical experiment. We discuss and explore some of these results, especially those related to vugular porosity in carbonate samples.

The results presented in Sections 4.1 and 4.2 are also presented in the article *Image-based simulation of molecular diffusion on NMR Pulsed-Field Gradient experiments: Feasibility to estimate tortuosity and permeability of porous media*, published at the 219th volume of the *Journal of Petroleum Science and Engineering* [67].

4.1 Validation

We validate our implementation comparing the simulated NMR response and the apparent diffusivity with the analytical solutions for isolated pores with simple geometry and periodic microstructures of spherical grains. We also investigate and discuss the effects in the accuracy of the results when varying the input parameters of the RW simulations.

4.1.1 Isolated pores with simple geometry

In the first validation experiment, we simulate the PFGSE NMR signal for isolated pores with simple planar, cylindrical, and spherical geometries. For the spherical pore, the magnetic field gradient is applied along the polar axis of the spherical polar coordinate frame. The analytic-numerical solutions for these problems are presented in [13] where the PFGSE amplitudes are recovered by computing the eigenstates of the diffusion propagator. We apply the standard approach of bisection method followed by Newton-Raphson iteration in order to recover the eigenvalues, roots of the transcendental equations arising from the application of relaxing boundary conditions. We reproduce these results in order to compare with the results from our simulations.

In Figure 4.1, the normalized PFGSE amplitudes of an isolated spherical pore are presented as a function of the magnitude of the wave vector $\mathbf{k}$. For this example, we adopted $D_0 = 2.5 \times 10^{-9} \text{ m}^2/\text{s}$, a value that is close to the bulk self-diffusion coefficient of water at room temperature. The spherical pore has a characteristic length, a , equivalent to its radius, r . In that sense, we adopt $a = 10 \mu\text{m}$, and a surface relaxivity, ρ , of $50 \mu\text{m}/\text{s}$. Notice that these values characterize the fast diffusion regime ($\rho a/D_0 = 0.2$). In [13], this high relaxivity value is adopted to illustrate the low dependency of the echo attenuation in respect to ρ . Here, we choose to mirror their results and, therefore, adopt the same value. A sample of 10 million walkers populated the simulations and the sphere was represented in a resolution, Δ , of $0.1 \mu\text{m}$.

The diffusion timescale, $t_D = a^2/D_0$, is the expected time for the majority of the particles to diffuse through the porous medium's characteristic length. In this context, each diagram in Figure 4.1 shows the amplitudes for four multiples of this value (0.2, 0.5, 1.0 and 2.0). This approach allows us to investigate the results for short and long values in this timescale. For the sake of clarity, we omit the amplitudes for isolated slabs and cylinders. Under the same conditions, their inclusion would be redundant and, thus, all the subsequent discussion regarding the results for spherical pores would also be applicable for these other simple geometries.

The PFGSE amplitudes computed from RW simulations have a fine agreement with the predicted theory over a large range of gradient magnitudes. We can observe that the trend of quasidiffraction peaks and valleys is well captured. This characteristic pattern can be clearly seen for times greater than $t = 0.5 t_D$ in Figure 4.1. As described in details at [13], these trends can be used to infer pore geometry information. For isolated spheres, the first valley appears at $(2\pi)^{-1}ka \approx 0.73$, with the echo attenuation effect caused by

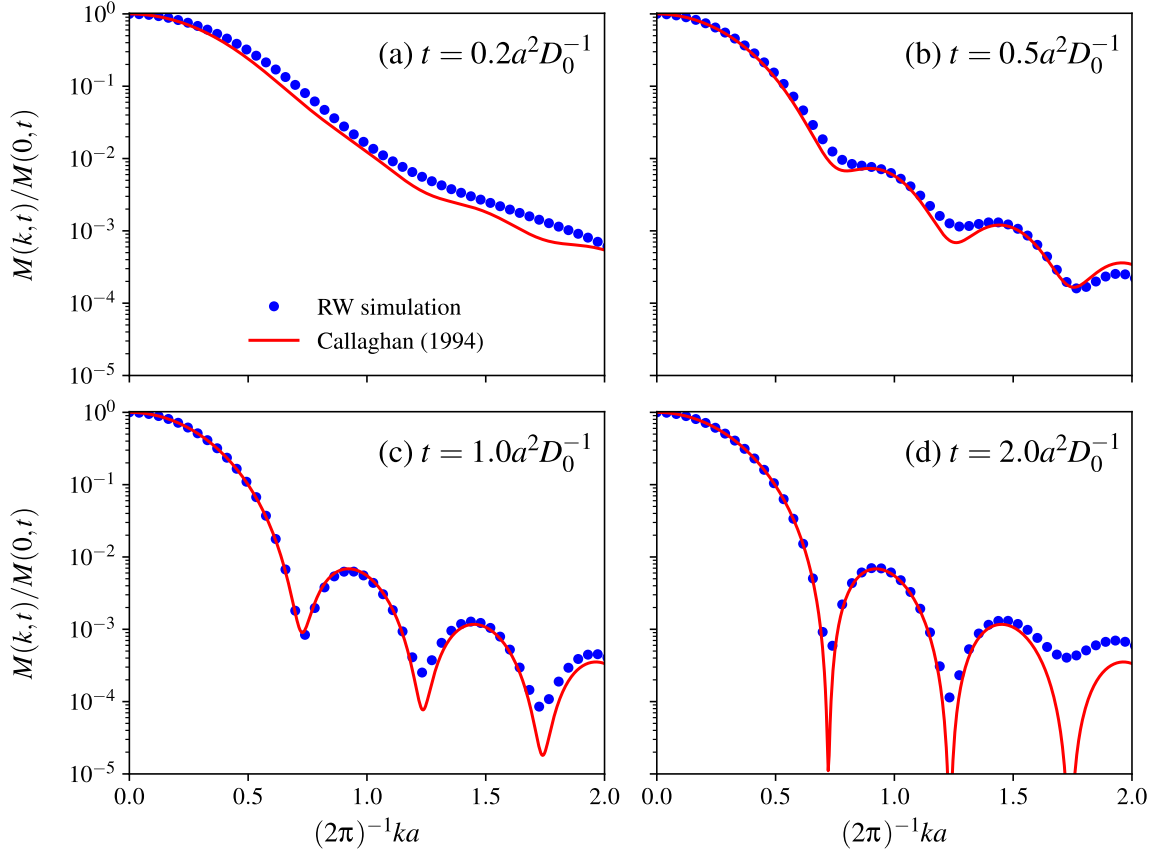


Figure 4.1: **Isolated spherical pore:** theoretical and simulated PFGSE amplitudes for spins trapped in a spherical pore of radius $a = 10 \mu\text{m}$, as a function of the wave vector magnitude, k . Results for four multiples of the diffusion timescale are displayed: (a) 0.2; (b) 0.5; (c) 1.0; and (d) 2.0.

pore wall relaxation weakly pushing to larger values.

A wider deviation can be observed in the range of large ka values for longer times. It suggests that the presence of higher wave vector $\mathbf{k}$ makes it harder to accurately reproduce the phenomenon, and a larger sample of walkers might be required. As a clear example, the third valley was not captured when $t = 2t_D$ in our simulations. However, when measuring the time-dependent diffusion coefficient, we are mainly interested in the beginning of the curve because Eq.(2.10) only holds when small wave vector $\mathbf{k}$ are applied. In this particular range, simulation and theory present a good fit, although we observe a slightly looser adjust when $t = 0.2t_D$, hinting that short-time computations may be less accurate.

In order to verify the influence that the number of walkers, n , has in the quality of the results, the same simulations are performed adopting different sizes n . Starting with a set of 10^3 walkers, we progressively increase this number by an order of magnitude until reaching 10^7 , the same amount used for validation. The normalized amplitudes for each walker sample size are presented in Figure 4.2, along with the theoretical prediction,

shown in solid lines. In Table 4.1, we show the coefficient of determination, R^2 , between simulated and analytical results.

Table 4.1: **Isolated spherical pore:** coefficient of determination, R^2 , between analytical and simulated predictions for the PFGSE amplitudes for varying ensemble sizes.

Coefficient of determination (R^2)				
n° of walkers	observation time			
	$0.2 t_D$	$0.5 t_D$	$1.0 t_D$	$2.0 t_D$
10^3	0.909	0.779	0.475	0.624
10^4	0.945	0.875	0.666	0.770
10^5	0.995	0.764	0.752	0.769
10^6	0.999	0.982	0.909	0.852
10^7	0.998	0.995	0.992	0.933

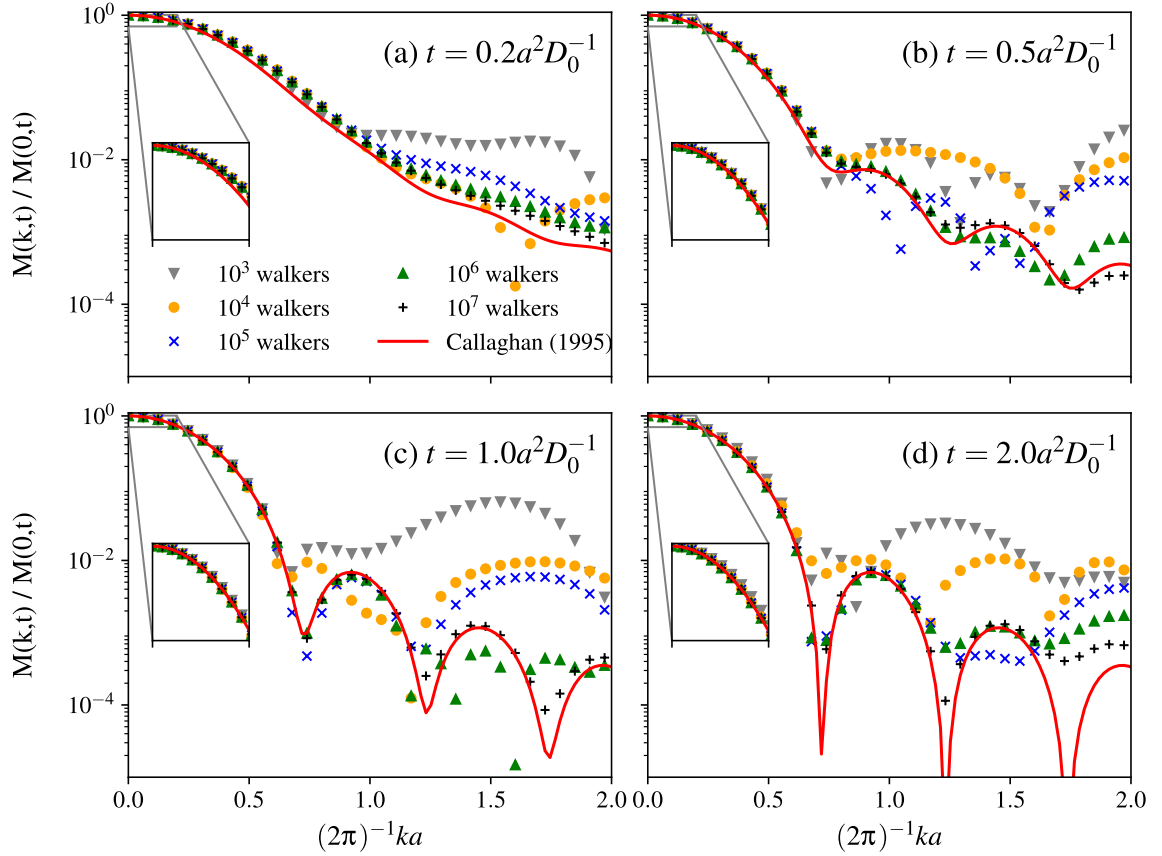


Figure 4.2: **Isolated spherical pore:** comparison of the PFGSE amplitudes for spins trapped in a spherical pore of radius $a = 10 \mu m$, shown as a function of the wave vector magnitude, k , for ensemble of walkers of varying sizes. The theoretical result (solid red line) is also shown. Surface relaxivity of $50 \mu m/s$ acts at the pore wall. Results for four multiples of the diffusion timescale are displayed: (a) 0.2; (b) 0.5; (c) 1.0; and (d) 2.0.

These results lead to interesting observations. First, we observe that a larger sample of walkers improves the fit between amplitudes from simulation and theoretical prediction. This is in fact expected because increasing the number of walkers actually imply in a better

representation of the diffusion propagator inside the porous region. Also, the number of walkers that is required for a good estimation depends not only on the magnitude of the applied gradient, but also on the observation time and, as a consequence, on the number of steps in the simulation. A good illustration can be seen in results for $t = 0.2 t_D$ and $n = 10^3$ where the amplitudes start to deviate when the gradient magnitude is such that $(2\pi)^{-1}ka$ is close to 0.9. However, when larger samples are adopted, deviations appear at higher $\mathbf{k}$ values. In addition, a simple observation exercise shows why we also need to be cautious in our assumptions: the amplitudes for $n = 10^6$ and $n = 10^7$ when $t = 0.2 t_D$ are remarkably close for all evaluated points, with the amplitudes of the larger of the samples slightly closer to the analytical curve. Table 4.1 also shows close $\mathbf{R}^2$ values for these results and, at this point, we could wrongly assume that 10^6 walkers are enough to simulate the experiment for any value of t or $\mathbf{k}$. However, results for other observation times rebut this assumption. On the other hand, amplitudes for short wave vector $\mathbf{k}$ are detailed in the insets at the lower left corner of each diagram in Figure 4.2. Results seem to be much less sensitive to the number of walkers in this range, suggesting that appropriate $D(t)$ estimations can be obtained with fewer walkers than the 10 million used for validation.

Finally, we investigate the effects of poor geometrical representation due to low image resolutions. For this purpose, we represent the same spherical pore in different resolutions. In the previous results, we use a resolution of $0.1 \mu m$, which takes 100 voxels to represent the characteristic pore length. We progressively worsen the original image so that one voxel in the new image represents a cube with 64 voxels of the previous image ($4^3 = 64$). Doing so, we acquire new coarser images with resolutions of $0.4 \mu m$, $1.6 \mu m$, and $6.4 \mu m$. Again, we use 10 million walkers in the simulations in order to achieve good amplitudes for higher wave vector $\mathbf{k}$. Also, for this experiment, regardless of the adopted resolution, we fix the length of the random step at $0.1 \mu m$, ensuring that only the geometrical representation of the porous medium shows variation.

The PFGSE amplitudes in Figure 4.3, and the coefficient of determination, $\mathbf{R}^2$, in Table 4.2, suggest good agreement between theoretical and simulated results for a large range of observation times, wave vector $\mathbf{k}$ magnitudes and image resolutions. Once again, the insets detailing the small wave vector $\mathbf{k}$ range highlight this agreement. For the image with $\Delta = 6.4 \mu m$, however, the results differed completely and the aforementioned characteristic quasi-diffraction peaks and valleys occur at unexpected wave vector $\mathbf{k}$ values. On a brighter note, even when $\Delta = 1.6 \mu m$, the amplitudes show a sensible adjust.

Table 4.2: **Isolated spherical pore:** coefficient of determination, R^2 , between analytical and simulated predictions for the PFGSE amplitudes for varying image resolutions.

Coefficient of determination (R^2)				
resolution (μm)	observation time			
	$0.2 t_D$	$0.5 t_D$	$1.0 t_D$	$2.0 t_D$
6.4	0.988	0.874	0.569	0.403
1.6	0.998	0.926	0.947	0.782
0.4	0.997	0.987	0.935	0.924
0.1	0.998	0.995	0.992	0.933

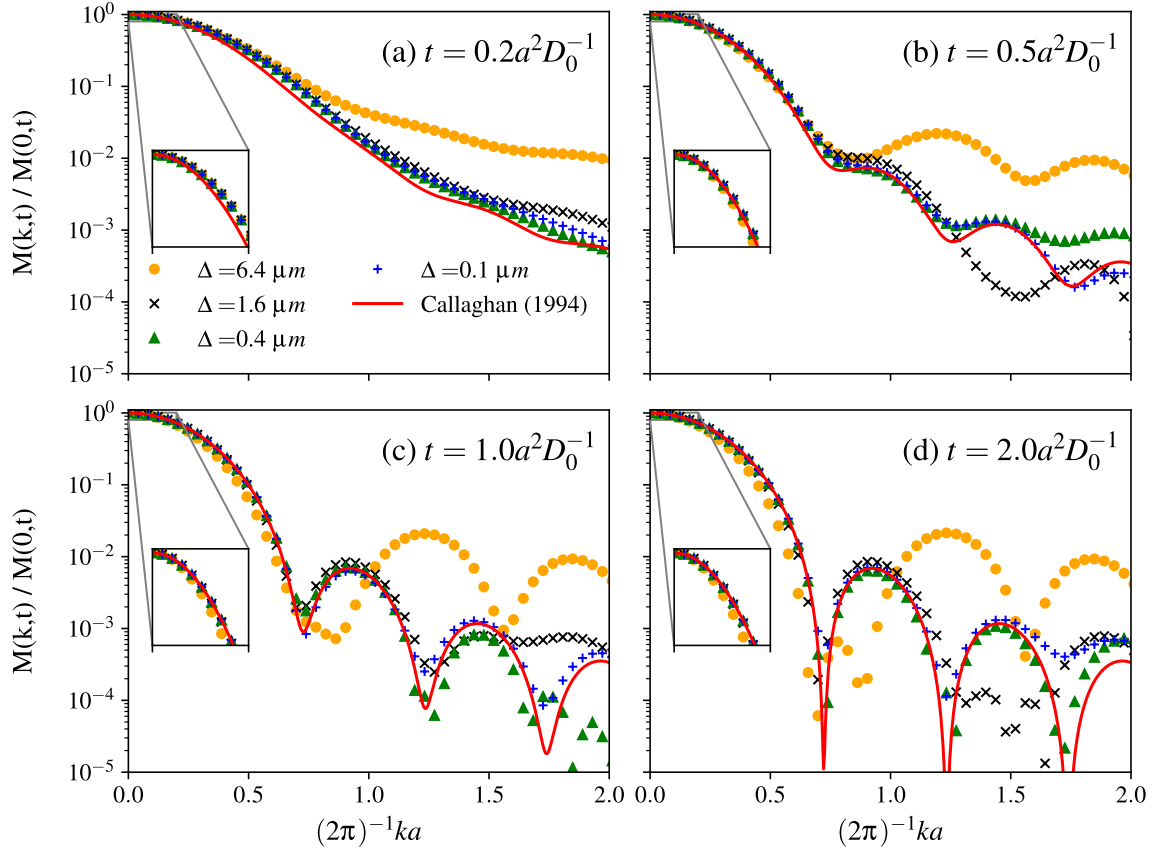


Figure 4.3: **Isolated spherical pore:** comparison of the normalized PFGSE amplitudes, $M(k,t)/M(0,t)$, for spins trapped in a spherical pore of radius $a = 10 \mu m$, shown as a function of the wave vector magnitude, k , for varying image resolutions. Also shown is the theoretical result (solid red line). Results for four multiples of the diffusion timescale are displayed: (a) 0.2; (b) 0.5; (c) 1.0; and (d) 2.0.

4.1.2 Periodic microstructure

In the second validation experiment, we simulate the PFGSE experiment in a periodic microstructure of cubic unit cells with centered spherical grains and estimate the time-dependent diffusion coefficient. Once again, we compare the estimations to established analytical predictions for validation purposes and discuss the results.

The PFGSE amplitudes of this periodic media can be expressed as a function of the eigenstates, $\bar{\psi}_{n\mathbf{q}}$, and eigenvalues, $\mu_{n\mathbf{q}}$, of the diffusion propagator as [8]

$$M(\mathbf{k}, t) = \frac{1}{\phi} \sum_n e^{-\mu_{n\mathbf{q}} t} |\bar{\psi}_{n\mathbf{q}}(\mathbf{g}_{\mathbf{k}})|^2 |_{\mathbf{q}=\mathbf{k}-\mathbf{g}_{\mathbf{k}}}, \quad (4.1)$$

where ϕ is the porosity of the medium and $\mathbf{g}_{\mathbf{k}}$ is the reciprocal lattice vector that is closest to $\mathbf{k}$.

In Bergman's work, it is shown how to compute the diffusion propagator and the PFGSE amplitudes [9]. Their main idea is to move the diffusion problem, usually set from a perspective that is only valid in the porous region and is based on mass density, to a more general perspective that is valid everywhere and is based on the chemical potential. This approach allows the original differential problem to be rewritten as a matrix eigenvalue problem defined by an infinite set of algebraic linear equations. Standard numerical methods can thus be applied in order to find $\bar{\psi}_{n\mathbf{q}}$ and $\mu_{n\mathbf{q}}$ that solve the truncated version of this system of linear equations. With the diffusion propagator in hands, this method allows computing $M(\mathbf{k}, t)$ for small wave vector $\mathbf{k}$ and estimating $D(t)$ for a given observation time.

Three scenarios of the periodic problem appear by varying the characteristic length of the the cubic unit cell edge, a , and the spherical grains radius, r . Intuitively, as larger grains imply in a more restrictive medium, $D(t)$ must assume smaller values. If $r < a/2$, neighbor grains do not touch, and the configuration resembles fluid immerse obstacles. If $r = a/2$, the arrangement portrays a simple cubic packing of hard spheres, upon which each grain touch each of its neighbor grains exactly at the center of the faces of the periodic unit cell. Last, grains overlap if $r > a/2$, setting a regular microstructure of strangled pore channels. The maximum r/a ratio is $1/\sqrt{2}$ and, beyond this value, $D(t \rightarrow \infty) = 0$ because the percolation limit of the medium is reached, i.e., the porous medium do not establish connected channels, but isolated pore regions.

In Figure 4.4, we show the apparent diffusivity, $\frac{D(t)}{D_0}$, estimated from our simulation as a function of the diffusion timescale. For these results, we adopted a periodic microstructure with $a = 20 \mu m$ and $r = 12 \mu m$, resulting in a theoretical porosity of 0.202. Once again, we adopt $D_0 = 2.5 \times 10^{-9} m^2/s$ and the pulse width, δ , is of $0.1 ms$. The theoretical prediction is presented along with the simulation results obtained directly from the mean squared displacement (msd) by combining Eq. (3.4) with Eq. (2.4), or via Stejskal-Tanner equation (S&T), using results from Eq. (3.5) in Eq. (2.11). Notice that in the PFGSE procedure, it does not make sense to measure the diffusion coefficient for

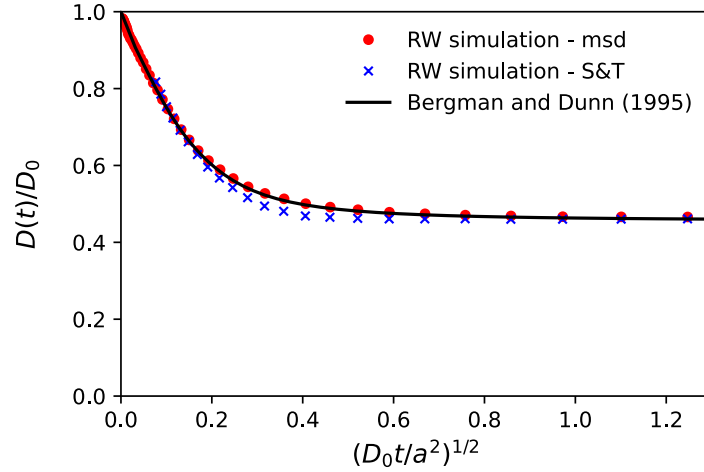


Figure 4.4: **Periodic microstructure:** apparent diffusivity as a function of the diffusion timescale in a periodic microstructure of spherical grains with porosity of 0.202. $D(t)$ is estimated by the mean squared displacement of random walkers (msd), and by the PFGSE amplitudes for small wave vectors $\mathbf{k}$ using the Stejskal-Tanner equation (S&T). Also shown in solid lines the analytical prediction.

observation times lower than the pulse width, δ [14]. Therefore, the method based on the Stejskal-Tanner equation is not adequate for $t < O(\delta)$ and, these results for $t < O(\delta)$ are omitted. For observation times outside this range, however, both methods provide similar results. After a careful analysis of the PFGSE amplitudes for small wave vector $\mathbf{k}$, we opted to use 10^5 walkers to simulate the experiment in a image with resolution of $0.1 \mu m$. The results also show a good agreement between simulated and theoretical estimations in the short and long time ranges. Furthermore, both methods show a similar D_e of 0.46 which is the value predicted by the analytical solution showed here, and by other estimations such as [7], based on the bulk effective electrical conductivity. For the sake of clarity, from here only the method based on the Stejskal-Tanner equation will be shown in this work.

Figure 4.5 presents the time-dependent diffusion coefficient estimations for the same microstructure represented in four different resolutions ($\Delta = 0.5, 1.0, 2.0$ and $4.0 \mu m$). As before, the aim here is to investigate the effect of geometrical representation in our simulations and, once again, we show the theoretical prediction for comparison purposes. Also, we vary the step length, ϵ , to four different values in respect to the adopted resolution, which gives estimations for $D(t)$ adopting $\epsilon = \Delta, \Delta/2, \Delta/4$ and $\Delta/8$.

Results for $\Delta = 1.0 \mu m$ and $0.5 \mu m$ show a good agreement between simulated estimations and theoretical predictions. Furthermore, for $\Delta = 0.5 \mu m$, $D(t)$ estimations for

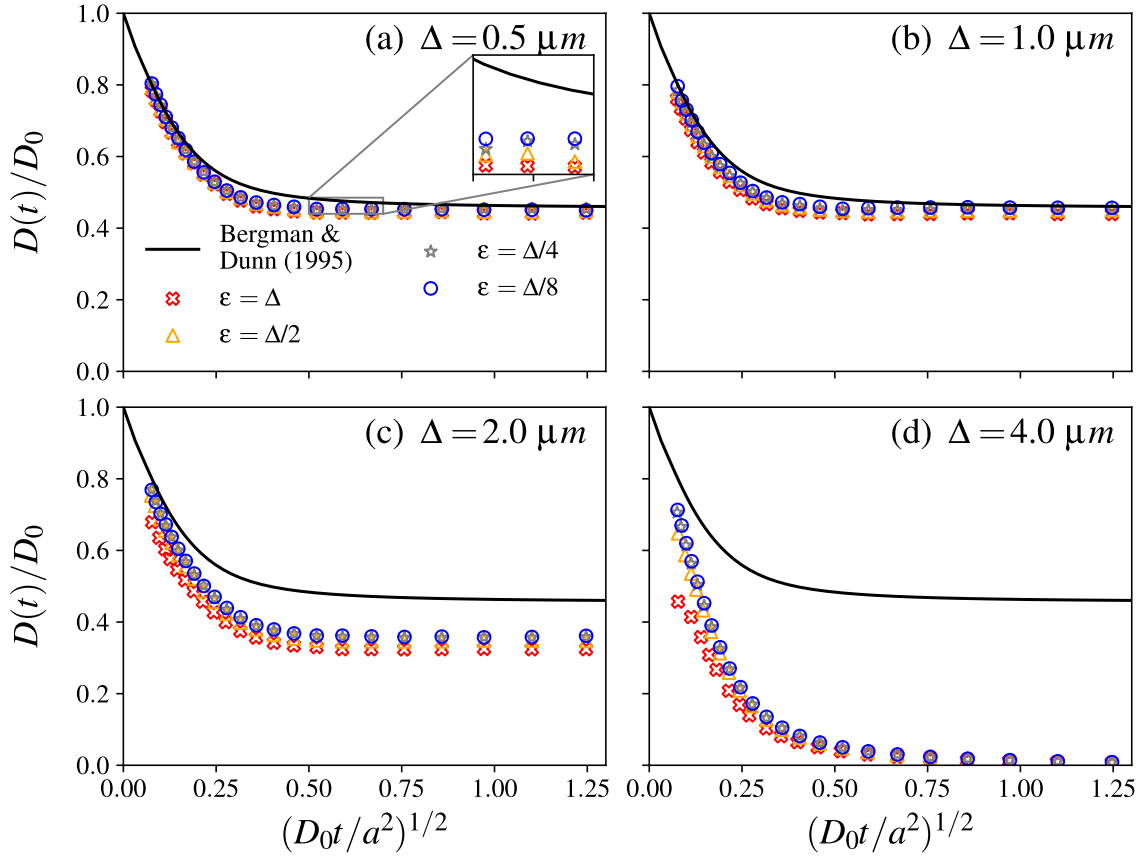


Figure 4.5: **Periodic microstructure:** estimations for the apparent diffusivity as a function of the diffusion timescale in a periodic microstructure of spherical grains with theoretical porosity of 0.202 represented in the resolutions (a) $0.5 \mu m$, (b) $1.0 \mu m$, (c) $2.0 \mu m$ and (d) $4.0 \mu m$. In each diagram, simulations were performed with varying step lengths.

long times recovers a D_e of around 0.46, which is the value predicted by the analytical prediction. For $\Delta = 1.0 \mu m$, it is slightly underestimated, presenting a relative error of less than 5%, which suggests it might be an adequate geometrical representation. On the other hand, for $\Delta = 2.0 \mu m$, we notice that the recovered D_e presents a relative error of around 25%, while, for $\Delta = 4.0 \mu m$, $D(t)$ asymptotically approaches zero in the long-time limit. This happens because the coarser geometrical representation of the theoretically well connected porous medium results in a voxelized representation composed essentially by isolated pores. Any tortuosity measurement in this medium is compromised regardless of the adopted step length. These observations illustrate the inability to correctly represent micro connectivity when coarser resolutions are adopted. Also, it might explain some diverging results when simulating diffusion in μ -CT images if an adequate resolution is not adopted.

In the top right corner inset of the $\Delta = 0.5 \mu m$ diagram, the observed trend that

appears when shorter steps are adopted is detailed. In general, this refinement results in $D(t)$ measurements closer to the theoretical prediction, partially offsetting the underestimation caused by coarse image discretization. This compensation is enhanced in coarser resolutions (this can be observed by comparison of results adopting $\Delta = 2.0 \mu m$ and $\Delta = 1.0 \mu m$). Still, as smaller steps are adopted, this improvement is gradually reduced. At the same time, smaller steps increase the simulation cost because the required steps per simulation are inversely proportional to ϵ^2 . Hence, while the step length is divided by half, our simulations takes around four times longer. This observation eventually leads to a point where it is no longer worth adopting smaller step lengths, because the relative improvement in the $D(t)$ estimation is minimal and does not make up for the increasing additional cost.

After investigation of the input parameters of the RW simulation, our experience suggests that, as an illustrative example, if we want to simulate the NMR diffusion experiment in a given medium with typical volume of around 500^3 voxels, $n = 10^5$ walkers is generally enough to yield reasonable results. Yet, notice that to get this number, we are assuming the image resolution is adequate to represent the micro connectivity of the porous medium and that only small $\mathbf{k}$ are needed in order to estimate $D(t)$. Therefore, depending on the level of heterogeneity of the medium or even the relaxivity strength acting at the pore walls, a larger n may be required. Furthermore, selecting an adequate step length is intimately related to the image resolution as it will be discussed from the results in synthetic media, presented in the next section.

4.2 Synthetic granular media

Once our code is validated, we can perform simulations in media with more complex pore spaces. First, we perform RW simulations from a set of procedurally generated synthetic media and compare the geometric parameters recovered from these simulations to the same parameters obtained from computational tools that use other established numerical methods. We also investigate these results with respect to absolute permeability measurements.

We create a set of synthetic 3D media that is procedurally generated by an adaptation of the *Spherical Growth* algorithm. A full description of this algorithm can be found in [42]. In a nutshell, it creates a central grain and with random insertions around this seed, it creates a packing of spherical grains. In our adaptation, in each insertion, it is allowed

for the grain to vary its radius within a defined range. Also, another input parameter of this algorithm is the allowed overlapping margin among neighbor grains. In that sense, 21 synthetic porous media were created by adopting varying radius and overlap lengths. Despite its simplicity, this method is able to create some diverse synthetic media with porosities ranging from 4.28% to 42.09% as shown in Figure 4.6.

For each medium, we run RW simulations in order to measure the apparent diffusivity and recover the surface-to-volume ratio of the pore space, S/V , and the tortuosity, τ . This is accomplished by finding the parameters that best fit the diffusivity measurements with Padé approximant, Eq. (2.7). Unlike the previously studied validation cases, there is no analytical prediction available in the literature for the apparent diffusivity in these synthetic media. We adopt a different approach, by analyzing and comparing our results

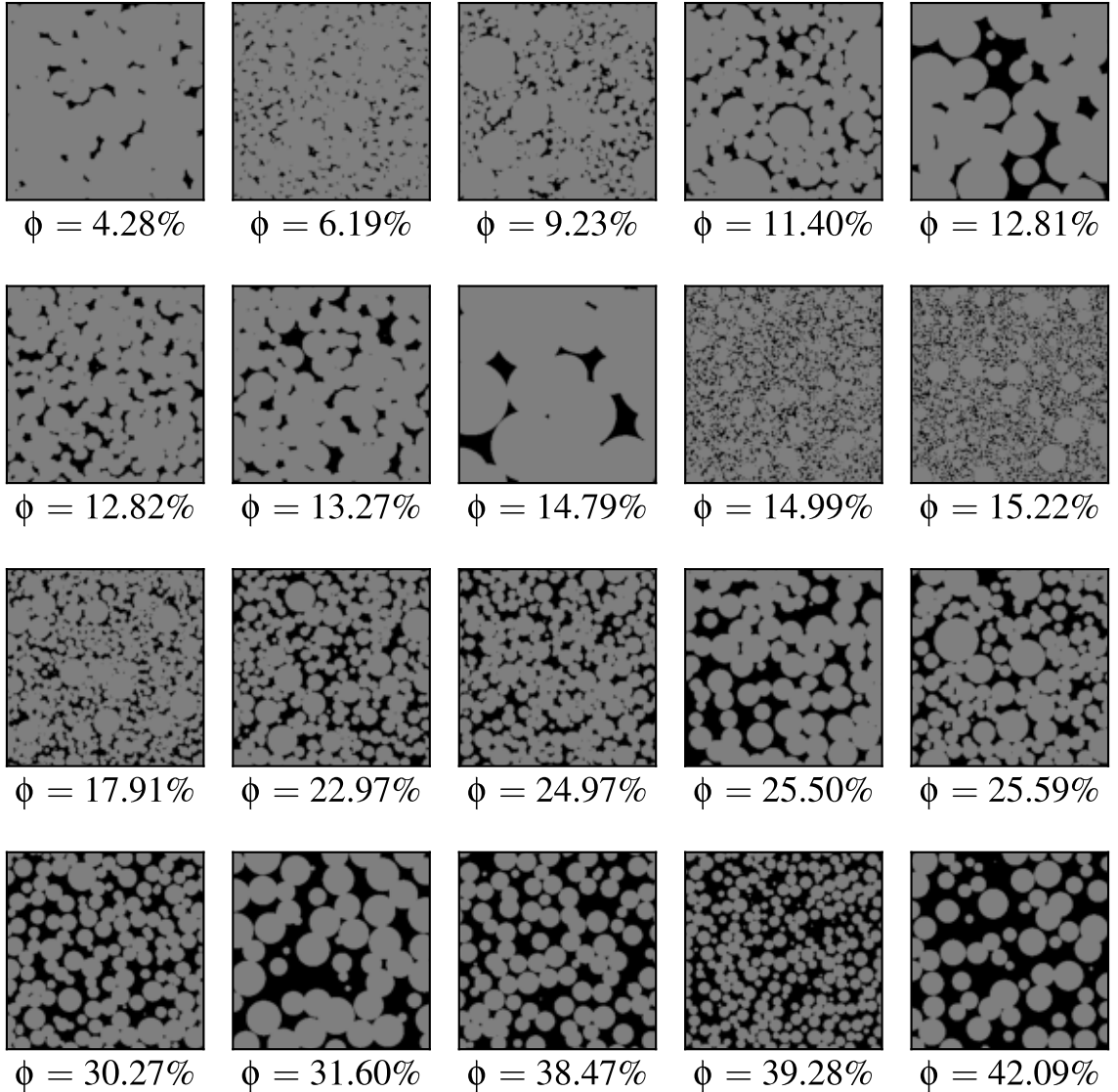


Figure 4.6: **Synthetic media:** example slices of procedurally generated porous media.

with surface-to-volume ratios and tortuosities acquired from other computational tools that use established numerical methods.

The present methodology is entirely based on digital images, which means that an average surface-to-volume ratio can be recovered by traversing the image and counting the pore-matrix interfaces in the immediate neighborhood of each pore voxel. In general, a porous medium represented by a digital image presents n_{pv} pore voxels and n_{pmi} pore-matrix interfaces where each voxel has a volume of Δ^3 and each interface has an area of Δ^2 . In that sense, the surface-to-volume ratio, S/V , can be therefore computed as

$$S/V_{image} = \frac{1}{\Delta} \frac{n_{pmi}}{n_{pv}}. \quad (4.2)$$

We show in Figure 4.7a the correlation between S/V_{RW} , recovered from our simulations, and S/V_{img} , acquired by straightforward application of Eq. (4.2) to the image data. Also, the results for simulations in the same media but adopting three different step lengths ($\epsilon = \Delta$, $\Delta/2$, and $\Delta/16$) are presented. The 1:1 relationship is represented by the dotted lines and, by assuming the value extracted from the image is the ground truth, it is clear that when shorter steps are taken, an improved S/V estimation is recovered from the simulations. The improved reading of the pore space is accomplished not only because adopting smaller steps imply in a finer discretization of time and space, but also because it enables $D(t)$ estimations in shorter observation times. In this context, the approximation in Eq. (2.5) is better captured when shorter observation times are considered.

In order to evaluate the tortuosity obtained from our RW simulations, we select 6 of the 21 synthetic porous media and compare the values recovered with values for the same parameter recovered from physical simulations that use image-based implementations of the Finite Difference Method (FDM) and Finite Element Method (FEM). A description of the implementations can be found in [18, 89], respectively. Comparing these diffusional tortuosities is not uncommon despite the intrinsic differences of each method. It is only possible though because the described fluid transport mechanism is essentially the same [30]. In that regard, both methods solve the diffusion differential problem, governed by Laplace's equation and with the same boundary conditions in which a constant mass concentration gradient acts between the opposite faces of the 3D image, while the other enclosing faces and the pore-matrix interface act as isolation barriers.

In comparison to the other two methods, the tortuosity provided by RW simulation seems to recover an intermediate value. This can be seen in Figure 4.7b. We observe that

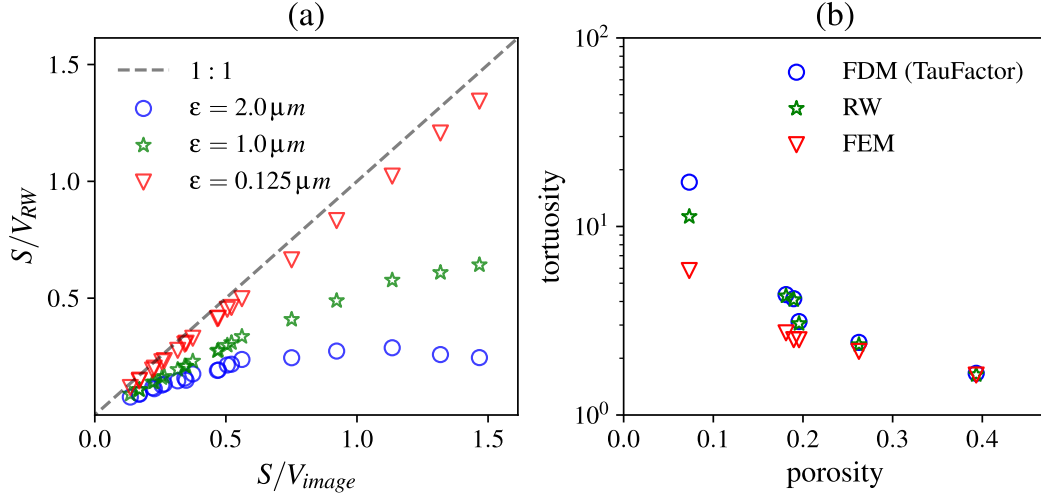


Figure 4.7: **Synthetic media:** (a) correlation between the porous surface-to-volume ratios obtained via RW simulations, S/V_{RW} , and directly from the digital image data, S/V_{Image} , using varying step lengths, ϵ . Dotted lines represent 1:1 relation; (b) estimated tortuosity vs porosity. Tortuosities are estimated from 3 numerical methods: Finite Differences Method (FDM - blue circles), Random Walk simulations (RW - green stars) and Finite Element Method (FEM - red triangles).

FDM and RW results present a good agreement, as the relative error is kept below 3% for every porous medium other than the smaller porosity that seems to be an outlier. In this regard, a possible explanation for the outlier result might be related to a bad image resolution or small REV sizes as reported in [30]. Moreover, this discrepancy may also be explained by numerical inaccuracies as small variations in media with lower effective diffusion coefficients, D_e , can represent bigger variations in their tortuosity due to the inverse relation between these quantities.

We measure the absolute permeability of the synthetic media by computing the fluid flow velocity field within the pore space. To accomplish this, we use a similar image-based FEM implementation that solves Stokes equation. Figure 4.8a shows τ^{-1} in respect to the porosities, and Figure 4.8b shows the tortuosities in respect to the surface-to-volume ratios. Data points are colored and sized according to the permeability in both diagrams: dark and small points represent media with low permeability, and light and big points are related to higher values.

Figure 4.8a visually suggests a linear relationship between the inverse of tortuosity and the porosity. Still, the synthetic media with porosities within the range of 12% to 18% present a wide deviation from this general trend. In this particular interval, τ^{-1} presents a variation that exceeds 400%, ranging from as low as 0.08 to over 0.4. This is a perfect illustration of how models solely based on porosity fail to accurately predict

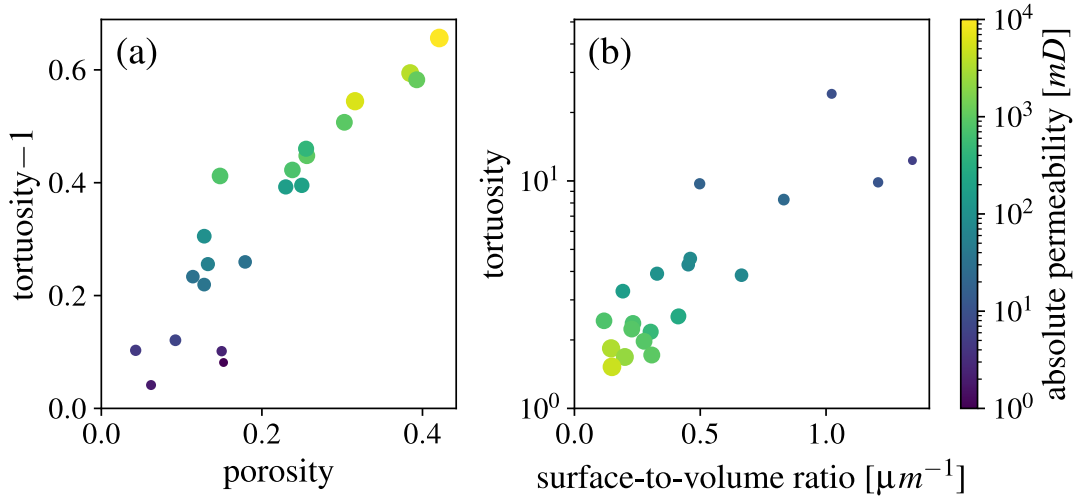


Figure 4.8: **Synthetic media:** (a) tortuosity^{-1} vs porosity; (b) tortuosity vs porous surface-to-volume ratio. In both diagrams, the color and the size of the points follow the color map scaled by the absolute permeability.

the tortuosity of a porous medium. On the other hand, the permeability of the media in this interval varies nearly three orders of magnitude, following the increase in τ^{-1} , i.e., media with lower tortuosity also present higher permeabilities. This fact may hint to reasonable explanation for the observed variation. It also brings the question if the parameters recovered from RW simulations can help predicting the absolute permeability measurements. In this context, Figure 4.8b present other interesting insights. In this diagram, we can observe that media with higher permeabilities occur close to the origin of the $(s/v, \log(\tau))$ coordinate system, i.e., it occurs in media with open and well-connected pore networks. On the other hand, media with more tortuous or strangled pore channels present lower permeabilities. In fact, some well-known permeability prediction models based on Cozeny-Karman equation use as input parameters porosity, pore-throat size and tortuosity measurements [30, 38].

The absolute permeabilities are presented in Figure 4.9 as a function of porosity, surface-to-volume ratio and tortuosity. For each quantity, dashed blue and solid red lines represent exponential curves ($y = ax^b$ and $y = ae^{bx}$, respectively) providing the best fit to the dataset by minimizing the least squared error. Also, in Table 4.3, we present the adjust parameters and some fitting metrics. The largest coefficient of determination, R^2 , and the smallest sum of squared residuals, ε , occurs for the tortuosity fitting curve, suggesting that, at least for these synthetic media, the tortuosity, in comparison to the other measured quantities, presents the strongest relationship with the absolute permeability.

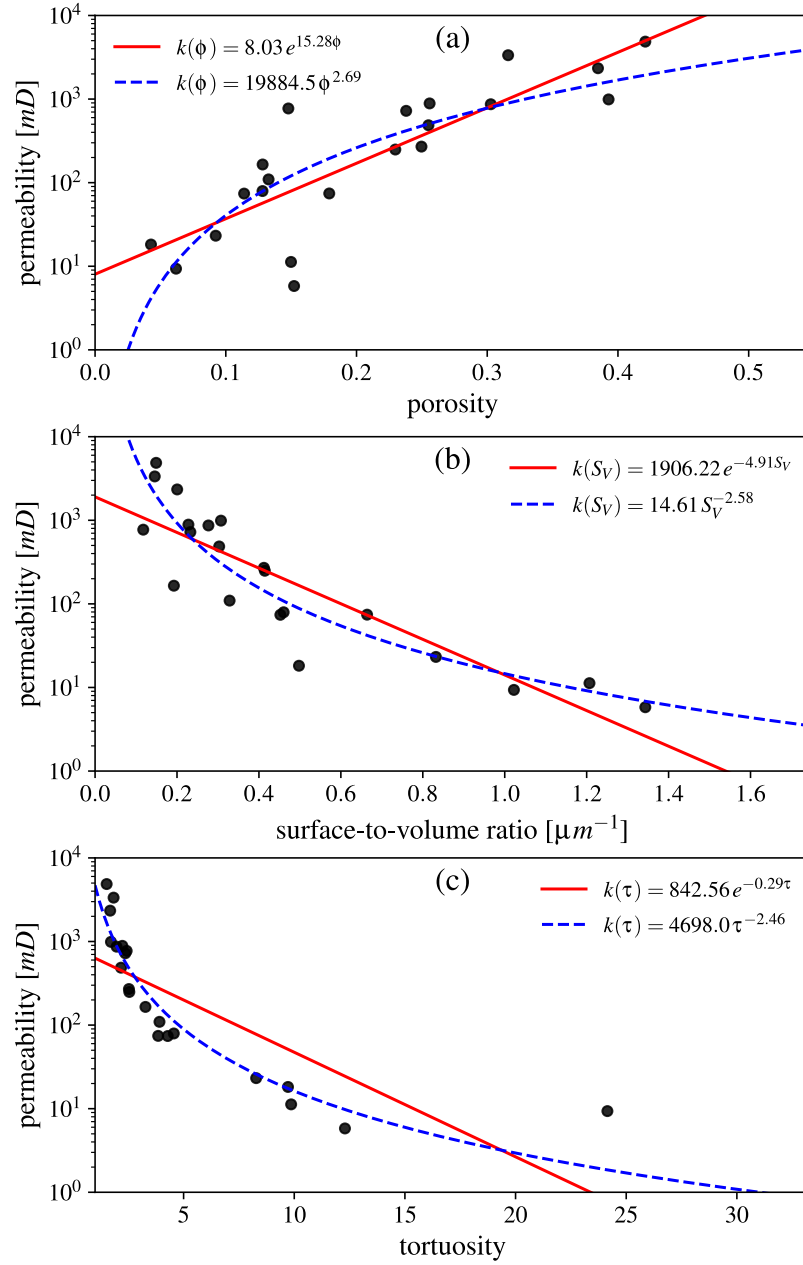


Figure 4.9: **Synthetic media:** (a) porosity, (b) surface-to-volume and (c) tortuosity exponential curves fitting absolute permeability data obtained via FEM fluid flow simulations.

Table 4.3: **Synthetic media:** absolute permeability exponential curves fitted with results from RW simulations.

x	$k(x) = ae^{bx}$				$k(x) = ax^b$			
	a	b	ϵ	R^2	a	b	ϵ	R^2
ϕ	8.03	15.28	23.44	0.68	19884.50	2.69	26.72	0.68
S/V	1906.22	-4.91	19.22	0.48	14.61	-2.58	14.68	0.35
τ	842.56	-0.29	31.44	0.39	4698.00	-2.46	8.58	0.74

k : absolute permeability;

ϕ : porosity; S/V : surface-to-volume ratio; τ : tortuosity.

ϵ : sum of residuals; R^2 : coefficient of determination.

4.3 Segmented Micro-computed Tomographic Images

In this section, we present the results of simulating NMR diffusion in a collection of segmented images of reservoir rocks. These images are acquired using 3D micro-computed tomography technique and appropriate post-processing, resulting in high-resolution images at the pore scale. Discussions regarding the results are also provided in this section. Firstly, we investigate the correlation between the computed diffusion results and the absolute permeability of the samples. In this context, we explore correlations for the geometric parameters inferred from the apparent diffusion curve, namely the surface-to-volume ratio and tortuosity. After that we compare the results obtained through computational simulation with those derived from laboratory NMR experiments.

4.3.1 Apparent diffusion in μ -CT images of reservoir rocks

Our collection comprises a total of 24 reservoir rock samples segmented in porous and granular phases, as detailed in tables 4.4 and 4.5. As shown, 14 of these samples correspond to sandstones, while the remaining 10 are carbonates. As we discuss in the following sections, we break the analyses and discussions into these two groups, following a common practice in both literature and industry, as each lithotype typically exhibits distinct properties. Additionally, the table provides important information about the images, including the acquisition resolution and the final porosity of the extracted cube used for simulations. The acquisition resolutions were chosen to strike a balance between accurately capturing the porous space and ensuring a representative volume that captures the sample's heterogeneity. For instance, if we were aiming to represent a 1mm^3 cube, a finer resolution would result in a larger image size and data volume. Achieving this balance posed a significant challenge. To enrich the analyses, we have also included experimental properties in the table, such as laboratory-measured porosity and surface

relaxivity, determined through the D - T_2 experiment [75]. The latter will be particularly useful when comparing computational and experimental results. For these simulations, accounting for surface relaxation effects is crucial. However, it is important to note that, initially, our focus is on estimating only the "pure" apparent diffusion computed in the simulations, based on the mean squared displacement of the walkers.

Table 4.4: **Sandstones micro-computed tomographic image properties.**

Δ_{image} : image resolution per voxel; ϕ_{image} : total porosity in the image volume; ϕ_{exp} : porosity measured experimentally in the lab; and ρ_2 : experimentally measured T_2 surface relaxivity.

Sample	Δ_{image} (μm)	ϕ_{image} (%)	ϕ_{exp} (%)	ρ_2 ($\mu m/s$)
Bandera Brown	2.0	12.2	20.8	8.6
Berea	1.0	15.3	20.2	11.4
Bandera Gray	2.0	12.8	21.7	10.0
Boise Idaho Gray	5.0	21.6	29.3	12.0
Bentheimer	5.0	18.6	23.2	7.6
Briarhill	3.0	16.4	24.3	16.5
Berea Stripe	2.0	19.3	21.2	12.1
Castlegate	5.0	20.2	26.3	20.0
Crab Orchard	2.0	2.7	6.5	24.2
Boise Idaho Brown	2.0	20.9	27.5	8.3
Kirby	1.0	21.7	21.8	19.4
Leopard	2.0	12.5	20.5	55.1
Sister Gray Berea	2.0	16.7	20.8	12.5
Torey Buff	1.0	7.4	16.9	10.0

Table 4.5: **Carbonates micro-computed tomographic image properties.**

Δ_{image} : image resolution per voxel; ϕ_{image} : total porosity in the image volume; ϕ_{exp} : porosity measured experimentally in the lab; and ρ_2 : experimentally measured T_2 surface relaxivity.

Sample	Δ_{image} (μm)	ϕ_{image} (%)	ϕ_{exp} (%)	ρ_2 ($\mu m/s$)
Austin Chalk	1.0	13.2	23.0	23.3
Desert Pink	0.9	31.0	30.5	12.3
Edwards Yellow	1.0	19.8	22.9	10.5
Guelph Dolomite	1.0	4.4	7.9	12.4
Indiana H.	1.5	3.1	18.9	35.8
Indiana L.	2.0	5.2	9.1	4.9
Indiana M.	2.0	3.8	13.8	3.3
Leuders	2.0	5.3	16.2	10.5
Sillurian Dolomite	2.0	15.2	12.4	9.5
Wiscosin	1.9	9.8	5.6	11.8

Example slices for each segmented sample are shown in Figure 4.10 (sandstones) and 4.11 (carbonates). In these images, the porous phase is depicted in black, while the grains are depicted in gray.

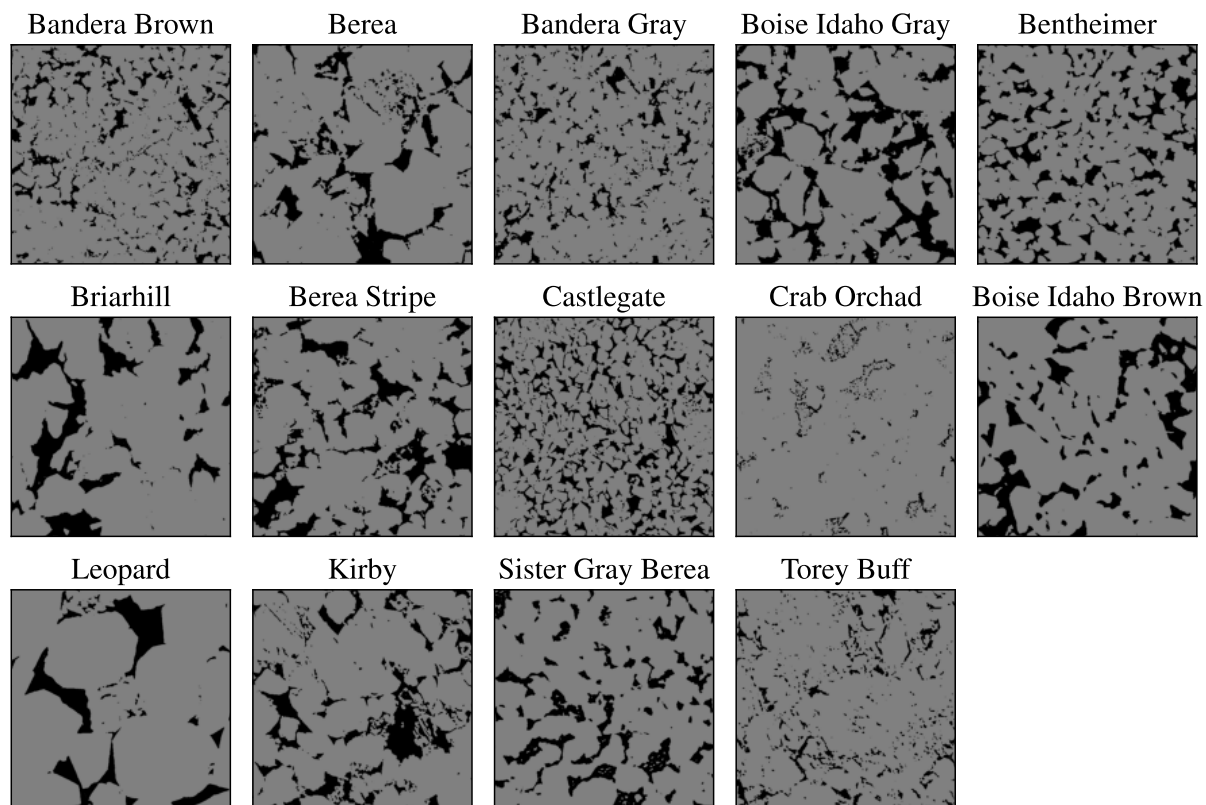


Figure 4.10: **Sandstones:** example slices of sandstones samples. In these images, the black pixels correspond to the porous phase.

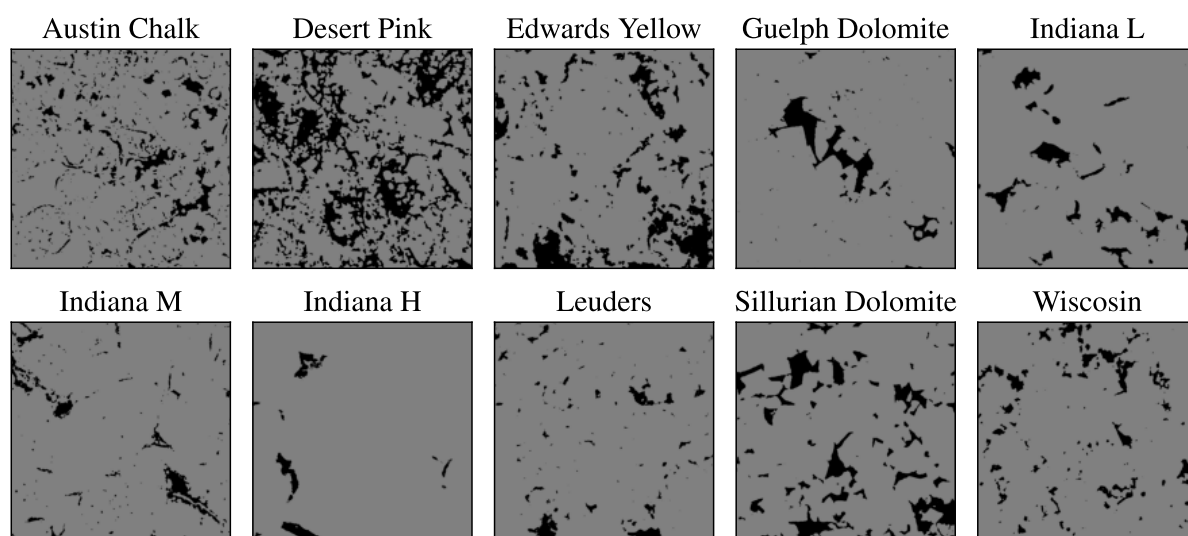


Figure 4.11: **Carbonates:** example slices of carbonates samples. In these images, the black pixels correspond to the porous phase.

Figures 4.12 and 4.13 showcase the results for diffusion simulation from the samples listed in Table 4.4 and 4.5, respectively. As observed, the characteristic behavior of apparent diffusion obtained in the synthetic porous media presented thus far in this study also emerges when simulations are conducted in media with significantly more intricate pore networks. In this investigation, we opted to allow the particles to traverse for a duration equivalent to approximately 5 seconds. It is worth noting that this time interval proved adequate for estimating a tortuosity limit based on the diffusion data for extended periods.

4.3.2 Absolute permeability correlations

Using the apparent diffusion data presented in the previous section, we utilize the Padé approximant, Eq. (2.7) to retrieve both the tortuosity and surface-to-volume ratio. Furthermore, for each sample, we conduct flow simulations to estimate their absolute permeability. These simulations were performed using the same computational code mentioned in Section 4.2, which employs the finite element method to solve the Stokes equation. For comparison, we also show the experimental results for absolute permeability measured in the lab, i.e., the routine petrophysics.

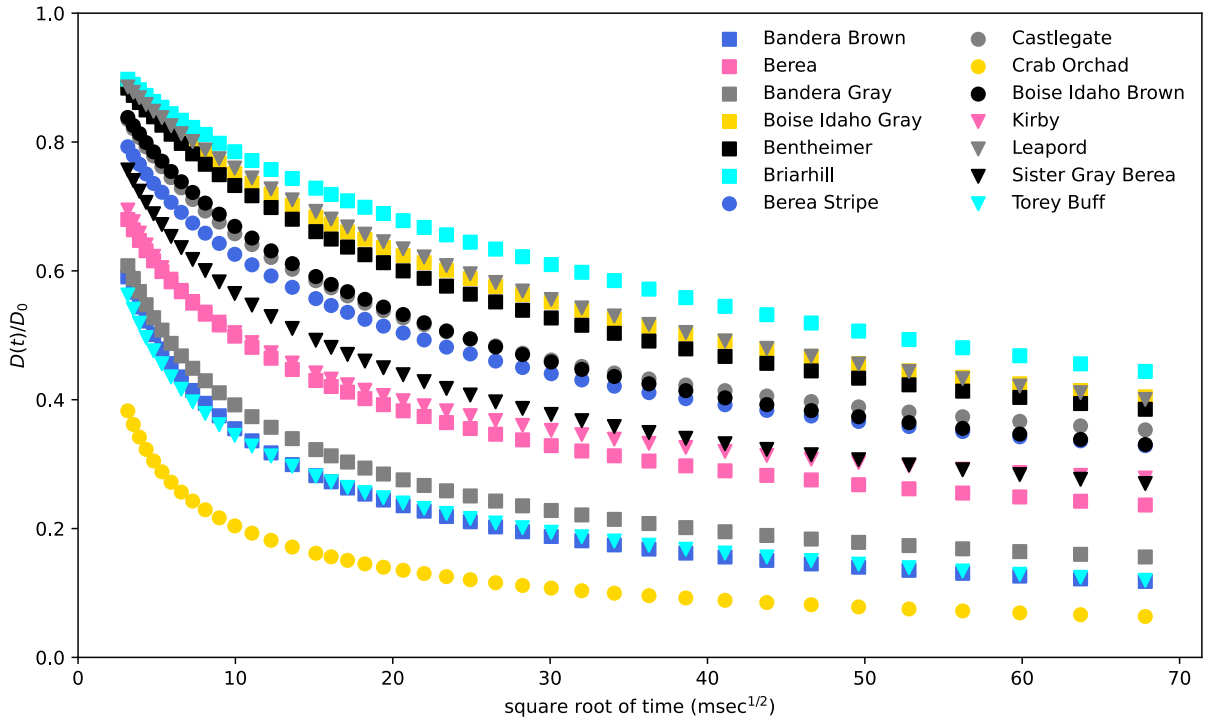


Figure 4.12: **Sandstones:** Results for the apparent diffusion from micro-computed tomographic images of sandstones samples.

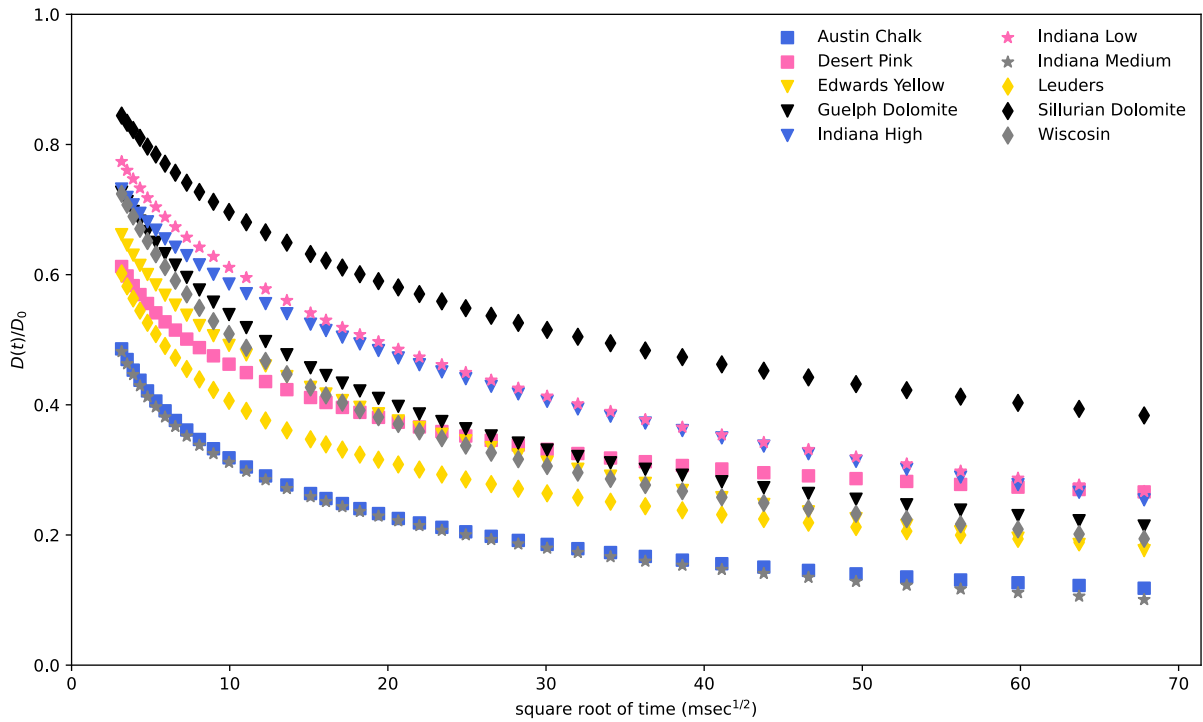


Figure 4.13: **Carbonates:** Results for the apparent diffusion from micro-computed tomographic images of carbonates samples.

4.3.2.1 Sandstones permeability correlations

The values for surface-to-volume ratio and tortuosity in sandstones, extracted from the diffusion simulations, along with the values of absolute permeability in the corresponding samples, obtained from the finite element simulation, are displayed in Table 4.6. Furthermore, we show the correlation between experimental and computational absolute permeability estimations in Figure 4.14. Last, we show the correlations between the inferred geometric parameters and the absolute permeability measured experimentally and from computational simulations in Figure 4.15 and 4.16, respectively.

Figure 4.14 show a good agreement between experimental and computational absolute permeabilities for most samples. We notice that the computational permeability of samples Bandera Brown, Crab Orchard and Kirby present a clear deviation from this pattern, around two orders of magnitude higher than its experimental measure. This may be a strong indication that the selected REV for these samples are not adequate to represent the geometry and heterogeneity of their porous space. For this reason, we exclude these samples from the experimental correlation analysis.

Tables 4.7 and 4.8 show the recovered values from the exponential fit of the geometric parameters and the experimental and computational absolute permeability, respectively.

Table 4.6: **Sandstones petrophysical properties.** S/V_P : average porous surface-to-volume ratio; τ : tortuosity of pore network; K_{EXP} : absolute permeability measured experimentally in the lab; and K_{FEM} : absolute permeability measured from Finite Elements fluid flow simulations.

Sample	S/V_P (μm^{-1})	τ (-)	K_{EXP} (mD)	K_{FEM} (mD)
Bandera Brown	0.99	13.59	1.21	63.38
Berea	0.65	5.00	149.00	43.02
Bandera Gray	0.91	8.65	22.60	69.17
Boise Idaho Gray	0.16	4.34	4310.00	3300.52
Bentheimer	0.18	4.59	2490.00	1829.98
Briarhill	0.13	3.67	3900.00	2391.09
Berea Stripe	0.35	3.92	415.00	479.60
Castlegate	0.28	4.05	1080.00	1106.10
Crab Orchard	2.38	18.60	0.09	6.49
Boise Idaho Brown	0.26	4.86	1510.00	691.92
Kirby	0.66	4.17	17.30	334.98
Leopard	0.16	4.60	1610.00	1075.29
Sister Gray Berea	0.45	4.88	149.00	373.81
Torey Buff	1.11	10.76	(-)	10.16

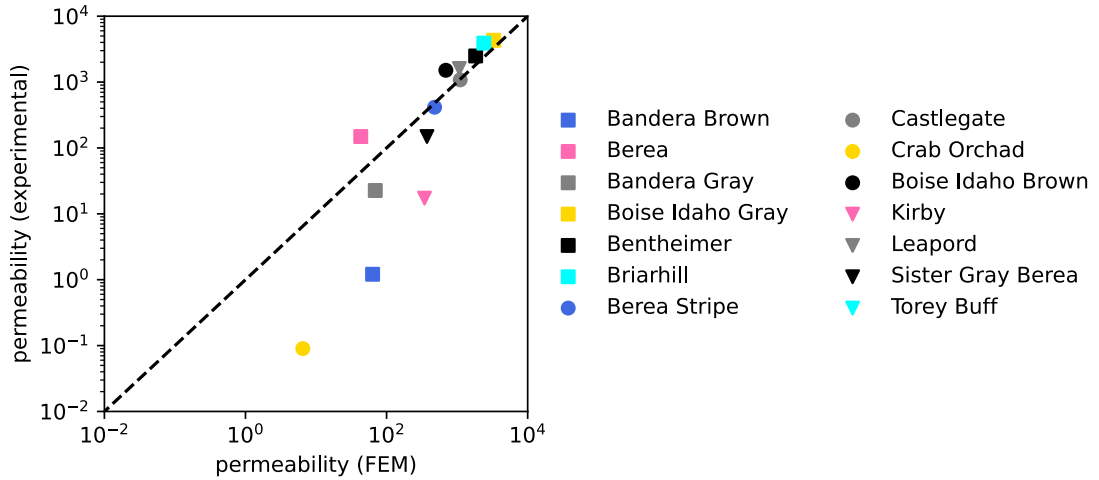


Figure 4.14: **Sandstones:** correlation between absolute permeability experimentally measured in the lab and computationally measured from FEM fluid flow simulations.

In both cases, the curve fitted correlating the surface-to-volume ratio to permeability present the higher R^2 ratio (0.93 and 0.90 for experimental and computational correlations, respectively).

The absolute permeability of sandstones exhibits a good correlation not only with the surface-to-volume ratio, but also, to some extent, with their tortuosity. These correlation are expected, given that these rocks typically possess homogeneous grains and well-defined pore spaces [62]. Consequently, it is not surprising that the industry already employs

Table 4.7: **Sandstones:** coefficients of the experimental absolute permeability exponential curves fitted with geometric parameters inferred from RW simulations.

x	$k(x) = ax^b$			
	a	b	ϵ	R^2
ϕ	22.42×10^6	4.59	11.91	0.27
τ	4.00×10^6	-5.59	8.99	0.60
S/V	-2.57	-4.91	3.74	0.93

ϵ : sum of residuals;

R^2 : coefficient of determination.

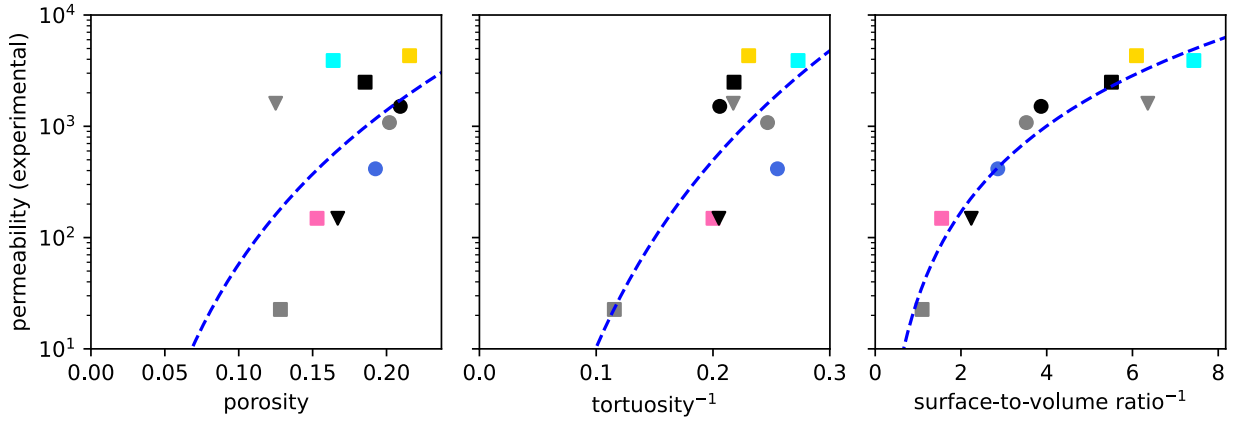


Figure 4.15: **Sandstones:** correlation between geometric parameters inferred from apparent diffusion curve and experimental absolute permeability.

surface-to-volume ratio results in alternative contexts, such as the transverse relaxation, T_2 , experiment, to build models for permeability estimation [41]. These models generally provide reasonably accurate outcomes. However, it is important to highlight that while these models tend to perform well for sandstones, they often yield less satisfactory results when applied to samples of different lithotypes with less predictable pore space behavior [93, 90]. Another noteworthy aspect is the influence of internal gradients on these results. For instance, when recovering the T_2 distribution of sandstones with a high content of ferromagnetic elements, the additional signal relaxation caused by these gradients can

Table 4.8: **Sandstones:** coefficients of the computational absolute permeability exponential curves fitted with geometric parameters inferred from RW simulations.

x	$k(x) = ax^b$			
	a	b	ϵ	R^2
ϕ	6.86×10^4	2.82	14.46	0.61
τ	8.69×10^4	-3.26	11.35	0.72
S/V	43.13	-2.16	6.84	0.90

ϵ : sum of residuals;

R^2 : coefficient of determination.

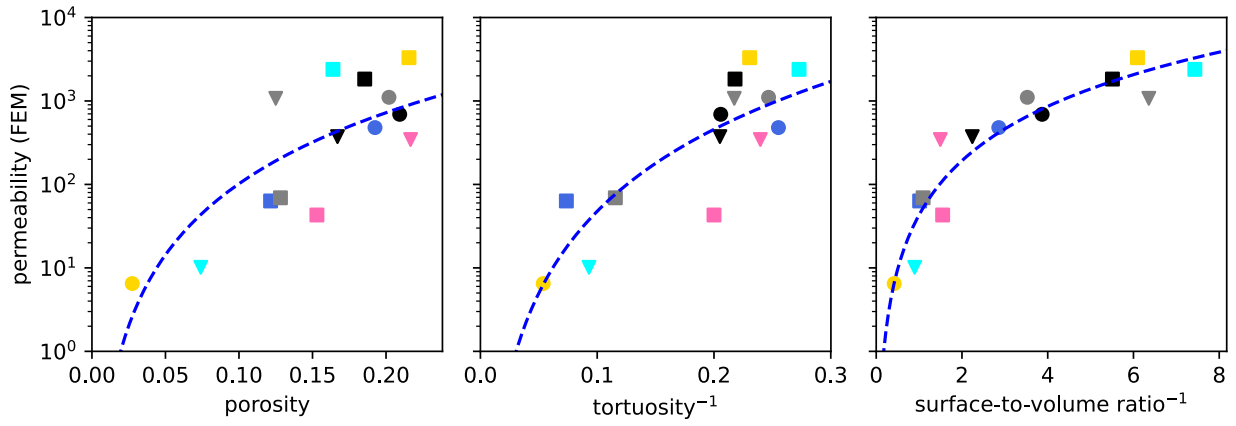


Figure 4.16: **Sandstones:** correlation between geometric parameters inferred from apparent diffusion curve and absolute permeability computationally estimated from FEM fluid flow simulations.

introduce a new element to the analysis that may not be directly related to the geometry of the porous medium [40, 29].

4.3.2.2 Carbonates permeability correlations

The values for surface-to-volume ratio and tortuosity in carbonates, obtained from the diffusion simulations, along with the values of absolute permeability in the corresponding samples, derived from the finite element simulation, are displayed in Table 4.9. Once again, we show the correlation between experimental and computational absolute permeability estimations (Figure 4.17), and the correlations between the inferred geometric parameters and the absolute permeability measured experimentally (Figure 4.15) and the absolute permeability estimated from computational simulations (Figure 4.16).

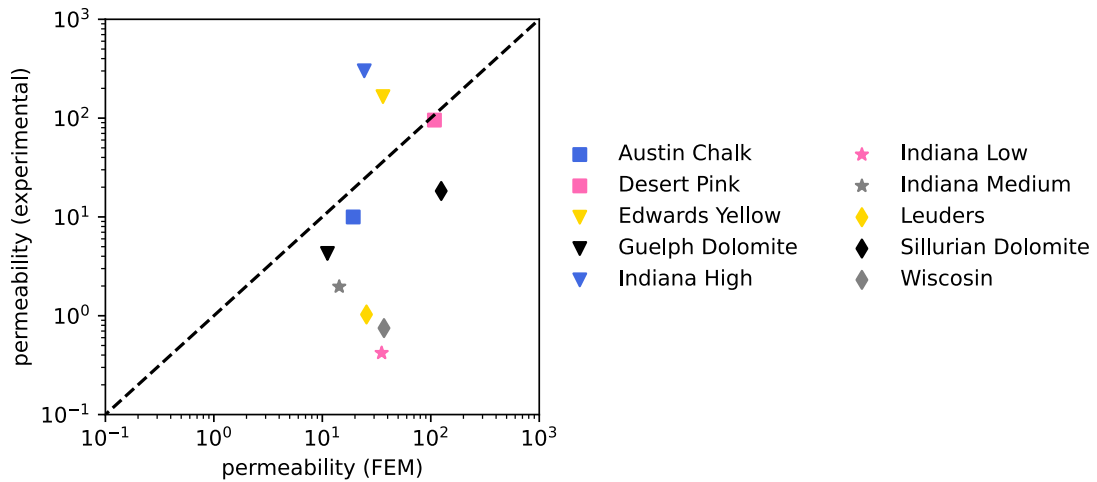


Figure 4.17: **Carbonates:** Correlation between absolute permeability experimentally measured in the lab and computationally measured from FEM fluid flow simulations.

Table 4.9: **Carbonates petrophysical properties.** S/V_P : average porous surface-to-volume ratio; τ : tortuosity of pore network; K_{EXP} : experimental absolute permeability; and K_{FEM} : FEM absolute permeability.

Sample	S/V_P (μm^{-1})	τ (-)	K_{EXP} (mD)	K_{FEM} (mD)
Austin Chalk	1.47	9.04	9.99	19.27
Desert Pink	0.96	3.96	95.50	108.05
Edwards Yellow	0.63	6.28	165.00	36.24
Guelph Dolomite	0.48	6.88	4.27	11.15
Indiana H.	0.43	4.43	301.00	24.37
Indiana L.	0.35	5.03	0.42	35.24
Indiana M.	1.49	9.84	1.97	14.32
Leuders	0.97	6.09	1.03	25.55
Sillurian Dolomite	0.24	3.46	18.30	124.81
Wiscosin	0.52	8.10	0.75	37.04

Unlike the sandstone results, Figure 4.17 do not show a good agreement between experimental and computational absolute permeabilities for most samples. We notice that the computational permeability of samples Indiana L., Indiana H., Leuders and Wiscosin present experimental and computational values disagreeing in around two orders of magnitude. Again, this is a strong indicator that the selected REV for these samples are not adequate to represent the geometry and heterogeneity of their porous space. For this reason, we exclude these samples from the experimental correlation analysis.

Tables 4.10 and 4.11 show the recovered values from the exponential fit of the geometric parameters and the experimental and computational absolute permeability, respectively. The experimental analysis show that porosity presents the highest correlation. However, this analysis is compromised by the small number of samples present in the data set. On the other hand, the curve fitted correlating the tortuosity to computational permeability present the higher R^2 (0.58) in comparison with the fit from other parameters.

Table 4.10: **Carbonates:** coefficients of the experimental absolute permeability exponential curves fitted with geometric parameters inferred from RW simulations.

$k(x) = ax^b$				
x	a	b	ε	R^2
ϕ	1.01×10^3	1.87	3.19	0.82
τ	1.19×10^3	-2.34	6.83	0.33
S/V	14.56	-0.50	8.12	0.04

ε : sum of residuals;

R^2 : coefficient of determination.

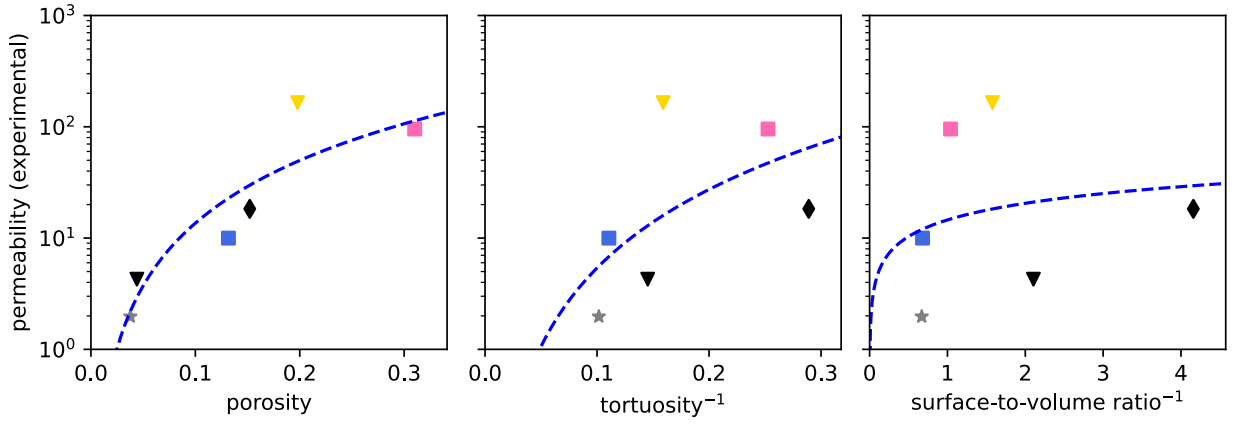


Figure 4.18: **Carbonates:** correlation between geometric parameters inferred from apparent diffusion curve and experimental absolute permeability.

Table 4.11: **Carbonates:** coefficients of the computational absolute permeability exponential curves fitted with geometric parameters inferred from RW simulations.

x	$k(x) = ax^b$			
	a	b	ϵ	R^2
ϕ	190.80	0.71	4.36	0.51
τ	658.83	-1.69	4.02	0.58
s/v	25.73	-0.51	4.81	0.15

ϵ : sum of residuals;

R^2 : coefficient of determination.

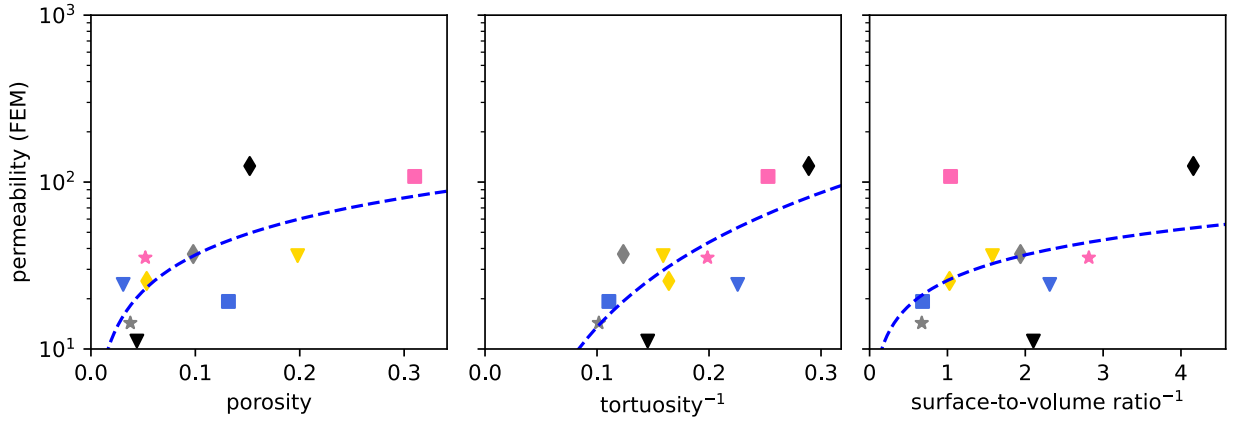


Figure 4.19: **Carbonates:** Correlation between geometric parameters inferred from apparent diffusion curve and absolute permeability computationally estimated from FEM fluid flow simulations.

As previously mentioned, results show that the absolute permeability of carbonates does not show a high correlation with the surface-to-volume ratio. However, exploring the relationship between absolute permeability and tortuosity seems to be a more promising approach. Unlike sandstones, carbonates exhibit a highly heterogeneous grain structure, leading to a complex pore network [62]. Therefore, investigating the behavior of apparent

diffusion in carbonates, especially for longer observation times, holds greater potential for estimating the permeability of these rocks.

4.3.3 Simulation vs. Experimental: results comparison

In this section, we compare the results obtained through simulation with those obtained by performing the NMR diffusion experiment in the laboratory. The diffusion experimental measurements were obtained using the 11-interval bipolar PFGSE sequence [77] and the 13-interval bipolar PFGSTE sequence [19]. For all the samples, 21 diffusion time values were used. Each acquisition was performed with 5 different values (ranging from 0.2 to 2.4 ms) for the duration of the pulsed gradients, thereby avoiding variations in ramp times and gradient pulse shapes. In the measurements, the intensity of the pulsed gradients was adjusted according to the observed diffusion times, ranging from 0.017 to 0.291 T/m. For the 13-interval sequence measurements, we used $\tau = 8$ ms. The apparent diffusion coefficient was determined following the methodology described in [74], and the data were fitted using the Padé approximant [14]. Measurements could not be performed on the Bandera Brown, Crab Orchard, and Torey Buff sandstones, as well as the Wisconsin carbonate, due to limitations in the signal-to-noise ratio. These limitations arose either from very short transverse relaxation times (Bandera Brown, Torey Buff), causing the signal to decay completely during the measurement time, or from very low porosities (Crab Orchard, Wisconsin).

As mentioned earlier, it is essential to incorporate the signal relaxation effect into the simulations for these comparisons [9]. To do so, we apply the boundary conditions presented in Section 3.1.1, using the surface relaxivity values contained in Table 4.4 and 4.5. Estimating the variation observed in the apparent diffusion curve due to the surface relaxation effect is not a trivial task. Factors such as the magnitude of surface relaxivity, as well as the size, shape, or connectivity of the pores in a sample, can significantly influence this effect. To illustrate, Figures 4.20 and 4.21 demonstrate the potential impact of this effect on the measured apparent diffusion on some of the samples of sandstones and carbonates, respectively.

Another notable effect is the introduction of noise to the obtained signal. This implies that estimating diffusion for very long times is no longer feasible. For instance, even if we had a sample where only bulk relaxation was active, this relaxation mechanism alone would render the signal unreliable for times exceeding around three seconds. After this point, the original signal would have decayed to a negligible fraction. Consequently,

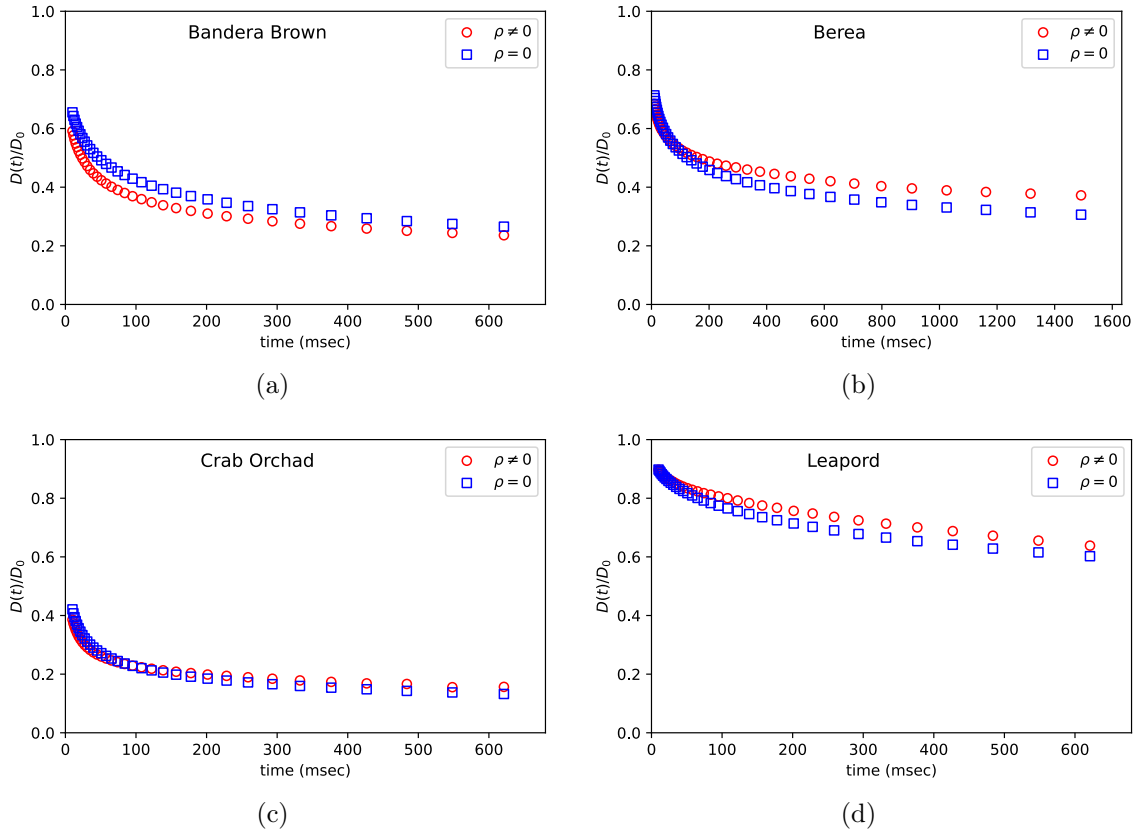


Figure 4.20: **Sandstones:** Examples of the influence of surface relaxation in diffusion simulations from sandstones. In each diagram, it is shown the apparent diffusivity from simulations with (red circles) and without (blue squares) the incorporation of surface relaxation effects.

estimating tortuosity in certain samples becomes unreliable since the signal acquisition range may not be sufficient for the diffusion curve to reach its characteristic plateau. Furthermore, in our comparisons between computational and experimental results, there is no need to extend the simulated signal beyond that obtained via the experiment. In addition, the introduction of relaxation and noise effects in the simulations typically introduces variability in the estimates of the apparent diffusion coefficient. This variability renders the estimates unreliable as the signal evolves.

4.3.3.1 Sandstones comparisons

The results for apparent diffusion in sandstones, obtained through computational and experimental measurements, are presented in Figure 4.22. The surface-to-volume ratio, S/V_P , and the tortuosity, τ , recovered for each sample using the Padé fit, Eq. (2.7), over the apparent diffusion coefficient measurements are presented in Table 4.13.

Overall, we observe good visual agreement between the data. This agreement also

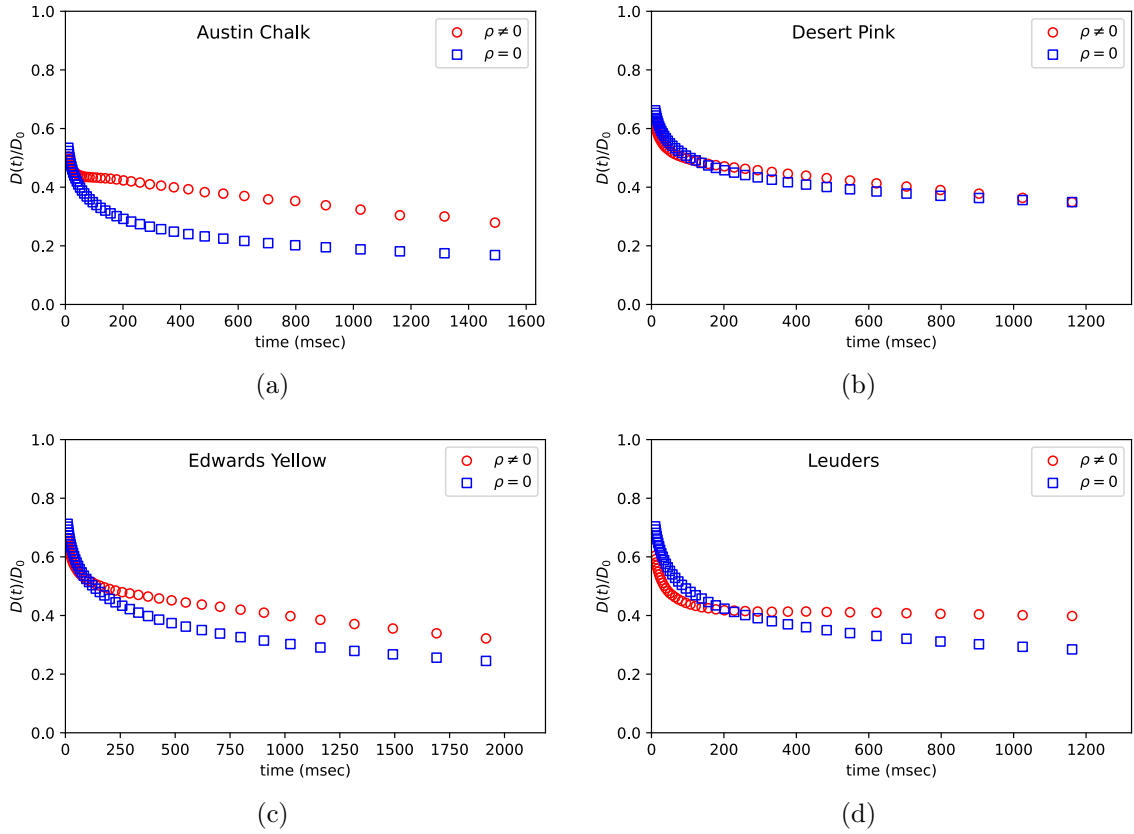


Figure 4.21: **Carbonates:** Examples of the influence of surface relaxation in diffusion simulations from carbonates. In each diagram, it is shown the apparent diffusivity from simulations with (red circles) and without (blue squares) the incorporation of surface relaxation effects.

highlights the successful execution of the segmentation protocol, supporting the idea that the chosen resolution adequately represents the porous space of the image for these samples.

We also highlight the results observed for the Bandera Gray sample that exhibit a very pronounced tortuosity. As shown in Figure 4.22, experimental measurements for this sample were limited to observation times of around 700 ms, the shortest among all the samples. For this reason, it is very unlikely that the apparent diffusion coefficient measurements reached anything close to the tortuosity limit. Therefore, the tortuosity parameter, fitted using the Padé approximant, becomes unreliable in this particular case.

4.3.3.2 Carbonates comparisons

The results for apparent diffusion in carbonates, obtained through computational and experimental measurements, are presented in Figures 4.18. The surface-to-volume ratio, S/V_P , and the tortuosity, τ , recovered for each sample using the Padé fit, Eq. (2.7), over

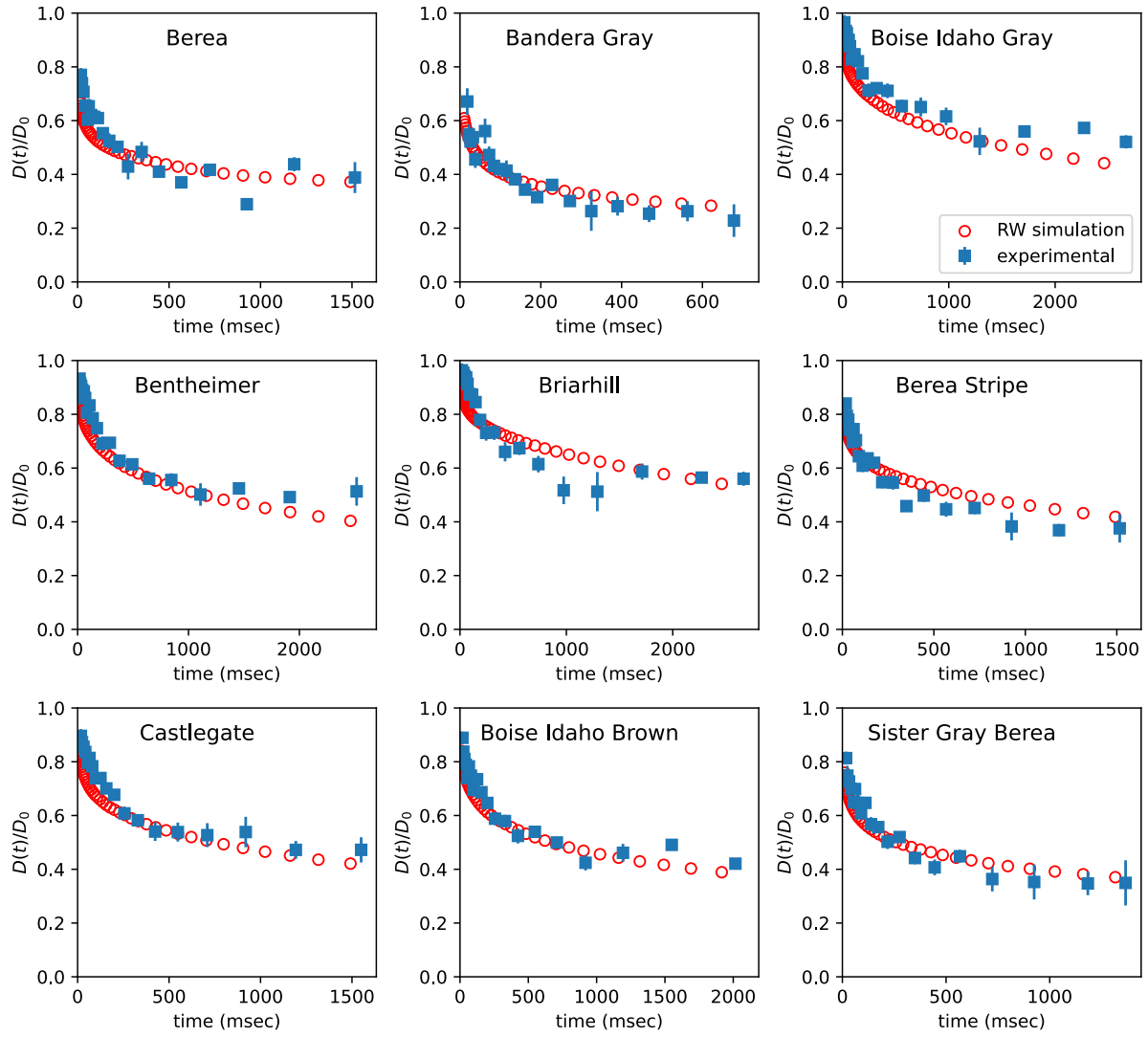


Figure 4.22: **Sandstones:** Comparison of experimental and computational results for the NMR apparent diffusion measurement for the sandstone samples.

Table 4.12: **Sandstones.** Recovered geometric parameters from the apparent diffusion coefficient experimental (exp) and computational (sim) measurements using Padé fit.

Sample	S/V_P		τ	
	exp	sim	exp	sim
Berea	0.29	0.77	3.20	2.99
Bandera Gray	0.72	1.04	17.98	5.16
Boise Idaho Gray	0.02	0.16	2.07	4.25
Bentheimer	0.02	0.17	2.28	5.36
Briarhill	-	0.13	2.11	2.36
Berea Stripe	0.21	0.37	3.73	2.87
Castlegate	0.07	0.27	2.44	3.50
Boise Idaho Brown	0.14	0.25	2.74	4.27
Sister Gray Berea	0.28	0.48	4.56	3.53

the apparent diffusion coefficient measurements are presented in Table 4.13. This time, however, we observe mixed results that require closer examination.

First, we observe good agreement for certain samples. Specifically, we highlight samples Desert Pink, Indiana M. and also Austin Chalk (at least, to some extent). For these samples, the apparent diffusion curves computed from simulations closely resemble their corresponding experimental results, both qualitatively and quantitatively. This agreement can be attributed to several factors. These particular samples may possess a more uniform porous structure, characterized by well-defined and open pores captured in the image. In addition, it may indicate that the segmentation process employed for these samples better represents their constitutive heterogeneity.

A second set of samples, however, exhibits a significant disparity between computational and experimental results. Specifically, in this second group, all samples display a distinctive recovery phase following the initial decay of the apparent diffusion coefficient. This behavior is not adequately described or predicted by current model and it is absent in the simulated results obtained from the available segmented images. For this reason, it was not possible to fit a surface-to-volume ratio for these samples, as exposed in Table 4.13.

Table 4.13: **Carbonates.** Recovered geometric parameters from the apparent diffusion coefficient experimental (exp) and computational (sim) measurements using Padé fit.

Sample	S/V_P		τ	
	exp	sim	exp	sim
Austin Chalk	0.49	2.54	5.24	2.91
Desert Pink	0.75	1.23	2.49	2.75
Edwards Yellow	-	0.86	1.72	2.92
Guelph Dolomite	-	0.48	2.15	3.86
Indiana L.	-	0.41	2.31	3.18
Indiana M.	1.15	2.14	4.11	4.25
Leuders	-	0.85	1.86	2.53
Sillurian Dolomite	-	0.28	2.12	2.21

The distinctive recovery phase is also observed in [61]. In this work, the authors use random walk simulations to study time-dependent apparent diffusion in digital rock alike porous media generated by stochastic methods. The authors show that this characteristic behavior is intimately related to high heterogeneity of the porous space.

A plausible explanation for this phenomenon could be the presence of vugular porosity that was not detected during the imaging acquisition process. Vugular porosity refers to a type of secondary porosity found in reservoir rocks, particularly in carbonates, that results

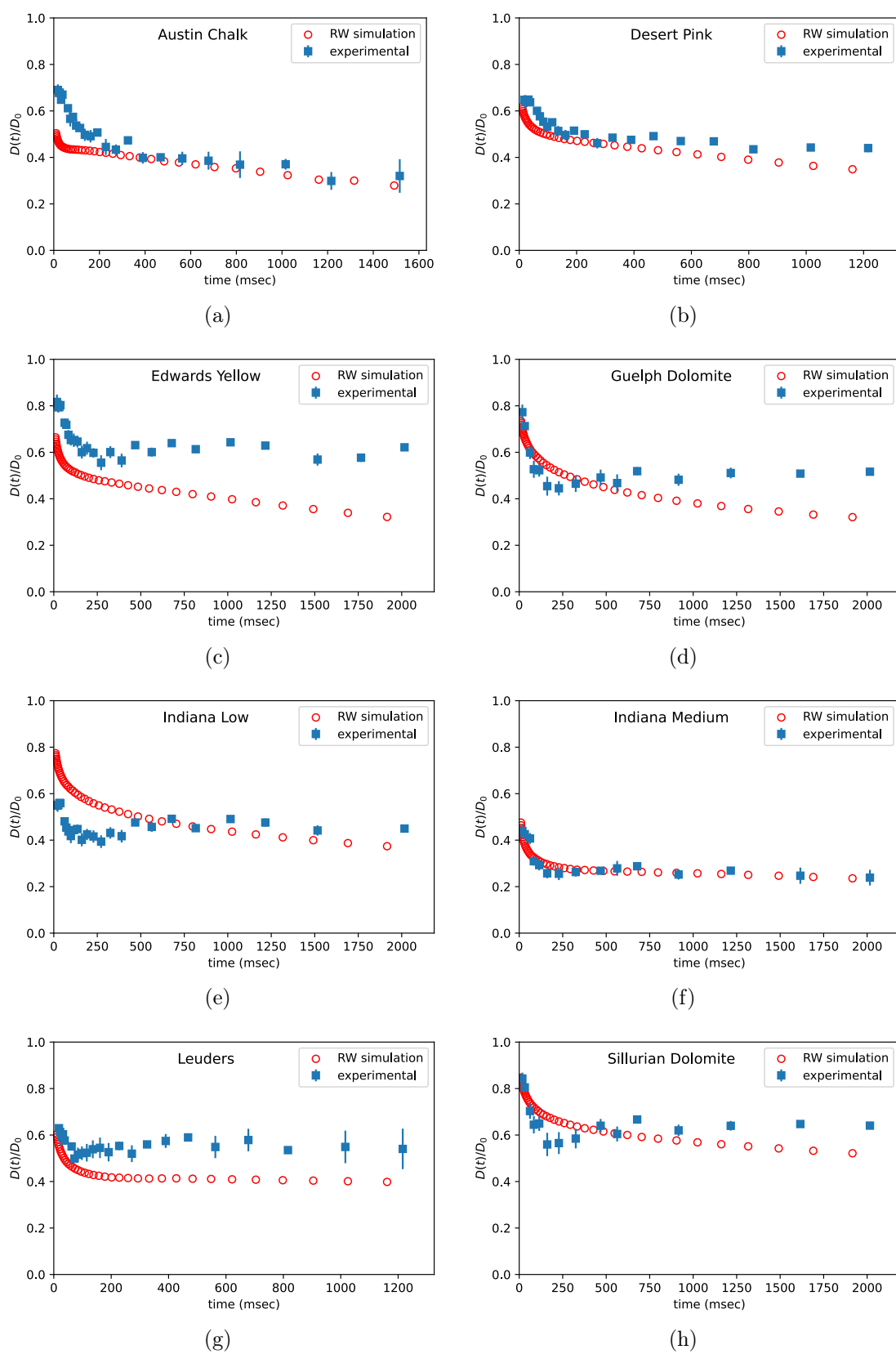


Figure 4.23: **Carbonates:** Comparison of experimental and computational results for the NMR apparent diffusion measurement for the carbonate samples.

from the dissolution of minerals, creating irregular, cavity-like voids known as vugs. Vugs form after the rock has already lithified, typically due to fluid movement that dissolves carbonate minerals. If well connected, vugular porosity can enhance permeability, improving fluid flow, and hydrocarbon recovery. If isolated, it contributes to storage capacity but does not aid in fluid flow, potentially creating reservoir heterogeneities [62, 53].

The presence of these vugs could explain the recovery phase found in the experimental results. In that sense, the fraction of magnetized spins present in the sample that are trapped in larger vugs would experience unrestricted diffusion, pushing the apparent diffusion coefficient to higher values. These measurements arise therefore from the "competition" of magnetized spins experiencing restricted and unrestricted diffusion [40].

Another plausible explanation could involve the influence of micro porosity on the results. These micro pores are expected to be so small that they cannot be captured in the images, despite the resolution employed being on the order of a few micrometers. Nevertheless, we consider their impact on the signal to be negligible, as the inherent surface relaxation effect would naturally filter out their contribution at the initial stage of the apparent diffusion curve, both in simulation and in the experiment.

4.3.3.3 On vugular porosity

Numerous studies in the literature explore the influence of vuggy pores in reservoir rock apparent diffusion measurements [50, 3, 21]. Considering the resolution of the image acquisition, it is highly probable that these pores exceed the depicted scale and therefore their direct acquisition in this scale is extremely challenging or even unfeasible. The high heterogeneity of these lithotypes often prevents finding the optimal REV size for carbonate samples. In this context, a dual-scale approach where images from the sample are acquired at different resolutions might be the best solution for depicting both scales in the analysis [58]. In addition, the presence of vugular porosity, characterized in most cases by pore sizes considerably larger than the scale of micrometers (the typical pore network of carbonate samples) would entail a substantial number of magnetized spins experiencing unrestricted diffusion or similar behaviors [40].

In order to test the hypothesis that the observed behavior in the experimental results may be explained by the presence of unrestricted diffusion experienced by magnetized spins trapped in vugs, we conducted a new simulation on the Edwards Yellow sample. To account for the effect of vugular pores, we added the recovered signal from the simulation on this sample to the recovered signal from the simulation on isolated spheres of different

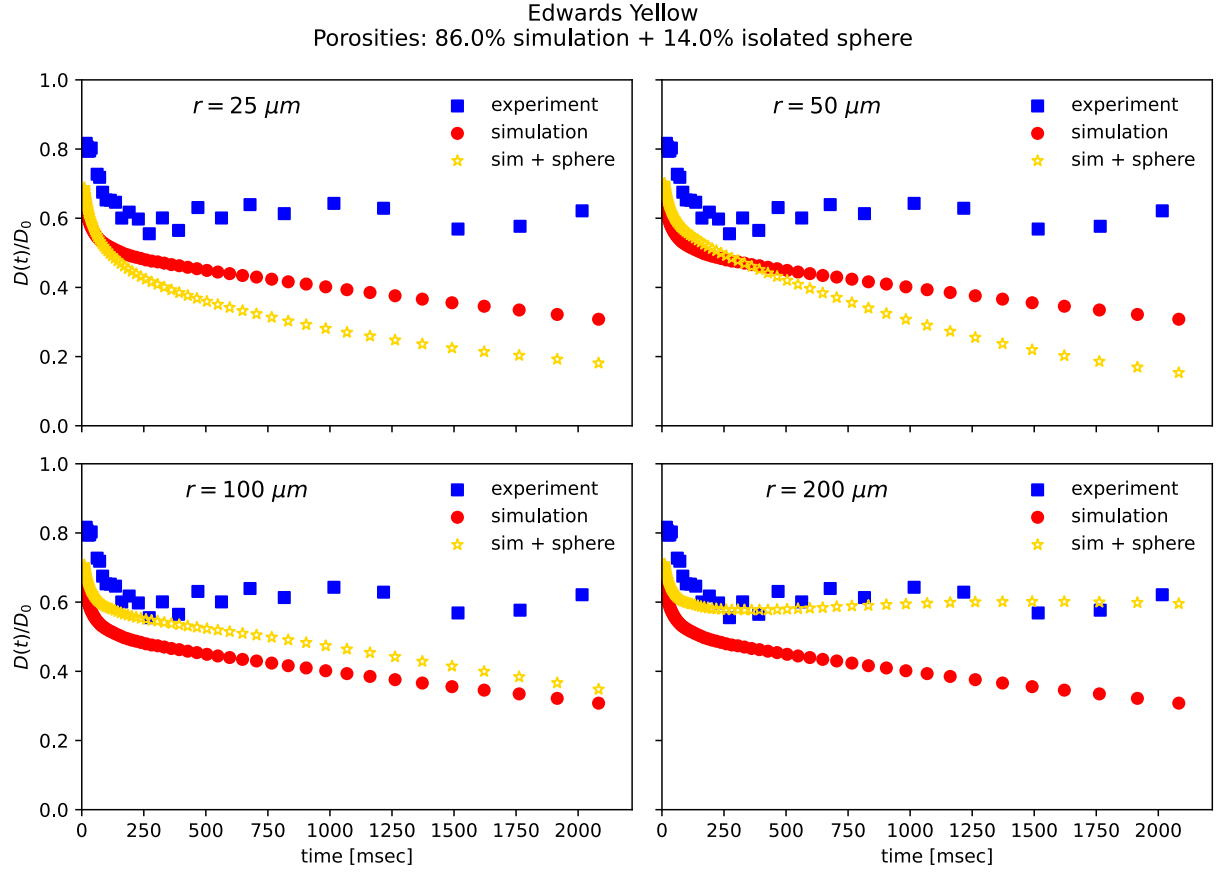


Figure 4.24: **Edwards Yellow:** vugular porosity recovery experiment

sizes. Furthermore, each signal is weighted according to its respective contribution to the overall porosity of the sample. This approach is feasible as the NMR signal used to recover the apparent diffusion coefficient is a sum of the independent contributions from each magnetized spin.

To estimate the proportion of vugular porosity, we assume that all the experimental porosity that cannot be explained by the segmentation corresponds to vugular porosity. It is important to note that this assumption is highly unlikely to be accurate. However, for the purpose of this experiment, we will proceed with this approach for its simplicity.

Thus, we have that:

$$\phi_{\text{vugular}} = \phi_{\text{experimental}} - \phi_{\text{image}}. \quad (4.3)$$

and

$$M(\mathbf{k}, t) = (\phi^* \times M(\mathbf{k}, t))_{\text{image}} + (\phi^* \times M(\mathbf{k}, t))_{\text{vugular}}, \quad (4.4)$$

where:

$$\phi_{\text{image}}^* + \phi_{\text{vugular}}^* = 1, \quad (4.5)$$

Another possible approach to estimate the proportion of vugular porosity in a sample is through deconvolution of the T_2 times distribution. In this approach, the component in the region of 2 seconds can be related to unrestricted diffusion and therefore vugular porosity.

The results for this experiment are presented in Figure 4.24. For the largest tested vugular pore (a 200-micrometer sphere), the simulated results exhibit a good fit and a behavior similar to that observed experimentally for the same sample. Notably, we observe the same recovery pattern of the apparent diffusion coefficient after the initial decay. This strongly supports the hypothesis that the presence of vugular pores in the sample introduces unexpected behavior, which is correlated with the unrestricted diffusion of a significant proportion of the magnetized spins. It also supports the idea that the resolution adopted for the carbonate samples that exhibit the deviant behavior is not capable of representing their pore space heterogeneity.

Chapter 5

Simulation of NMR diffusion relaxation

In this chapter, we present the results of simulations of NMR transverse relaxation times, T_2 , that incorporate the diffusion relaxation mechanism. As mentioned in Section 2.3, this particular mechanism arises from the interaction between magnetized spins and spatial variations in the magnetic field within the porous medium, when echoes times are not short enough. These variations occur due to the high contrast in magnetic susceptibility between the saturating fluid and the materials comprising the rock matrix [40, 17]. This effect is particularly pronounced in sandstones, which are typically rich in materials that can induce high magnetic susceptibility contrasts [27].

In the context of NMR, the Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence is the most commonly used experiment for measuring transverse relaxation. From the resulting decay of this sequence, it is possible to recover a relaxation time distribution and, after appropriate treatment, this resulting distribution can also be interpreted as pore size distributions [6]. However, signal attenuation caused by the diffusion relaxation mechanism can compromise the usual interpretation, particularly in samples with high internal magnetic field gradients. To mitigate this effect, the most common approach is using low external magnetic fields and reducing the echoes spacing, allowing less time for magnetized spins to experience magnetic field fluctuations between signal acquisitions. However, even with this methodology, it can still lead to misinterpretation of T_2 data [40]. Yet, this attenuation can provide insights into the rock mineral composition. For example, the identification of internal magnetic field gradients of higher magnitudes in a reservoir rock can indicate a high concentration of clay and other ferromagnetic materials in its composition [27, 17]. In this context, as described in section 2.3.2, the multi- τ technique, which involves repeating the CPMG sequence with a well-structured package of different

echo spacings, allows the estimation of internal gradients in a given sample. With appropriate numerical treatment, it is possible to obtain even more valuable information, such as estimating a quantitative distribution of the magnitude of internal gradients in the pore space [40].

In this regard, we begin by validating the methodology presented in Section 3.3 through a comparison of the results obtained from our simulations with analytical predictions found in the literature. We evaluate both the estimated magnetic field and the attenuation associated with the diffusion relaxation mechanism. Subsequently, we present the results of simulations using the multi- τ procedure on segmented micro-tomographic images. In this context, we utilize segmented images where each voxel of the solid phase is classified based on the constituent material. We showcase the results from simulations conducted on sintered media as well as the results from simulations on selected reservoir rock samples of particular interest. Furthermore, we include the experimental results obtained for both cases. We then discuss these results, with a particular focus on the magnetic field internal gradient distributions. Finally, we propose and investigate an alternative methodology that considers the effect of restricted diffusion in these measurements.

5.1 Validation

As described in Section 3.3, incorporating the diffusion relaxation mechanism into the simulations requires a two-step workflow:

- i) Computing the residual magnetic field resulting from the contrast between the magnetic susceptibility of the constituent minerals in the grains of the solid phase and the fluid inside the pore space;

- ii) Adding a routine into the T_2 relaxation simulation code that considers the magnetic field, computed in the previous step, and applies relaxation to magnetized particles as they travel through regions with fluctuations in the magnitude of this field.

Thus, each step requires a specific validation detailed as follows. Also, we validate the modeling of surface relaxation with analytical predictions found in the literature.

5.1.1 NMR T_2 surface relaxation

Validation of the implementation of surface relaxation in RW NMR T_2 simulation is made by comparing the results with analytical prediction of simple spherical pores. In this

sense, according to the theory, for a spherical pore of radius, r , and surface relaxivity, ρ_2 , if we isolate the surface relaxation phenomena, the magnetization follows a simple exponential decay as

$$M(t) = M_0 e^{-\frac{t}{T_2}}, \quad (5.1)$$

where, in the fast diffusion regime [12], the decay rate due to surface relaxation, $T_{2,\text{surface}}$, is given by:

$$\frac{1}{T_{2,\text{surface}}} = \rho_2 \frac{S}{V} = \rho_2 \frac{3}{r}. \quad (5.2)$$

Figure 5.1 shows the comparison between the analytical prediction and simulation results for the surface relaxation decay in distinct scenarios. Note that the signal decays presented are in log scale for the time dimension. In all the portrayed scenarios, with varying spherical pore radius and surface relaxivities, both decays are in perfect agreement. Thus, our simulation is correctly capturing the surface phenomena, i.e., the interactions that arise from walker collisions with the pore walls in the images.

We also show Figures 5.2 and 5.3 for a better view of the isolated effect of varying either the pore size or the surface relaxivity in the RW simulations, respectively. In Figure

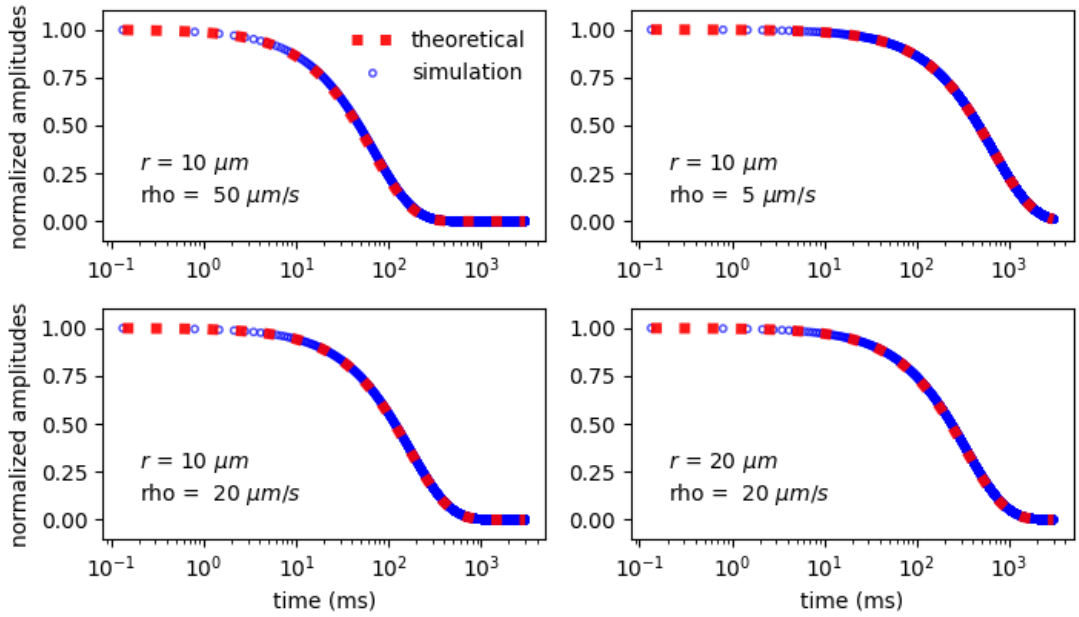


Figure 5.1: **NMR T_2 - surface relaxation**: comparison between simulation results (blue circles) and analytical prediction (red squares) for an isolated spherical pore. Each diagram represents a different scenario for the pore radius, r , and the surface relaxivity, ρ_2 .

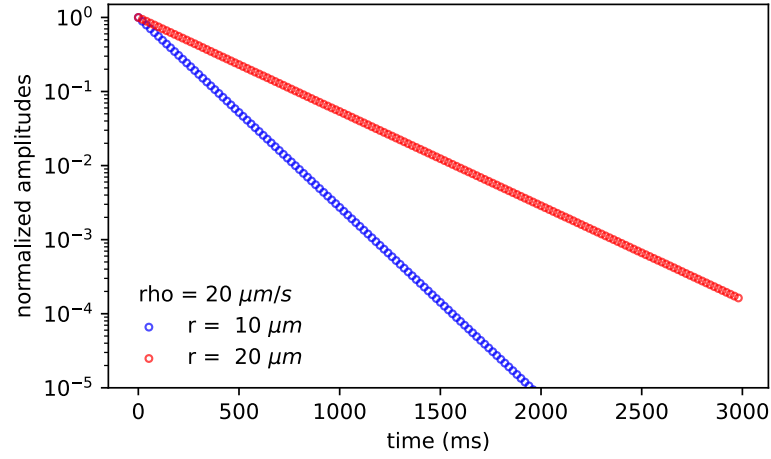


Figure 5.2: **NMR T_2 - surface relaxation**: comparison between simulation results for isolated spherical pores with varying radius, r , for a fixed surface relaxivity, ρ_2 , of $20 \mu\text{m/s}$. Blue dots correspond to the results for a sphere of radius $10 \mu\text{m}$, while red dots correspond to the results for a sphere of radius $20 \mu\text{m}$.

5.2, we show that by fixing the surface relaxivity at $20 \mu\text{m/s}$, and choosing a smaller spherical pore of radius $10 \mu\text{m}$ in comparison with a pore of radius $20 \mu\text{m}$, the resulting magnetization signal presents a significant shorter decay as the curve is considerably shifted towards shorter times. The same analysis is applied when we choose to fix the pore size at $10 \mu\text{m}$ while applying varying relaxivity factors. As expected, as stronger relaxation forces are applied when walkers interact with the pore walls, we observe that the curve reaches shorter decay times. Notice that, this time, for a better comparison, the y-axis is shown in log-scale so that the resulting decays are simple straight lines.

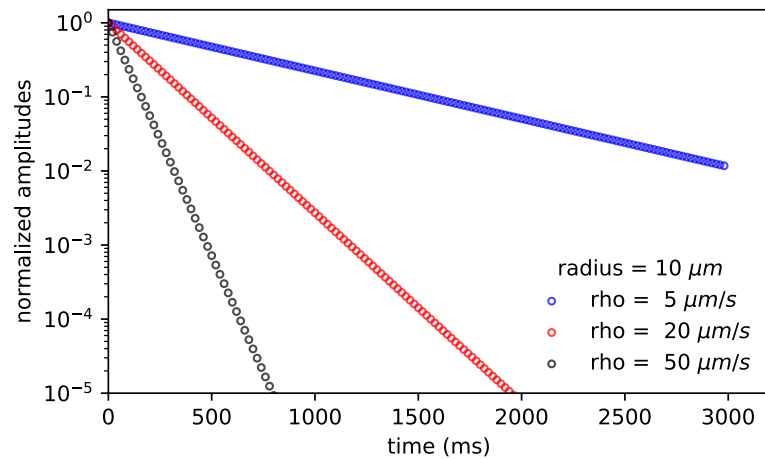


Figure 5.3: **NMR T_2 - surface relaxation**: comparison between simulation results for isolated spherical pores with a fixed radius, r , of $10 \mu\text{m}$ and varying surface relaxivity, ρ_2 , are applied. Blue dots correspond to the results when a $5 \mu\text{m/s}$ surface relaxivity is applied. Red and black dots, respectively, correspond to the results when surface relaxivities of 20 and $50 \mu\text{m/s}$ are applied.

5.1.2 Incorporation of diffusion relaxation mechanism in NMR T_2 simulations

In this section, we validate the implementation of the routine that incorporates diffusion relaxation into the computational code for RW NMR T_2 relaxation simulations. Our approach involves comparing the simulated results with the analytical prediction found in the literature. As described in [40], the predicted magnetization for an infinite unrestricted medium subjected to a constant magnetic field gradient, $M(t)$, is given by:

$$\frac{M(t)}{M_{t=0}} = e^{-\frac{1}{12} D_0 \gamma^2 g^2 t_E^2 t}, \quad (5.3)$$

where: $M_{t=0}$ is the initial magnetization signal, D_0 is the bulk self-diffusion coefficient of the fluid, γ is the gyromagnetic ratio of the 1H nucleus, g is the effective magnetic field internal gradient, and t_E is the adopted echo spacing between signal acquisitions.

Thus, by simulating the initial signal acquisition in an unrestricted infinite medium with a constant internal gradient, while also varying the inter-echo spacing, we obtained the results shown in Figure 5.4. Notably, a good agreement between the simulated data (blue circles) and the analytical prediction (dotted lines) is observed for all the simulated scenarios.

5.1.3 Static magnetic field calculation

In order to validate our methodology, we will refer to the work [23], which calculates the static magnetic field for an infinite periodic medium of spheres packed in a cubic array. This array is illustrated in Figure 5.5. In this context, we compare the calculated field, obtained by solving Maxwell's equations under the zero current condition using our Image-based Finite Element implementation, with the field predicted by the analytical-numerical method presented in the referenced work.

Figure 5.6 illustrates the internal gradients distribution for the arrangement depicted in Figure 5.5. Consistent with [23], the conditions assumed were that the spherical grains have a radius of 100 micrometers. The system is under an externally applied field of $H_0 = 239.4$ Gauss (i.e., a proton Larmor frequency of 1 MHz) and the magnetic susceptibility of the solid phase and the porous phase (fluid) correspond to 50×10^{-6} emu/cm³ and -0.7×10^{-6} emu/cm³, respectively. Notice that here, for easier comparison, we use the same measurement units used in the reference work.

The gradients computed using the finite element method exhibit a reasonable fit

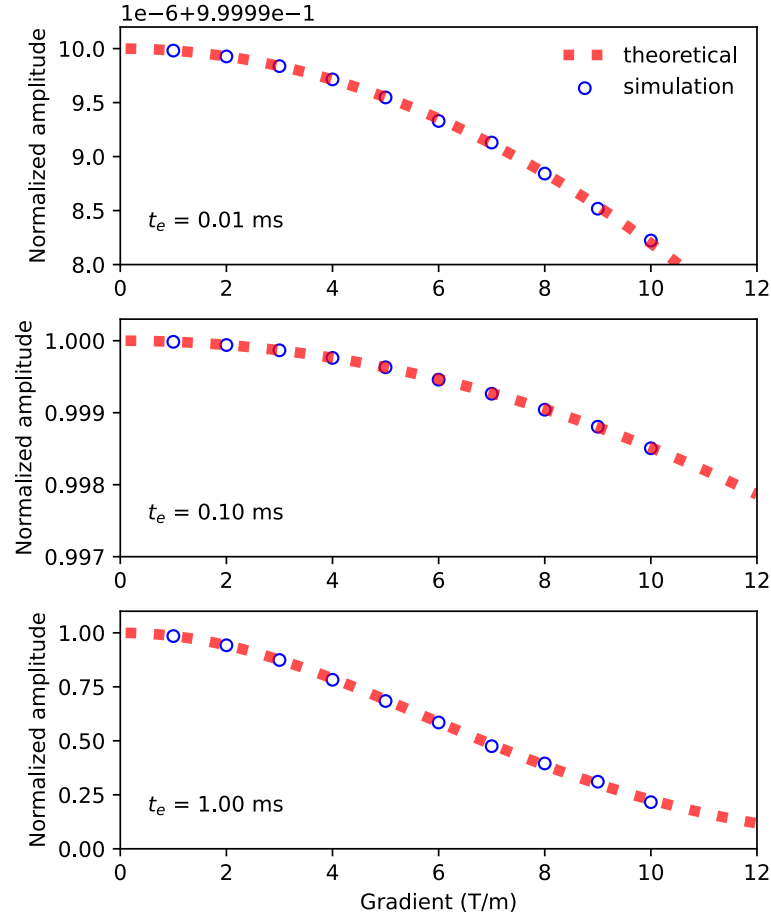


Figure 5.4: Computed results for the normalized amplitude of the magnetization signal of the first echo for spins experiencing unrestricted diffusion under uniform magnetic field gradients in the z-direction. The results are shown in varying scenarios of echo spacing (0.01, 0.10 and 1.00 ms) and gradient magnitudes (1 to 10 T/m). Also shown in dotted blue lines is the analytical prediction computed from Eq. 5.3.

when compared to the analytical-numerical results presented in [23]. Notably, both sets of results indicate a significant concentration of values within the same region of around 200 Gauss/cm. However, we observe an underestimation of higher magnitude gradients in the results from FEM simulation. This limitation may be related to and constrained by the resolution of the image as it will be seen in later results.

5.2 Simulations from segmented μ -CT images

In this section, we present the results of simulations performed on micro-computed tomographic images, where the solid phase is segmented based on its geochemical composition. Each segmented mineral has a magnetic susceptibility, which is used as an input to calculate the magnetic field within the porous medium. The values for the magnetic

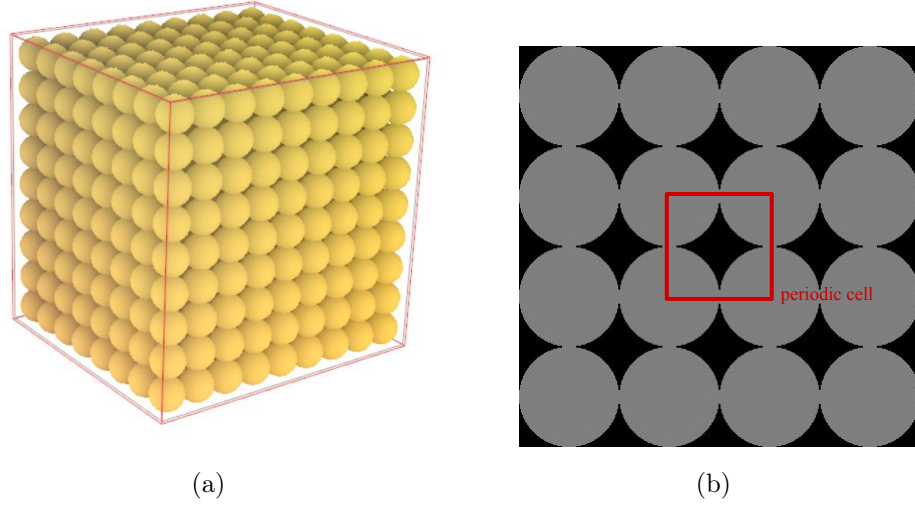
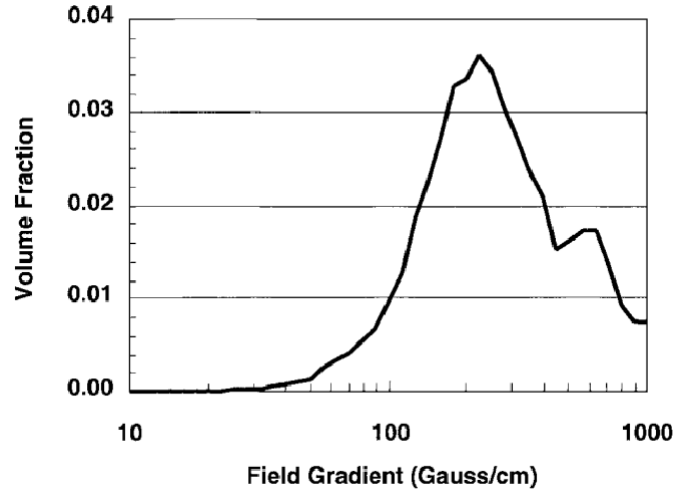


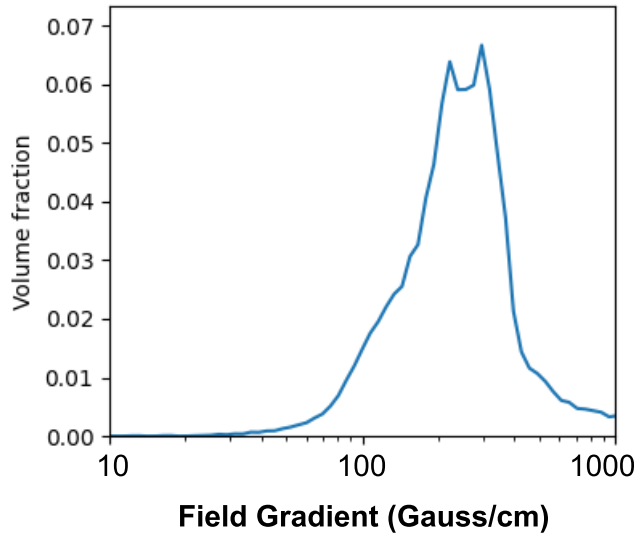
Figure 5.5: Illustration of the periodic array of touching spheres. (a) three-dimensional model of a small portion of the infinite porous medium; (b) central slice (two-dimensional representation) of the infinite medium with the periodic cell highlighted by the red solid lines.

susceptibilities were obtained from [39, 63]. It is noteworthy that the calculated magnetic field is a spatial map, i.e., each point on the mesh has a unique value. This image-based method allows us to assign a distinct field value to each voxel. Once this map is computed, it serves as an input for the NMR T_2 simulations.

Following the multi- τ procedure, a series of T_2 decay simulations are conducted varying the echo spacing between signal acquisitions, t_E . A longer interval leads to a stronger diffusion relaxation effect. As outlined in 3.3.4, the multi- τ curve is constructed by collecting multiple acquisitions for each t_E value and normalizing the signal using the acquisition with the shortest t_E . The resulting signal is then subjected to Tikhonov regularization processing, generating a quantitative distribution of magnitudes for the internal gradient of the sample. The regularizer is chosen based on the L-curve criterion which simultaneously tries to minimize both the residual and the solution's norm [36]. To compare and evaluate the methodology, the described procedure is also applied to experimental data from CPMG sequences performed in the laboratory. Experimental results are obtained from CPMG decays at 0.05 T (corresponding to a *r.f.* of 2.2 MHz for 1H) for 15 distinct echo intervals: 100, 132, 175, 231, 395, 404, 535, 707, 935, 1236, 1635, 2162, 2859, 3781 and 5000 ms. Finally, we present a discussion on the obtained results.



(a)



(b)

Figure 5.6: Comparison of the magnetic field internal gradient distribution in a periodic array of touching spheres (where $r = 100 \mu m$). An externally applied magnetic field of $H_0 = 239.4$ Gauss and the magnetic susceptibility of the solid phase and the porous phase (fluid) correspond to 50×10^{-6} emu/cm³ and -0.7×10^{-6} emu/cm³, respectively. (a) analytic-numerical prediction from [23]; (b) Our FEM prediction under the same circumstances.

5.2.1 Sintered models

Table 5.1 provides the mineral composition and magnetic properties of the constituent materials in a sintered porous sample. The sample is composed of glass beads with iron oxide intrusions, and its image was acquired at a resolution of 20 micrometers per voxel.

Figure 5.7 shows a representative slice of the sintered sample used in this study, along

Table 5.1: **Sintered models:** phase segmentation.

material	%	$\chi (\times 10^6)$
pores	44.2	-9.2
glass	55.7	-16.4
iron oxide	0.1	20200

with the corresponding magnitude of the calculated magnetic field within the porous space. Figure 5.8 shows the multi- τ curve extracted from experimental measurements and RW simulations. Figure 5.9 illustrates the internal gradients distributions from three perspectives: i) internal gradients calculated based on the magnetic field obtained through FEM (solid black line); ii) internal gradients computed using the multi- τ methodology and RW simulation, where the FEM-computed magnetic field was used as input (dashed red line); and iii) internal gradients from experimental measurements using the multi- τ procedure (dotted blue line).

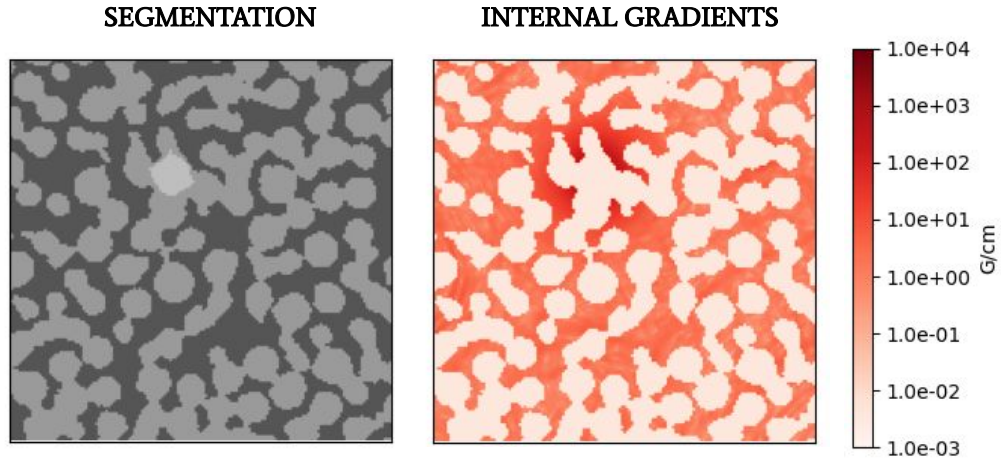


Figure 5.7: **Sintered model:** left image: example slice of image segmentation containing glass (gray), pores (darker gray) and iron (lighter gray); right image: internal magnetic field gradient color map. In this color map, darker regions represent higher gradient magnitudes.

5.2.2 Sandstone Berea Stripe

Table 5.2 and 5.3 present the mineral composition and image phase segmentation of the Berea Stripe sandstone, respectively. We also show the magnetic properties of the constituent materials of the sample. The image of the sandstone was acquired at a resolution of 2.0 micrometers per voxel. Once again, we provide both the composition obtained from the mineralogy study of the sample and the composition of the phases in the final

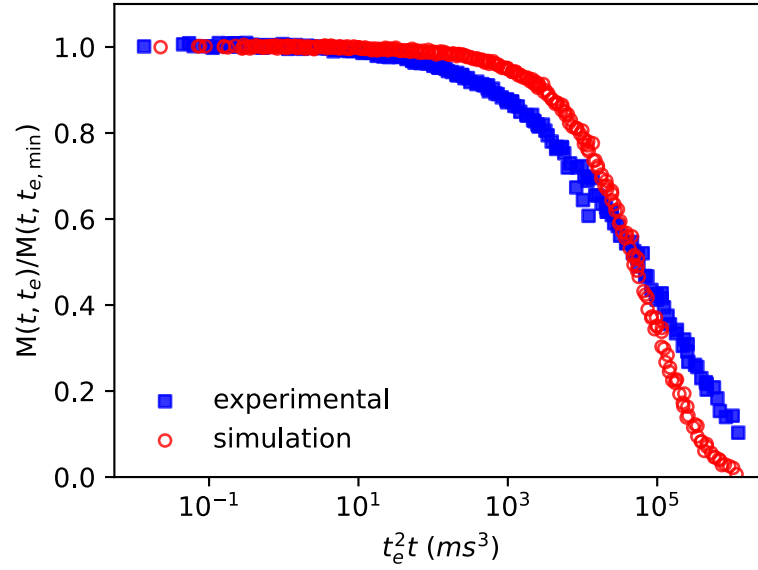


Figure 5.8: **Sintered model**: NMR Multi- τ decay data from experimental measurements (blue) and RW simulations (red).

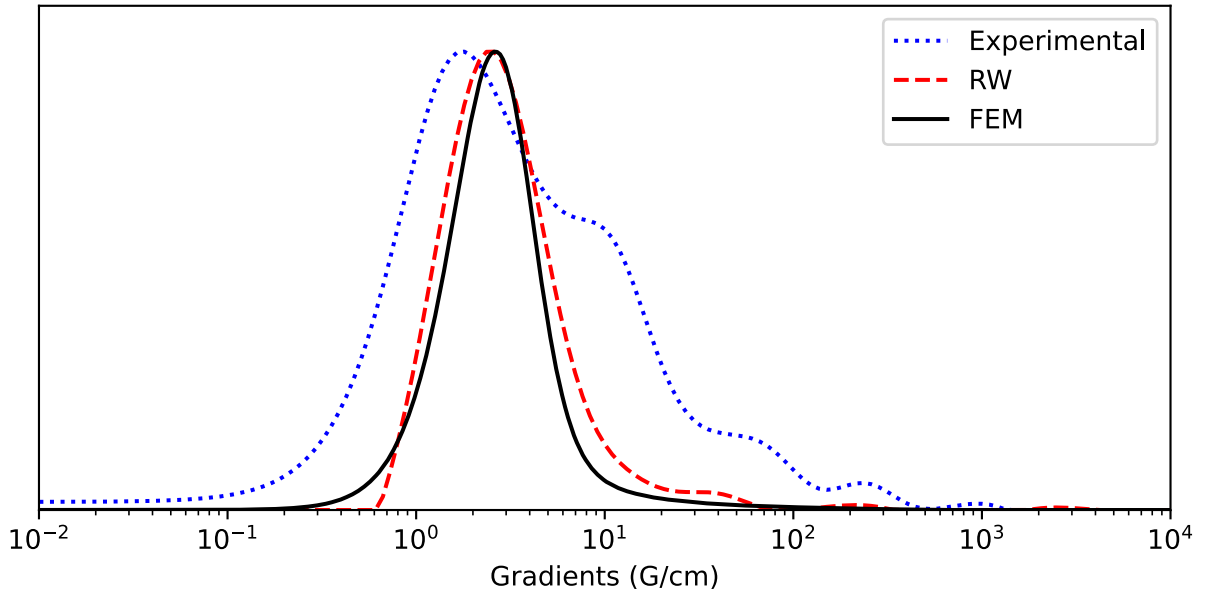


Figure 5.9: **Sintered model**: Internal magnetic field gradient distribution estimated from three different approaches: experimental NMR Multi- τ (blue dotted lines), simulated NMR Multi- τ (dashed red lines), and Finite Elements Method (solid black curve).

image segmentation. The predominant components of this sample are Quartz, with some presence of Kaolinite, Illite, Feldspar, and Ankerite. It is worth noting the inability to distinguish Quartz and Feldspar in the segmentations, and therefore, this phase was assumed to be Quartz, with its magnetic susceptibility used in the calculations. Furthermore, it is important to highlight the lower representation of Illite in the acquired image, a clay mineral with high magnetic susceptibility.

Table 5.2: **Berea Stripe**: x-ray diffraction mineralogy.

material	%	$\chi (\times 10^6)$
pores	21.2	-9.2
quartz	72.1	-16.4
kaolinite	2.7	-49.4
ankerite	2.2	145.0
K-feldspar	0.9	-14.9
illite	0.9	410.0

Table 5.3: **Berea Stripe**: phase segmentation.

material	%	$\chi (\times 10^6)$
pores	19.4	-9.2
quartz + feldsp.	72.7	-16.4
ankerite	4.4	145.0
kaolinite	4.1	-49.4
illite	0.1	410.0

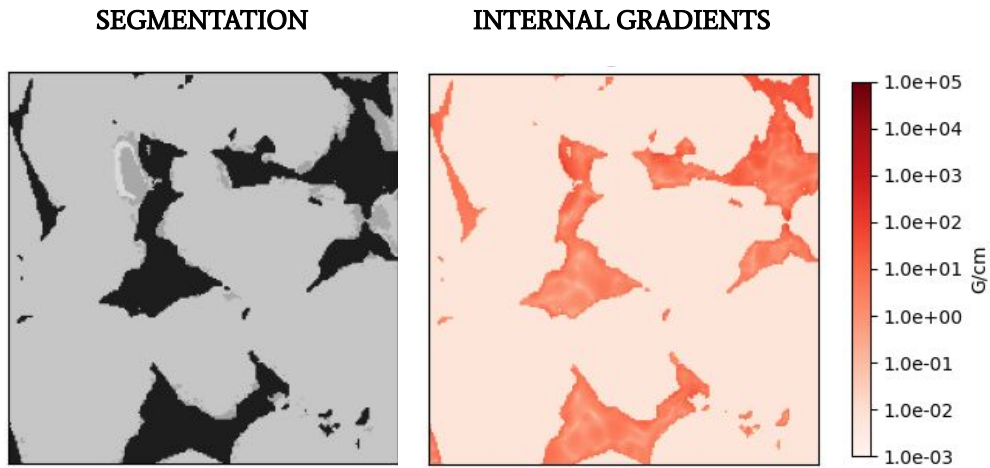


Figure 5.10: **Berea Stripe**: left image: example slice of image segmentation containing pores (black), quartz/feldspar (gray), kaolinite (darker gray) and ankerite (lighter gray); right image: internal magnetic field gradient color map. In this color map, darker regions represent higher gradient magnitudes.

Figure 5.10 illustrates a representative slice of the Berea Stripe sample along with the corresponding magnitude of the calculated magnetic field within the porous space. Figure 5.11 shows the multi- τ curve extracted from experimental measurements and RW simulations. Figure 5.12 showcases the distribution of internal gradients from three perspectives.

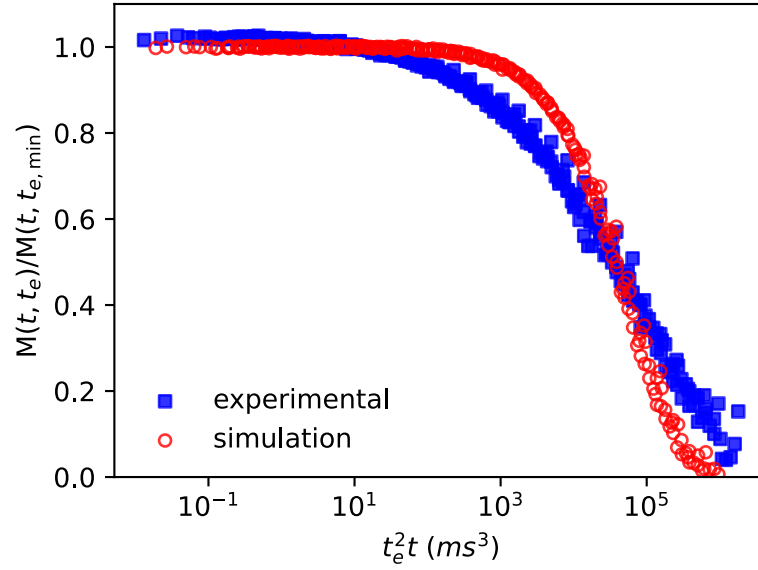


Figure 5.11: **Berea Stripe**: NMR Multi- τ decay data from experimental measurements (blue) and RW simulations (red).

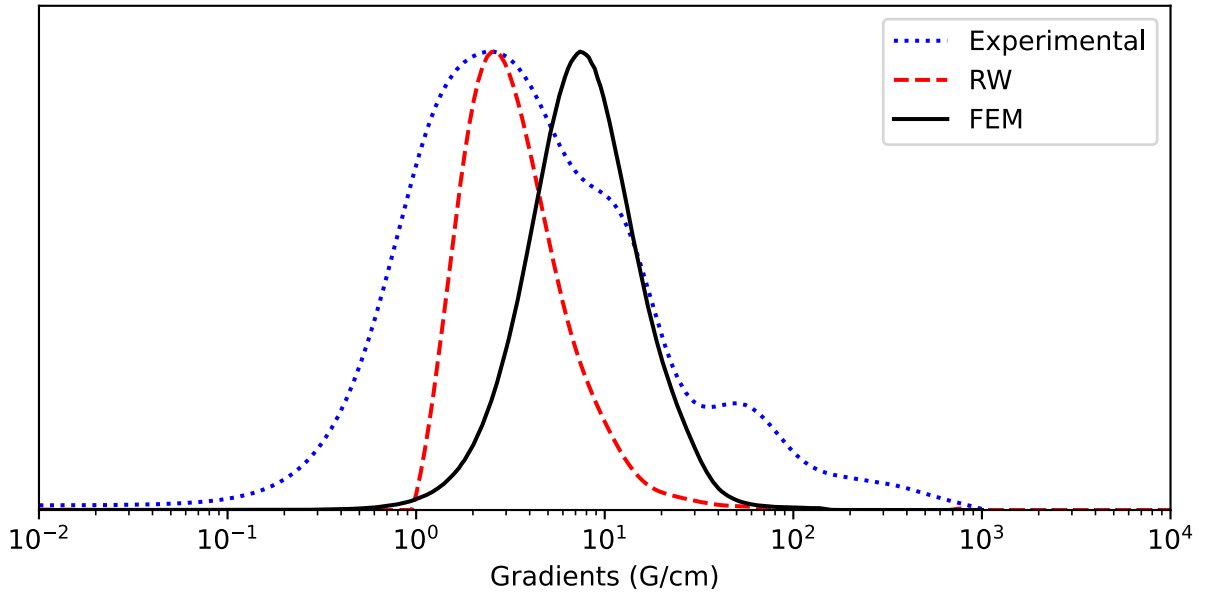


Figure 5.12: **Berea Stripe**: Internal magnetic field gradient distribution estimated from three different approaches: experimental NMR multi- τ (blue dotted lines), simulated NMR multi- τ (dashed red lines), and finite elements method (solid black curve).

5.2.3 Sandstone Boise Idaho Brown

Table 5.4 and 5.5 present the mineral composition and image phase segmentation of the Boise Idaho Brown sandstone, respectively. We also show the magnetic properties of the constituent materials of the sample. The image of the sandstone was acquired at a resolution of 2.0 micrometers per voxel. Once again, we provide both the composition obtained from the mineralogy study of the sample and the composition of the phases in

the final image segmentation. The predominant components of this sample are Quartz and Feldspar, with smaller proportions of Mica/Illite and Zeolite. It is worth noting that it was not possible to distinguish Quartz, Feldspar, and Zeolite in the segmentation of this sample. As in previous cases, this phase was assumed to be Quartz, with its magnetic susceptibility used in the calculations.

Table 5.4: **Boise Idaho Brown:** x-ray diffraction mineralogy.

material	%	$\chi (\times 10^6)$
pores	27.9	-9.2
quartz	28.0	-16.4
K-feldspar	36.0	-14.9
mica/illite	2.2	410.0

Table 5.5: **Boise Idaho Brown:** phase segmentation.

material	%	$\chi (\times 10^6)$
pores	20.9	-9.2
quartz + feldsp.	76.8	-16.4
mica/illite	2.2	410.0

Figure 5.13 illustrates a representative slice of the Boise Idaho Brown sample along with the corresponding magnitude of the calculated magnetic field within the porous space. Figure 5.14 shows the multi- τ curve extracted from experimental measurements and RW simulations. Figure 5.15 showcases the distribution of internal gradients from three perspectives.

5.2.4 Sandstone Bandera Gray

Table 5.6 and 5.7 present the mineral composition and image phase segmentation of the Bandera Gray sandstone, respectively. We also show the magnetic properties of the constituent materials of the sample. The image of the sandstone was acquired at a resolution of 2.0 micrometers per voxel. Once again, we provide both the composition obtained from the mineralogy study of the sample and the composition of the phases in the final image segmentation. The majority of this sample is composed of Quartz, with smaller proportions of Feldspar, Ankerite, Illite, Chlorite, and small amounts of Calcite and Kaolinite. It is worth noting that distinguishing Quartz, Feldspar, and Calcite in the segmentations was not feasible. As in previous cases, this phase was assumed to be Quartz, using its magnetic susceptibility in the calculations. Furthermore, there was an observed underrepresentation of clay minerals such as Illite and Chlorite, which have

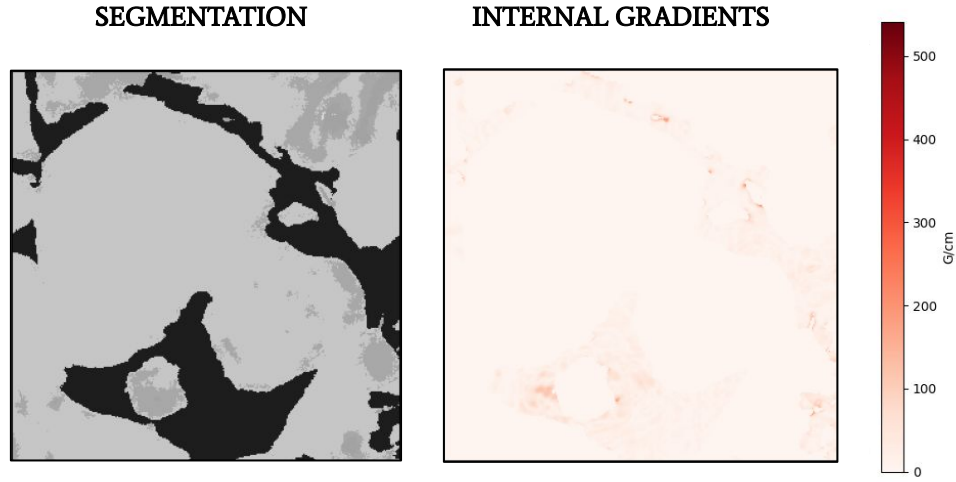


Figure 5.13: **Boise Idaho Brown**: left image: example slice of image segmentation containing pores (black), quartz/feldspar (gray), mica/illite (darker gray); right image: internal magnetic field gradient color map. In this color map, darker regions represent higher gradient magnitudes.

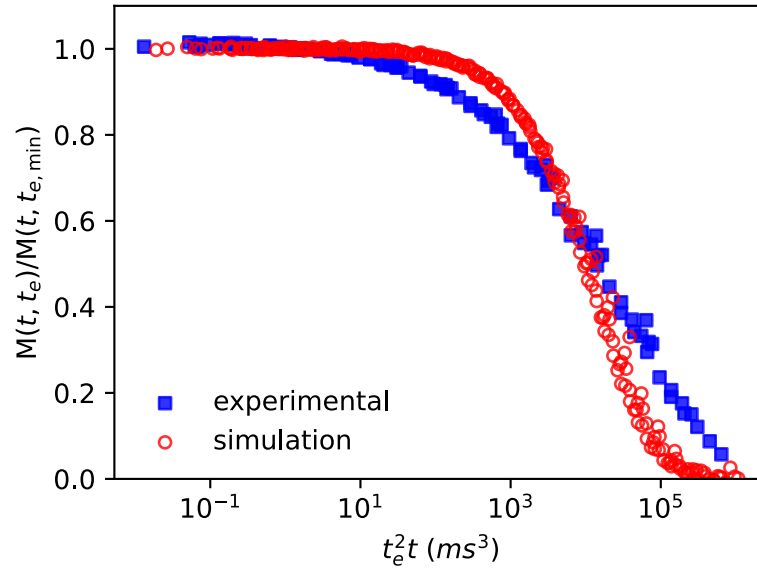


Figure 5.14: **Boise Idaho Brown**: NMR Multi- τ decay data from experimental measurements (blue) and RW simulations (red).

higher magnetic susceptibility and consequently generate stronger internal gradients in their proximity.

Figure 5.16 shows a representative slice of the Bandera Gray sample, along with the corresponding magnitude of the calculated magnetic field within the porous space. Figure 5.17 shows the multi- τ curve extracted from experimental measurements and RW simulations. Figure 5.18 illustrates the distribution of internal gradients from three different

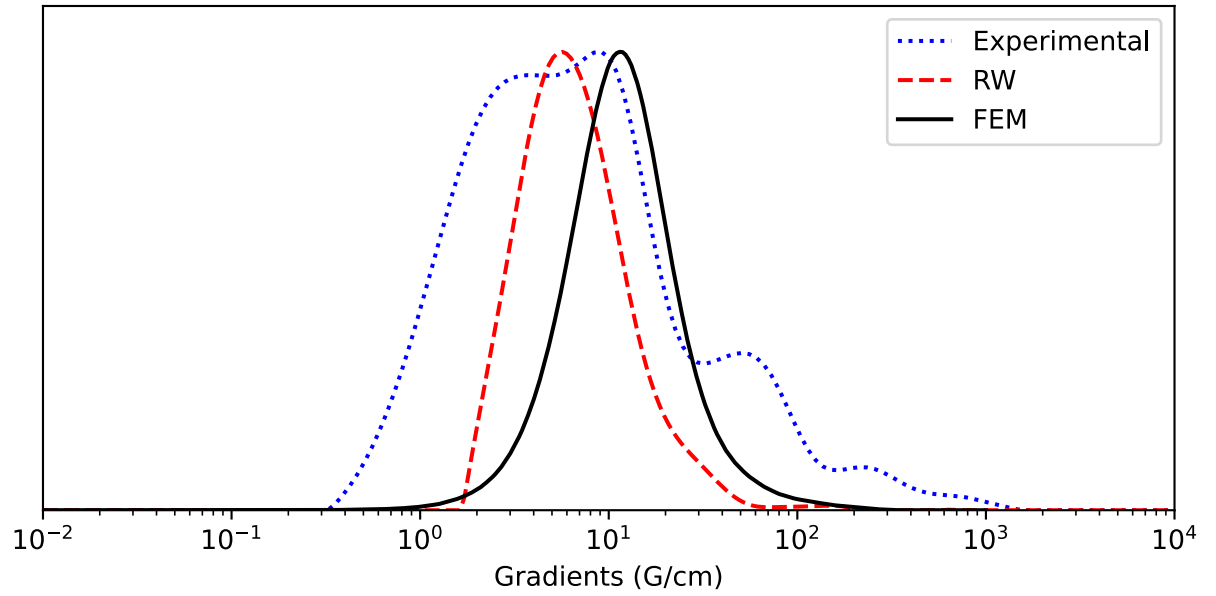


Figure 5.15: **Boise Idaho Brown:** Internal magnetic field gradient distribution estimated from three different approaches: experimental NMR multi- τ (blue dotted lines), simulated NMR multi- τ (dashed red lines), and finite elements method (solid black curve).

Table 5.6: **Bandera Gray:** x-ray diffraction mineralogy.

material	%	χ ($\times 10^6$)
pores	21.1	-9.2
quartz	48.8	-16.4
K-feldspar	11.3	-14.9
ankerite	6.4	145.0
mica/illite	5.0	410.0
chlorite	4.1	490.0
kaolinite	1.8	-49.4
calcite	1.0	-13.8

Table 5.7: **Bandera Gray:** phase segmentation.

material	%	χ ($\times 10^6$)
pores	12.8	-9.2
quartz + feldsp. + calcite	82.1	-16.4
ankerite	5.0	145.0
mica/illite	< 0.1	410.0

perspectives.

5.2.5 Sandstone Kirby

Table 5.8 and 5.9 present the mineral composition and image phase segmentation of the Kirby sandstone, respectively. We also show the magnetic properties of the constituent

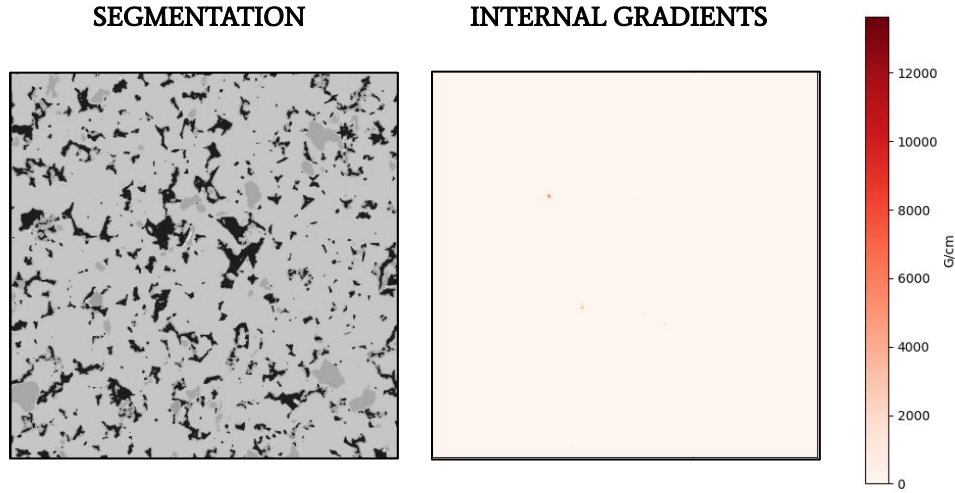


Figure 5.16: **Bandera Gray**: left image: example slice of image segmentation containing pores (black), quartz/feldspar/calcite (gray), ankerite (darker gray) and mica/illite (lighter gray); right image: internal magnetic field gradient color map. In this color map, darker regions represent higher gradient magnitudes.

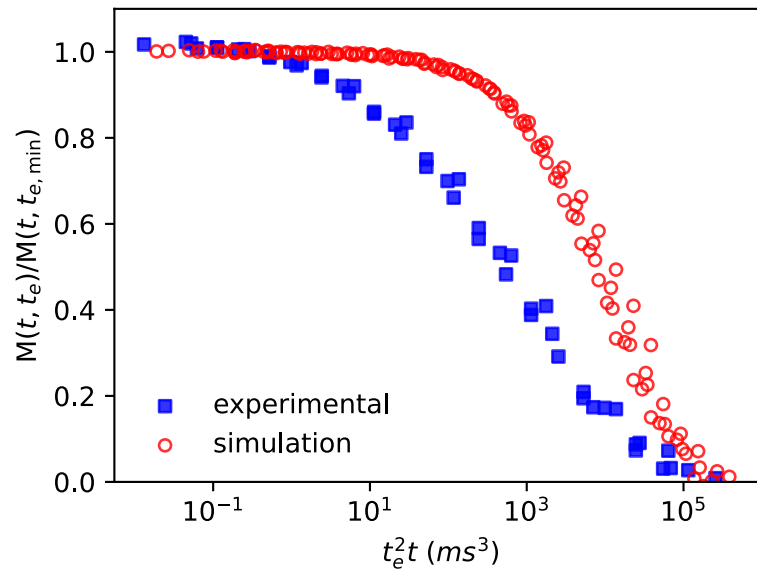


Figure 5.17: **Bandera Gray**: NMR Multi- τ decay data from experimental measurements (blue) and RW simulations (red).

materials of the sample. The image of the sandstone was acquired at a resolution of 2.0 micrometers per voxel. Once again, we provide both the composition obtained from the mineralogy study of the sample and the composition of the phases in the final image segmentation. The majority of this sample is composed of Quartz, with smaller portions of Feldspar, Kaolinite, and Illite. It is worth noting that it was not feasible to distinguish Feldspar and Calcite in the segmentations. As in previous cases, this phase was assumed to be Feldspar, using its magnetic susceptibility in the calculations. Furthermore, it was

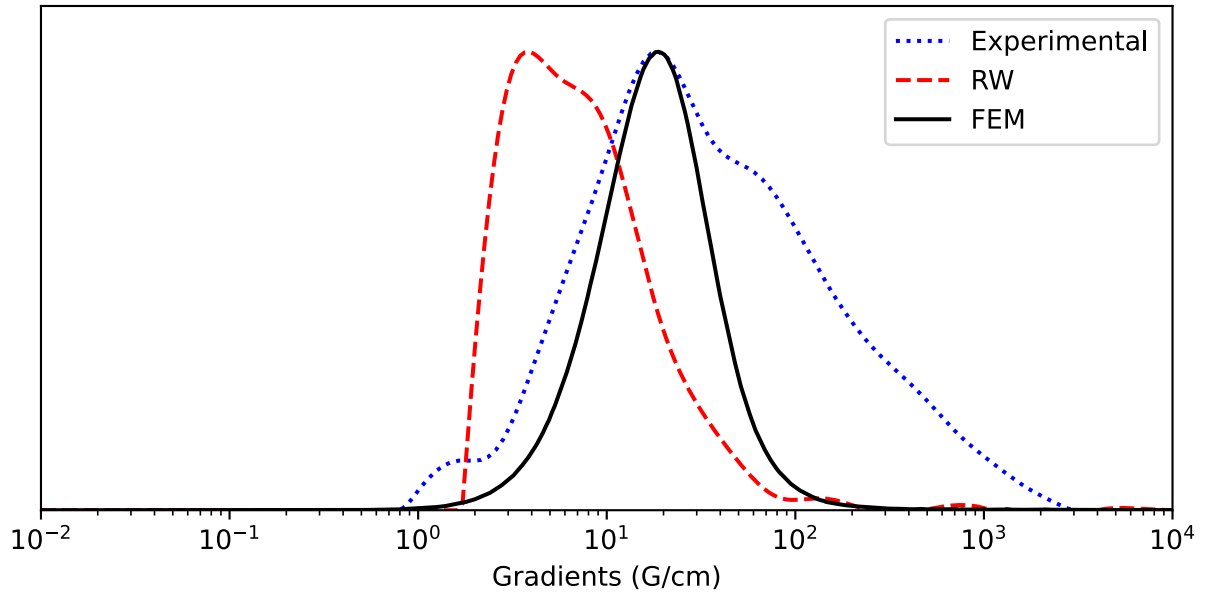


Figure 5.18: **Bandera Gray**: Internal magnetic field gradient distribution estimated from three different approaches: experimental NMR multi- τ (blue dotted lines), simulated NMR multi- τ (dashed red lines), and finite elements method (solid black curve).

observed that Illite has a low representation in the segmentation. This component has the highest magnetic susceptibility of the constituent materials and its absence could potentially lead to weaker internal gradients in the sample.

Table 5.8: **Kirby**: x-ray diffraction mineralogy.

material	%	$\chi (\times 10^6)$
pores	22.7	-9.2
quartz	61.6	-16.4
K-feldspar	9.8	-14.9
kaolinite	3.3	-49.4
illite	2.6	410.0

Table 5.9: **Kirby**: phase segmentation.

material	%	$\chi (\times 10^6)$
pores	21.7	-9.2
quartz	62.8	-16.4
K-feldspar + kaolinite	15.2	-14.9
illite	0.4	410.0

Figure 5.19 shows a representative slice of the Kirby sample, along with the corresponding magnitude of the calculated magnetic field within the porous space. Figure 5.20 shows the multi- τ curve extracted from experimental measurements and RW simulations. Figure 5.21 illustrates the distribution of internal gradients from three different perspectives.

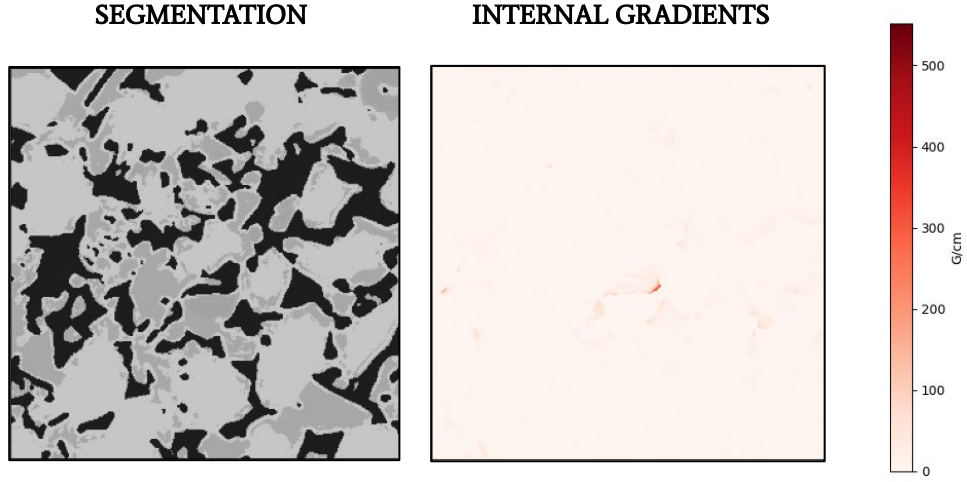


Figure 5.19: **Kirby**: left image: example slice of image segmentation containing pores (black), quartz (gray), feldspar/kaolinite (darker gray) and illite (lighter gray); right image: internal magnetic field gradient color map. In this color map, darker regions represent higher gradient magnitudes.

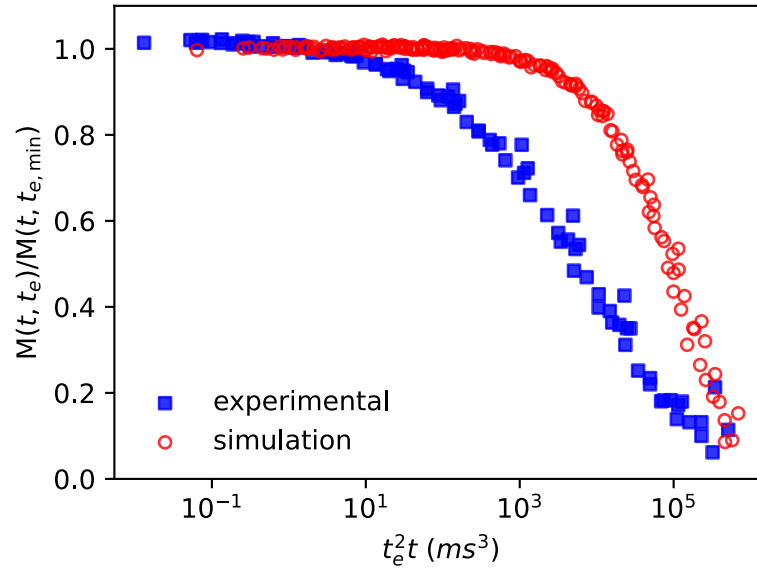


Figure 5.20: **Kirby**: NMR Multi- τ decay data from experimental measurements (blue) and RW simulations (red).

5.3 Results discussion

The first noteworthy point is the evident absence of high-magnitude internal gradients in the results from simulations compared to the experimental results. This pattern is consistently observed across all presented results and can be attributed to the image's inability to represent the porous region below the resolution of acquisition. According to the model, it is expected that the internal magnetic field gradients with the highest

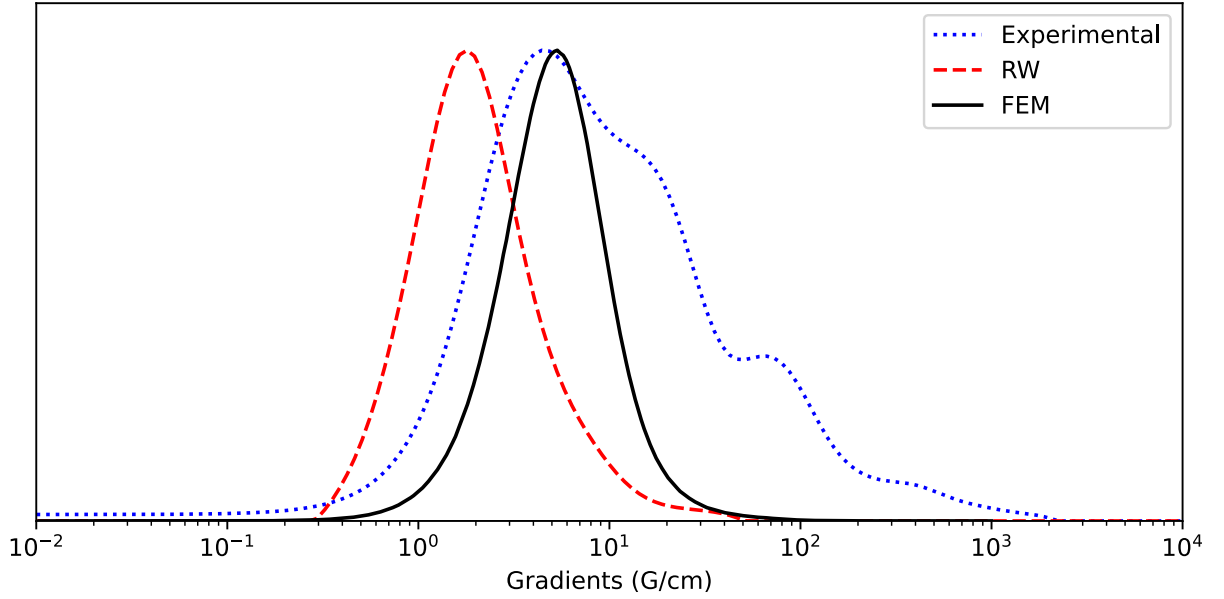


Figure 5.21: **Kirby**: Internal magnetic field gradient distribution estimated from three different approaches: experimental NMR multi- τ (blue dotted lines), simulated NMR multi- τ (dashed red lines), and finite elements method (solid black curve).

magnitudes occur in the more confined regions, i.e., smaller pores, typically related to clay minerals [27, 20]. Thus, the unrepresented pores are the natural candidates where we would expect to observe these occurrences.

Taking this into account, another significant point to highlight is that, for the sintered sample and for the sandstones Berea Stripe and Boise Idaho Brown, a favorable agreement between simulation and experimental results was observed for low-magnitude internal gradients estimations. Notably, these results align particularly well in terms of the magnitudes exhibiting the highest occurrence of gradients. However, this agreement does not extend to the results of the Bandera Gray and Kirby. We observed that the simulations from these samples underestimated the internal gradients. As previously mentioned, it can be attributed to the insufficient representation of materials with high magnetic susceptibility in the segmentations, such as Illite and Chlorite. The correct representation of these materials in the segmentation would likely induce high gradients in their surroundings. Due to the size of their grains, the presence of clay materials result in smaller pore sizes in the samples. Therefore, the sample will present a microporosity below the resolution of the acquired images. To address this challenge, one potential strategy involves calculating the magnetic field inside this microporosity region and employing upscaling techniques to transfer these results to the pore scale of micro-computed tomographic image. This approach is out of the scope of the present work, but can be found in [20].

One final noteworthy aspect is the significant disparity between the distribution of internal gradients estimated through the multi- τ methodology and the distribution of internal gradients obtained directly from the FEM-calculated magnetic field in the rock samples. Notably, for certain samples like Berea Stripe and Bandera Gray, the peaks of the distributions appear to be shifted by up to an order of magnitude, indicating that the simulations experience smaller field variations compared to the actual field. It is important to remember that for the RW simulation, the FEM-calculated field represents the actual field present in the sample, i.e., these are the field fluctuations the simulated spins are actually experiencing. The ideal methodology to estimate the internal gradients distribution should yield overlapping curves, recovering the *true* gradient distribution. However, this "gap" is to some extent expected as discussed in [29, 1]. In fact, the internal gradients estimated by the multi- τ methodology are effective measures, i.e., the average gradient experienced by magnetized spins during the inter-echo interval.

It is worth highlighting however that the curves do indeed exhibit significant overlap in the results obtained for the sintered sample. This sample not only presents grains and pores with considerably simpler geometry, but also features a fairly open porous space. This results in the majority of spins experiencing unrestricted diffusion during the experiment and simulation. Since the multi- τ methodology assumes the asymptotic short-time regime as a premise, it is only natural to obtain consistent results for this sample. On the other hand, we believe that it is not reasonable to expect that the majority of spins experience unrestricted diffusion for samples with complex pore networks, such as reservoir rocks. These samples will naturally exhibit a higher heterogeneity in its pore space. With that in mind, in the next section, we explore an approach to try overcome this limitation in the methodology.

5.3.1 On the effect of restricted diffusion

In this section, we propose making certain adjustments to the multi- τ modeling in order to incorporate the effect of motion restriction into this methodology. Our suggestion is to simply replace the term that utilizes the coefficient of free diffusion, D_0 , with the apparent diffusion coefficient, $D(t)$. This modification was recommended in [29], although, to the best of our knowledge, it has not been applied in other studies within the literature.

As described in Section 2.3.2, each data point on the multi- τ curve is modeled as Eq. 2.18 to obtain a distribution of gradients, $f(g_{\text{eff}})$. Therefore, the problem of finding the internal magnetic field gradient distribution that best fits the collected data points can

be framed as a minimization problem and solved for g_{eff} as described by Eqs. 3.16 and 3.17.

Note that in this approach each row of matrix A corresponds to a specific observation time, t_i , and echo spacing interval, $t_{E,i}$, while each column corresponds to a $g_{\text{eff},j}$. The proposed approach involves replacing the value of D_0 with $D(t_{E,i})$. Thus, we have:

$$a_{i,j}^* = \exp\left\{-\frac{1}{12}\gamma^2 D(t_{E,i}) g_{\text{eff},j}^2 t_{E,i}^2 t_i\right\}, \quad (5.4)$$

here each row is also associated with a specific apparent diffusion coefficient, $D(t)$, which can be measured, for example, using the method presented in Section 3.2. The rest of the procedure remains unchanged. It is also worth highlighting that replacing D_0 with $D(t)$ effectively directs the regularization towards higher values of g_{eff} , as the new input value is necessarily smaller than the previous one (i.e., $D(t) < D_0$, for $t > 0$). The results presented in Section 5.3 indicate that this is indeed a desirable outcome, since, based on our hypothesis, there is an underestimation of the internal gradients due to the restricted diffusion experienced by the magnetized spins during the echoes interval.

5.3.2 Comparing different approaches in isolated spherical pores

The first step in our analysis involves testing this novel approach in a porous medium with a simple geometry. With that in mind, we conducted simulations in an isolated spherical pore with a radius of 100 micrometers under a uniform internal gradient. In Figure 5.22, the results for estimating the distribution of internal gradients are presented when the magnetic field is uniform in the Z-direction and assumes values of 10, 100, and 1000 G/cm.

Remarkably, unlike the previous methodology that underestimated the true gradient at lower magnitudes, the new approach accurately recovers the specified gradient value in all cases. An interesting point to note is that both approaches seem to recover the correct estimation for the underlying internal magnetic field gradient as higher magnitudes of this gradient are applied. This can be explained by the faster relaxation imposed by such strong magnetic field fluctuations. In this scenario, most spins lose coherence before truly experiencing motion restriction. In that sense, we observe that the higher gradient scenario better fits the original hypothesis of unrestricted diffusion, upon which traditional multi- τ modeling is based. It is worth mentioning that the new approach is versatile enough to also incorporate this configuration.

In the second step, we analyze the effect of changing the pore size. In Figure 5.23, the same experiment was repeated for isolated spherical pores of radius 1 and 10 micrometers. As shown, the new approach fails to recover the correct value as smaller pores are introduced. This modification can affect the relaxation regime experienced by the spins. Under the same gradient conditions, a smaller pore pushes the system to the motional averaging regime.

5.3.3 Applying new approach in sandstone micro-computed tomographic images.

Finally, we applied the new approach to the segmented images of reservoir rocks previously presented. The results are depicted in Figures 5.24, 5.25 and 5.26 for sample Berea Stripe, Boise Idaho Brown and Bandera Gray, respectively. Additionally, for each sample, the

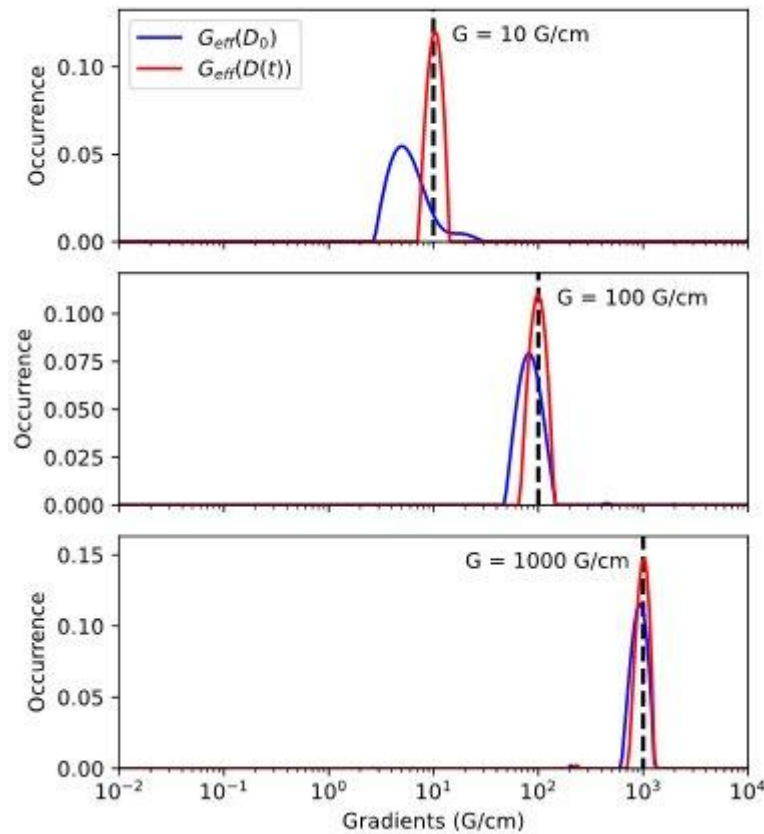


Figure 5.22: Internal gradients distribution for a spherical pore of radius 100 μm under three distinct uniform magnetic field gradients. Results are estimated from two different approaches: traditional multi- τ method (blue) and adapted multi- τ (red). The adapted method incorporate the apparent diffusion coefficient to account for restricted diffusion experienced by magnetized spins in the system.

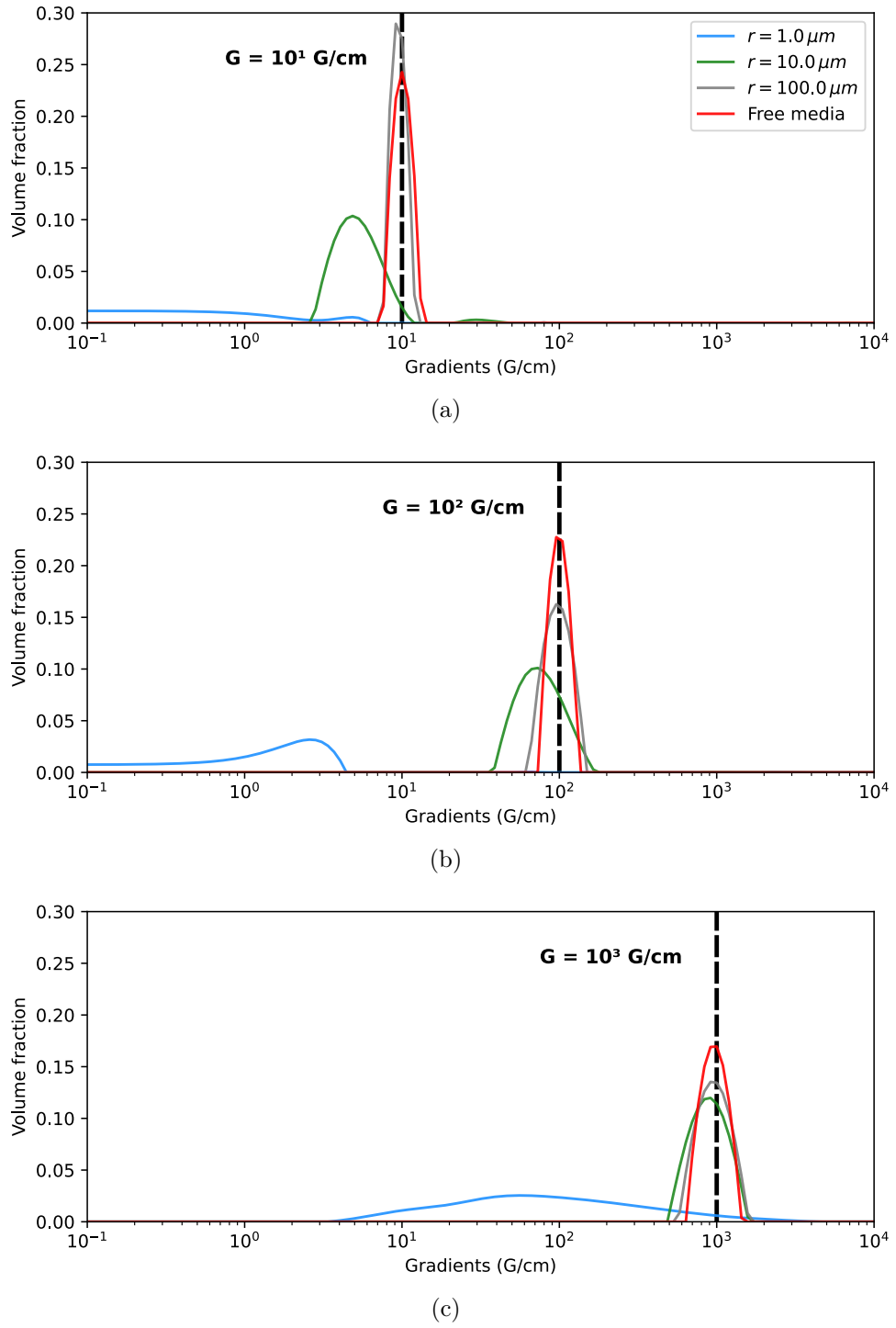


Figure 5.23: Internal gradients distribution for a spherical pore of varying radius under three distinct uniform magnetic field gradients. Results are estimated using the adapted multi- τ methodology that incorporates restricted diffusion.

results of the estimated internal gradient distributions using the previous methodology and the distribution obtained directly from the FEM-calculated magnetic field are also presented.

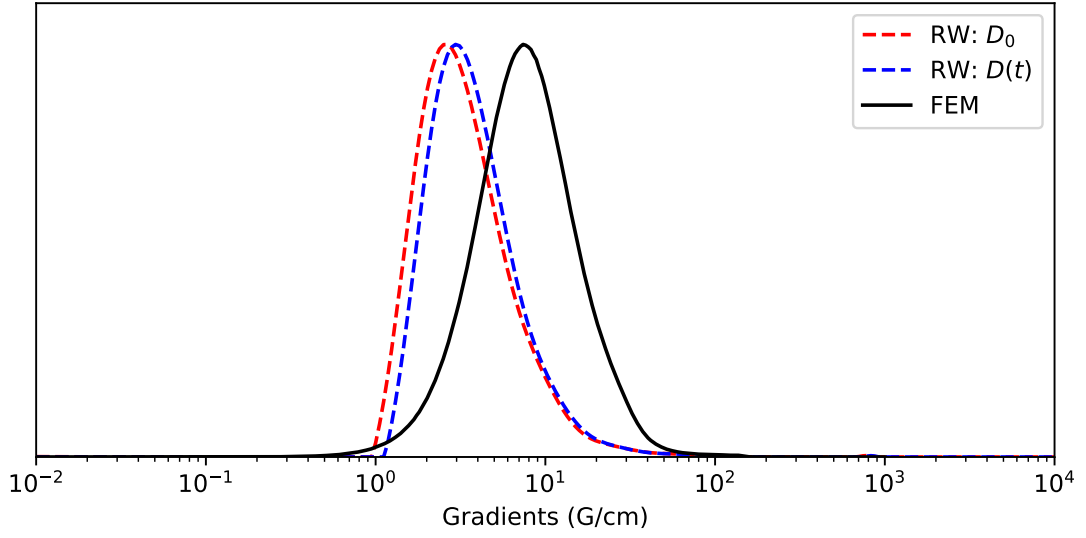


Figure 5.24: **Berea Stripe:** internal gradients distribution from three computational methods: traditional multi- τ (dashed red); multi- τ adapted for restricted diffusion (dashed blue); finite elements method (solid black).

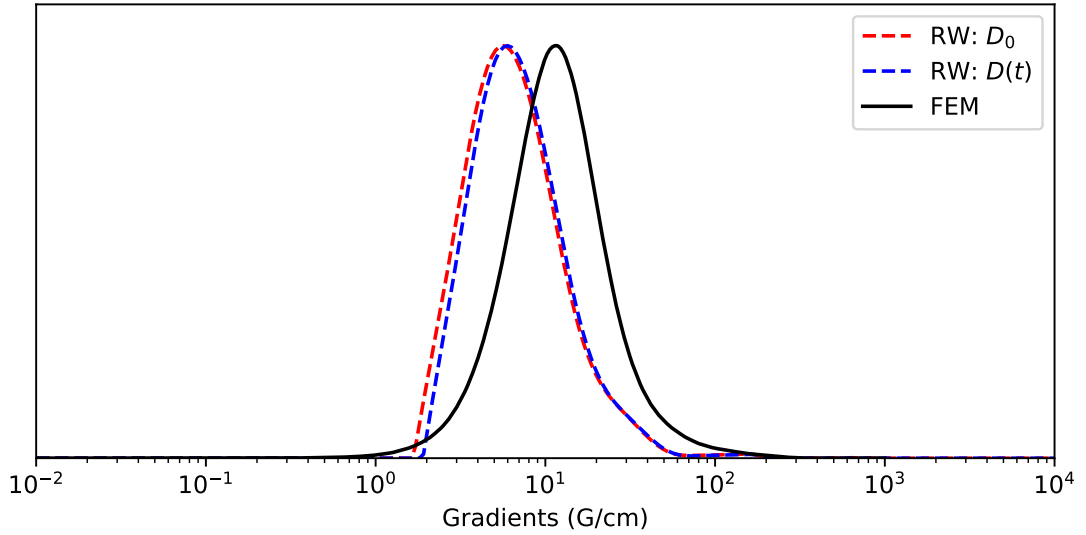


Figure 5.25: **Boise Idaho Brown:** internal gradients distribution from three computational methods: traditional multi- τ (dashed red); multi- τ adapted for restricted diffusion (dashed blue); finite elements method (solid black).

In all cases, we observe that the internal gradient distributions estimated from the new approach are closer to the distribution derived directly from the background magnetic field (i.e. the FEM internal gradient distribution). However, here we observe a limited relative improvement for this new approach. The new approach essentially corrected the problem in simpler geometries, so we can hypothesize that the primary reason for this low relative gain lies on the complex geometry of the pore network in reservoir rocks.

Also, the results show that Hürlimann's hypothesis of considering unrestricted diffu-

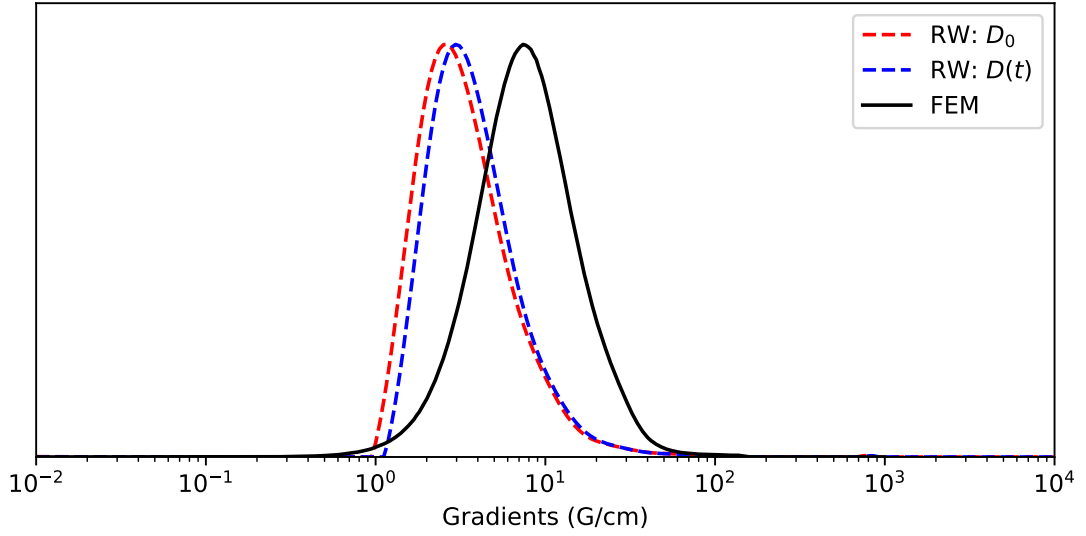


Figure 5.26: **Bandera Gray**: internal gradients distribution from three computational methods: traditional multi- τ (dashed red); multi- τ adapted for restricted diffusion (dashed blue); finite elements method (solid black).

sion by adopting D_0 is actually very reasonable [40]. The marginal gain observed from results in sandstone samples when the dynamic approach proposed here is adopted may not justify the additional required complexity. On the other hand, if we were able to include pore size and restriction information in the formulation, better results would be expected.

Another point to consider is the presence of regions where the magnetic field exhibits more pronounced variations due to the concentration of materials with higher magnetic susceptibility. The formulation of the multi- τ methodology does not specifically address this aspect, which can result in such inaccuracies.

Chapter 6

Summary and conclusions

In this study, we presented an efficient implementation for image-based computer simulations of the NMR diffusometry and relaxometry experiments in porous media. The diffusion and relaxation implementations are inspired by the Pulsed Field Gradient Spin Echo and Carr-Purcell-Meiboom-Gill pulse sequence techniques. To simulate Brownian molecular motion, we adopted the well-established Random Walk method. Our approach leverages massive parallel computing at a thread-level granularity, where each particle, or magnetized spin, is simulated on a dedicated GPU core. Modern GPU architectures typically feature thousands of cores working concurrently, enabling highly parallelized execution of computationally demanding tasks such as particle movement, handling image borders, collisions with pore walls, and evaluation of spin dephasing due to interactions with fluctuations on the underlying field. Additionally, we employed standard reduce techniques on the GPU to compute the mean square displacement, PFG amplitudes, and signal summarization, as these tasks involve summations across the entire set of walkers. In order to optimize memory access, we also propose representing the porous volume in a data structure organized by cubic subsets of 4^3 voxels. We show that in such a configuration, walkers tend to remain in the same region 75% of the time. The achieved performance improvement through parallelization and data structure optimization allowed us to conduct simulations consisting of hundreds of thousands of steps on a large volume of particles within seconds on a personal computer.

Regarding diffusion in porous media, we validated our implementation by comparing simulation results with well-established analytical predictions for porous media with simple geometry. We also conducted sensitivity analyses by varying simulation parameters, including the number of particles, step size, and image resolution, to assess their impact on result accuracy. Despite significant variations in the size of the particle set populating

the porous space, our simulations consistently showed minimal deviations in the PFG NMR signal within the small wavevector range. This range is particularly important as it allows us to recover the time-dependent apparent diffusion coefficient, indicating that a large number of particles is not necessary to achieve accurate results for this measure. However, our investigations revealed that image resolution plays a critical role in capturing the intricate details of the pore network, especially the smallest pore throats that are vital for accurate diffusion modeling. Inadequate resolutions can lead to unreliable diffusion results. Addressing this challenge, we found that adopting shorter step sizes partially compensates for the resolution limitation, but it does not fully mitigate the issue. For instance, in a typical simulation on a 500^3 voxel image with a resolution of 1.0 micrometer per voxel, reasonable results can be obtained with a set of one hundred thousand particles and a step size of 0.5 micrometers. However, in cases of highly heterogeneous porous media or strong surface relaxation effects in the magnetization signal, these values may prove insufficient.

We then curated a dataset comprising 21 procedurally generated synthetic media. Through simulations of the diffusion experiment, we obtained the surface-to-volume ratio and tortuosity for each synthetic medium. Upon visual evaluation of the results, we noted an exponential relationship between both quantities and absolute permeability. To investigate these correlations further, we applied an exponential curve fitting to the data. Subsequently, we conducted diffusion simulations on segmented micro-computed tomographic images of a set of reservoir rock samples, including 14 sandstones and 10 carbonates. By using the Padé approximant, we extracted the surface-volume ratio and tortuosity from the observed apparent diffusion phenomena. For the sandstone samples, we established a strong correlation between both measures and absolute permeability. However, in the case of carbonates, we found that the absolute permeability showed a weak correlation with the surface-volume ratio, and only the tortuosity showed promising results.

We also performed simulations incorporating surface relaxation effects to compare the results with experimental measurements. In this study, we observed excellent agreement on sandstone samples. However, in the case of carbonates, only a small subset of the samples exhibited a similar level of agreement. In this context, a significant portion of our carbonate dataset exhibited experimental data that deviated from theoretical predictions. We speculate that this behavior may come from the presence of vugular pores, which can be orders of magnitude larger than the average pore size and may not be accurately represented in the acquired images. The existence of these vugs, along with

the fraction of magnetized spins confined within this region, experiencing unrestricted diffusion, could explain the observed signal recovery. To simulate these conditions, we conducted an experiment by combining the simulated signal from the image with the diffusion signal within an isolated spherical pore. The results of this experiment in the Edwards Yellow sample suggest that this may indeed be one of the factors contributing to this phenomenon and could account for the discrepancy between the apparent diffusion recovered from simulation and experiment. In this context, as indicated by our findings, measurements of apparent diffusion have the potential to aid in permeability prediction. We anticipate that this work will serve as a valuable reference not only for those interested in simulating NMR diffusion experiments but also for guiding the design of new permeability prediction models and enhancing the interpretation of NMR diffusion signals, which inherently possess numerous intricacies.

We also expanded the effects of diffusion in the context of the NMR transverse relaxation experiment (T_2 decay) in this work. Specifically, we incorporated the relaxation mechanism resulting from the interaction between diffusing magnetized spins and spatial variations of the background magnetic field into T_2 decay simulations. These variations arise from the pronounced contrast in magnetic susceptibility between the saturating fluid and the constituent materials of the rock matrix, particularly notable in clay-rich sandstones. To incorporate this effect, a two-step workflow was implemented. Firstly, the background magnetic field was computed by solving Maxwell's equations under the zero current condition. An image-based finite element method was employed for this purpose, using segmented images in which the solid phase was segmented into mineral phases representing the rock composition, along with the magnetic susceptibility of each material. Subsequently, the T_2 decay was simulated using the calculated magnetic field as input. The second step consists of the addition of a routine to account for the interaction between spins and the background magnetic field within the conventional simulation code for this decay.

The workflow was validated against analytical predictions found in the literature. Notably, both the implementation employed to calculate the magnetic field and the routine utilized to account for signal attenuation resulting from diffusion relaxation exhibited excellent agreement with these predictions. Subsequently, T_2 decay simulations were conducted on segmented micro-computed tomographic images of reservoir rock samples. Adhering to the multi- τ procedure, these simulations were performed for varying echo spacings, i.e., the interval between decay acquisitions. In that sense, after normalizing the signal, we build a curve combining various decays and apply regularization to re-

cover an estimate of the internal gradient distribution within the samples. A comparative analysis was carried out between the simulation results and experimental data. This comparative study was performed in a sintered sample, as well as in four distinct sandstones, namely Berea Stripe, Boise Idaho Brown, Bandera Gray and Kirby. The findings revealed a predominant agreement between the experimental and simulation estimates of gradient distribution, especially when comparing the region of peak occurrences for the majority of the studied samples. However, this was not observed for sandstones Bandera Gray and Kirby. The disparity in the results for these particular samples may be attributed to the insufficient quality of their segmented images, as they failed to accurately capture constituent materials with higher magnetic susceptibility that are present in the mineralogy of these samples. We also noted that all the simulations were unable to capture internal gradients of higher magnitudes. According to the theory, these gradients will generally manifest at smaller pores which may not be well represented in the resolution of acquisition.

Lastly, we compared the internal gradient distributions recovered through the multi- τ simulation with those directly derived from the magnetic field calculated using FEM. We noted a systematic underestimation of the gradients estimated via simulations in sandstones, leading us to hypothesize on the influence of restricted diffusion, which is neglected by the traditional methodology. In this context, we proposed a novel approach that accounts for the restriction effect and incorporates the apparent diffusion coefficient in the calculations. Encouraging results were obtained through experiments conducted on isolated pores featuring simple geometries and uniform internal gradients. For these cases, the new approach resulted in significantly improved agreement with the applied gradients. However, despite its promise, when this new approach was tested on the data obtained from simulations in sandstones, we only got marginal improvements. We believe that this outcome can be attributed to factors such as the increased complexity of the pore network geometry and the presence of regions characterized by more pronounced gradients, particularly in regions close to materials with high magnetic susceptibility.

6.1 Future work

6.1.1 Machine Learning models for permeability prediction

The development of permeability prediction models using the data obtained from diffusion simulations represents a natural next step based on the results presented in this

study. In this regard, our objective is to explore machine learning models, which typically require a large datasets for effective training and validation. Relevant examples of similar approaches can be found in the literature [43, 66, 79, 81].

Furthermore, we have already compiled a digital rock library containing over 40,000 procedurally generated synthetic porous media. Flow simulations using the FEM method have been conducted yielding estimates for absolute permeability. A subset of these digital rocks was generated using the same algorithm [42] employed to generate the synthetic media presented in Section 4.2 of this work. Additional algorithms were utilized to generate synthetic media with more intricate porous structures. We anticipate that this increased variability will enhance the generalizability of our prediction models. Therefore the workflow entails: i) extracting geometric parameters, such as porosity, surface-to-volume ratio, and tortuosity, from the diffusion simulations presented in this study; ii) using these parameters as input variables of the prediction models; iii) training and fine-tuning data-driven models to predict permeability. Thus, absolute permeability serves as the target variable in a classic supervised learning problem.

6.1.2 Simulation of the DT_2 experiment

Another suggested approach is to simulate the NMR experiment that generates D - T_2 maps (NMR diffusion vs. transverse relaxation, T_2). This experiment involves combining the diffusion experiment with the CPMG sequence, which produces the T_2 decay. We believe we have all the necessary computational tools to develop the code for simulating this procedure.

6.1.3 Simulation of diffusion in multiphase systems

Simulating the apparent diffusion in multiphase systems, where two or more fluids fill the porous space, is another area of particular interest. We are currently developing a code to handle water-oil systems under different wettability scenarios. In addition, advanced imaging equipment and techniques enable studies that simulate conditions of confined multiphase fluid encountered in situ. Moreover, simulating these complex systems has the potential to provide estimations of relative permeability in reservoir rocks.

6.1.4 Multi-scale methods for addressing vugular porosity

As discussed in the section 4.3.3.3, the presence of vugular porosity can potentially affect the accuracy of diffusion simulations in carbonates. To overcome this challenge, applying multi-scale methods is a possible solution. In this approach, simulation results obtained from images acquired at the typical pore scale resolution can be combined with results from images obtained at coarser resolutions, which can adequately represent the vuggy pores.

6.1.5 Multi-scale methods for addressing internal gradients in microporosity

Another application of multi-scale methods suggested in the context of this study concerns the impact of microporosity on the estimation of internal gradients. This issue is particularly relevant in clay-rich sandstones. In this regard, we can refer to the work [20]. In this study, the magnetic field within these nanostructures is estimated using the Dipolar Sum method. A potential alternative at our disposal would be to calculate it using the same Finite Element implementation employed in other parts of the study. Once this field is calculated, we can "recover" it to the pore scale and proceed with the simulations following the presented workflow.

6.1.6 Inverse problem formulation for determining magnetic susceptibility of mineral components

Magnetic susceptibility is a parameter that involves some level of uncertainty. In this study, we adopted the values listed in [39, 63]. However, as discussed in [17, 45], in many cases, this value is not well-defined and entails some uncertainty. With the tools developed and presented in this study, we can formulate an inverse problem to determine the magnetic susceptibility that best fits the data of the internal gradient distribution or even the T2 relaxation distribution. Since this is inherently a complex and not well posed problem, we can use metaheuristics such as genetic evolution algorithms to solve the minimization problem, finding the value that optimally matches the experimental data. Similar to the approach employed in this study, we would follow the two-step workflow presented in Section 2.3. However, in this case, the magnetic susceptibility value associated with each phase identified in the image segmentation would be an adjustable parameter. Thus, the objective function would aim to minimize the "distance" between the simulated distribu-

tion curve and the target distribution curve, which may be experimental data obtained from laboratory measurements.

6.1.7 Inverse problem formulation for estimating the effective internal gradient as a function of pore size

Another potential application of inverse problem formulation that we have identified relates to the effective internal gradient experienced by magnetized spins. In this regard, we draw inspiration from the work [5, 6]. In their study, the authors leveraged simulations and experimental T_2 decay data in reservoir rocks to estimate surface relaxivity as a function of pore size. Once again, evolutionary metaheuristics were employed in the optimization problem. In our case, we propose using the distribution of internal gradients obtained through multi- τ analysis to estimate the effective internal gradient as a function of pore size.

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APPENDIX A – PFGSE RW Simulation Workflow

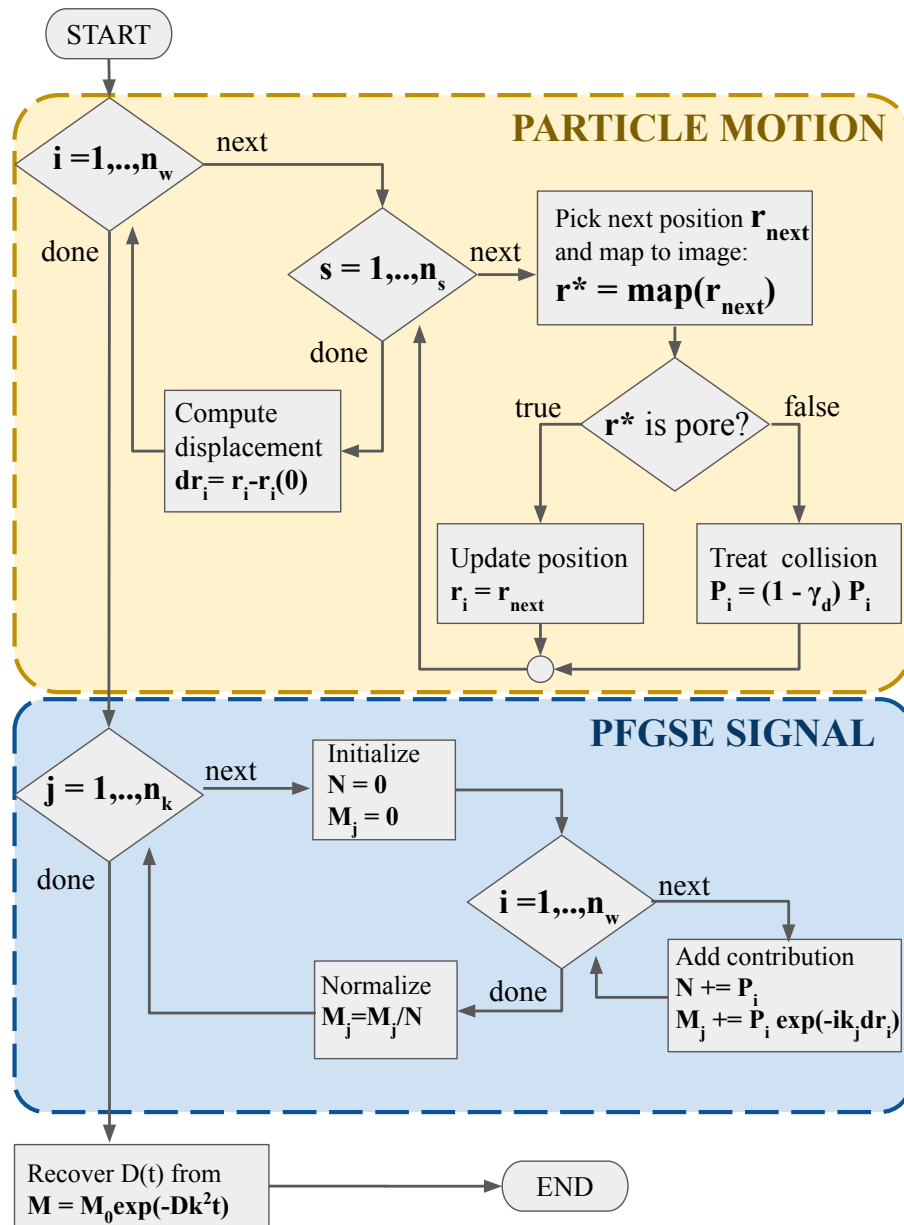


Figure A.1: **PFGSE RW simulation:** algorithm workflow.

APPENDIX B – CPMG RW Simulation Workflow

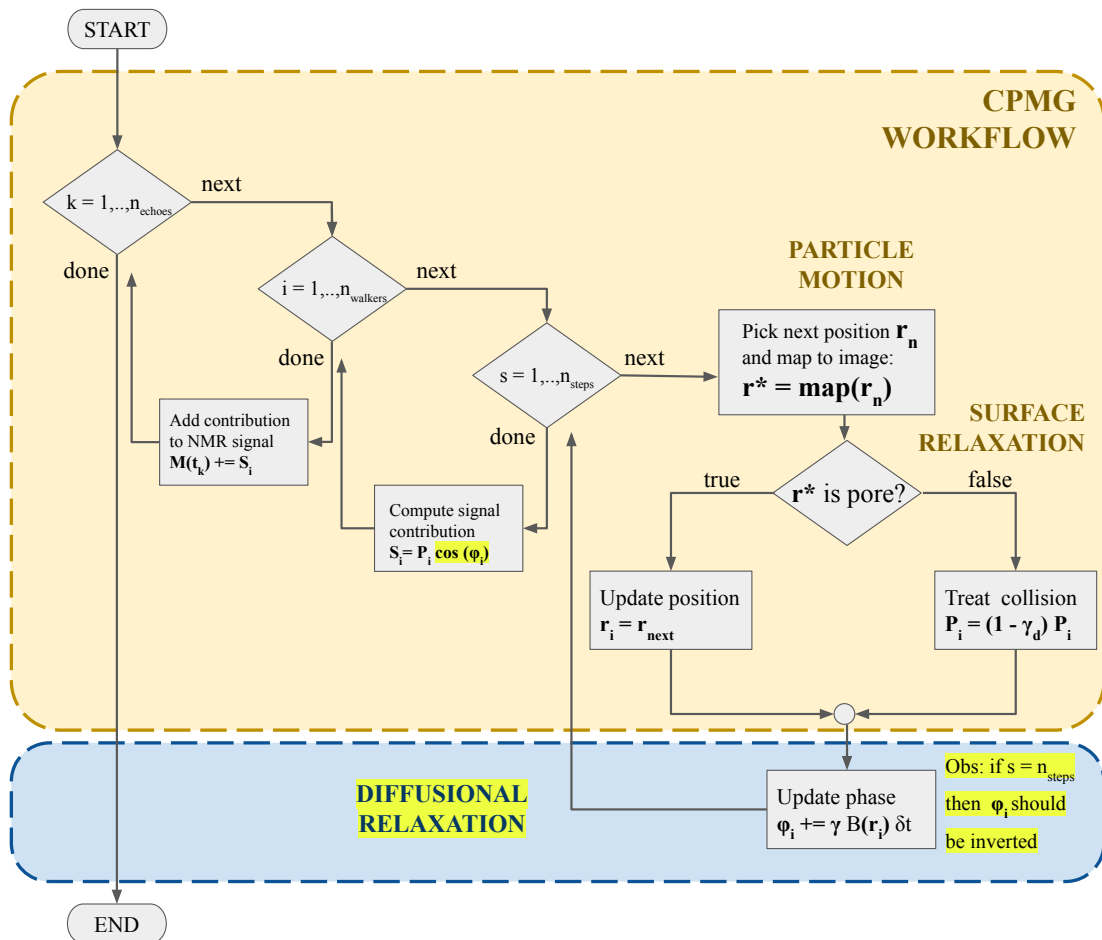


Figure B.1: CPMG RW simulation: algorithm workflow.